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AUTOTRONICS - AELZG533/AEZG533 -

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Engineering group



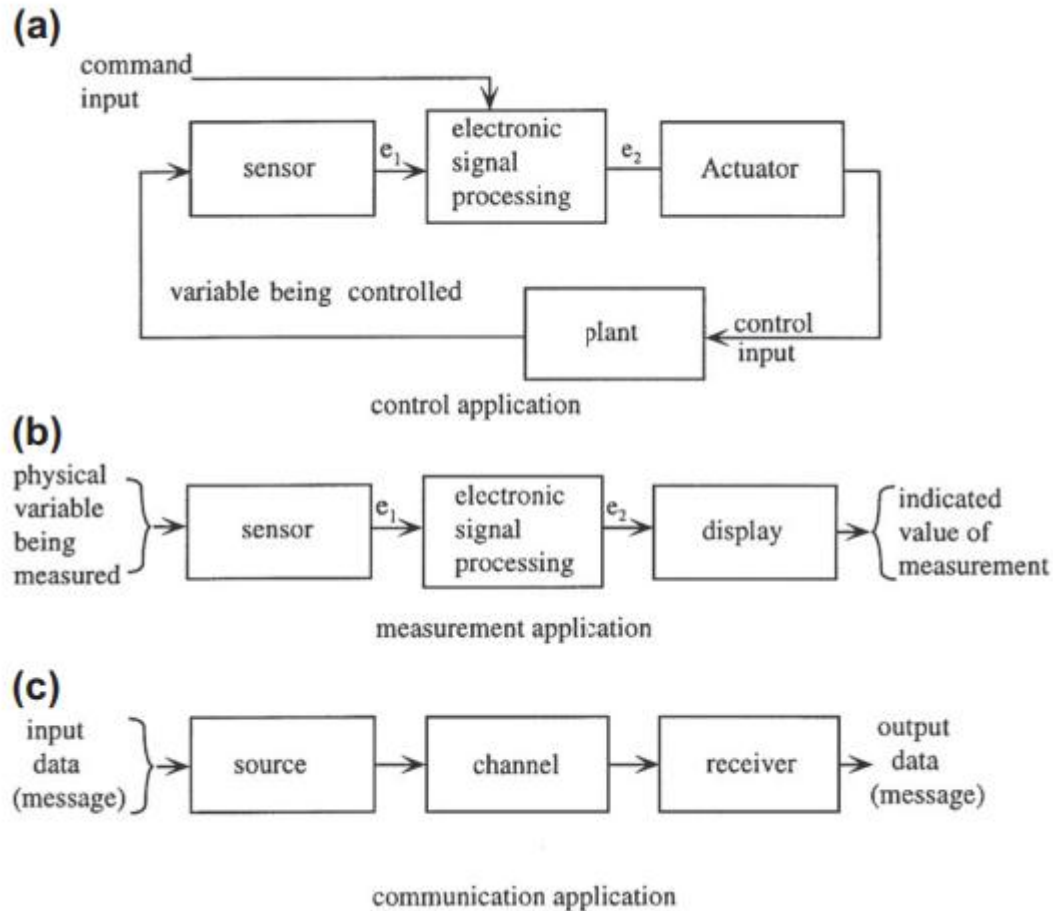
L2 & 3 – Electrical and electronics fundamentals

Outline

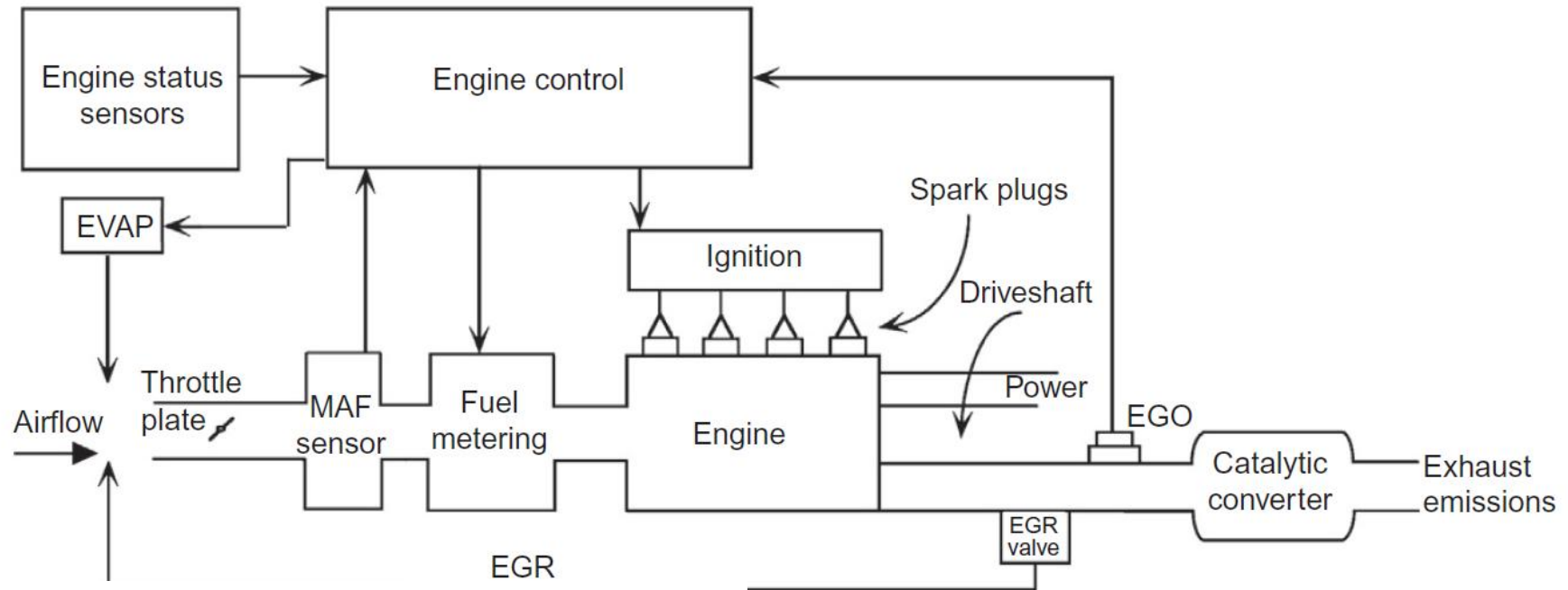


- Last Class
 - i. Introduction to the course
 - ii. Evaluation scheme
 - iii. Everything signal
- Today's class
 - i. Subsystem examples
 - ii. Fundamentals required
 - iii. Electrical and electronics in the vehicle

System flow

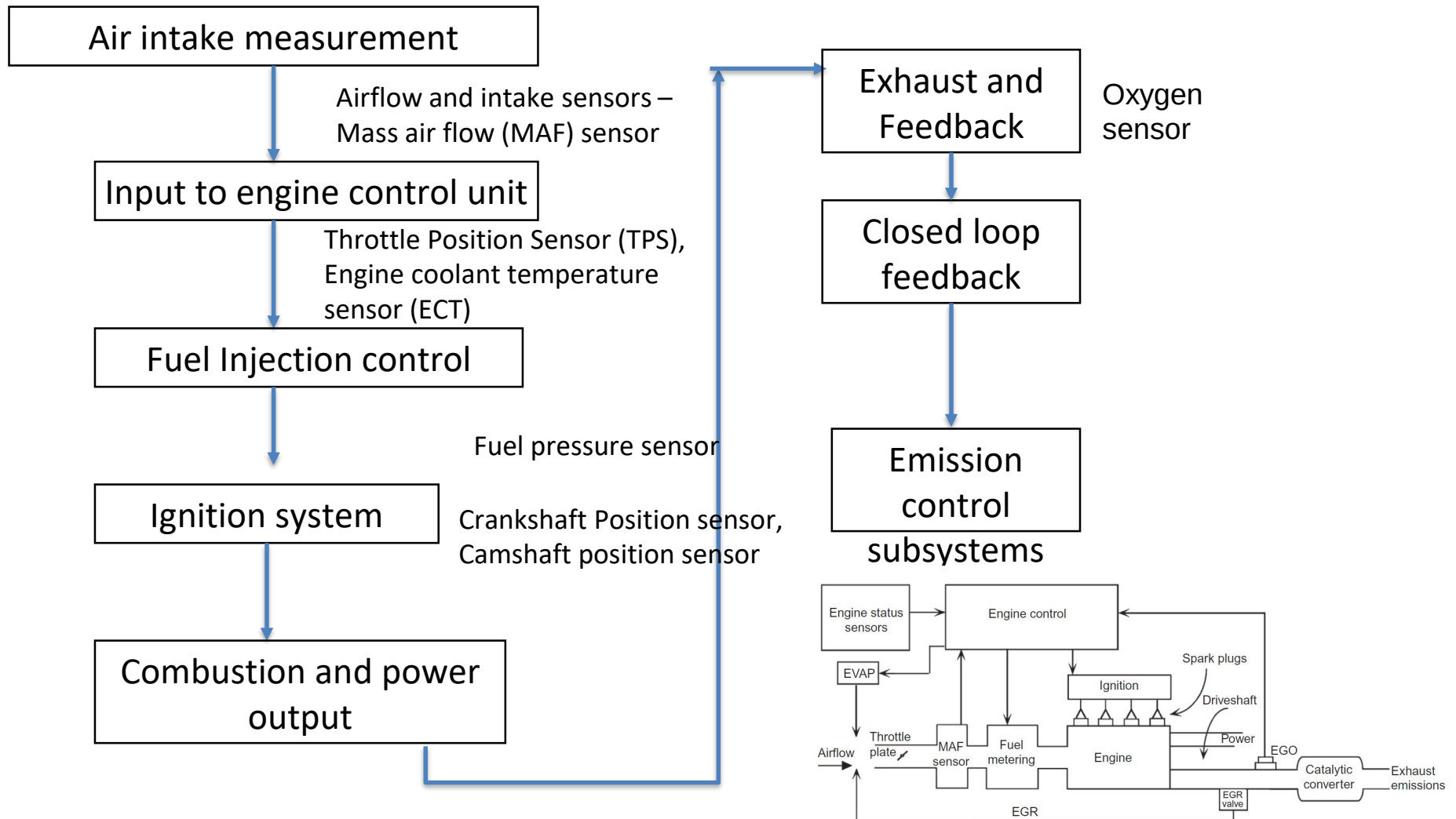


Building blocks - Systems



1. Throttle position sensor (TPS)
2. Mass air flow rate (MAF)
3. Engine temperature (coolant temperature) (CT)
4. Engine speed (RPM) and angular position
5. Exhaust gas recirculation (EGR) valve position
6. Exhaust gas oxygen (EGO) concentration

Engine control System

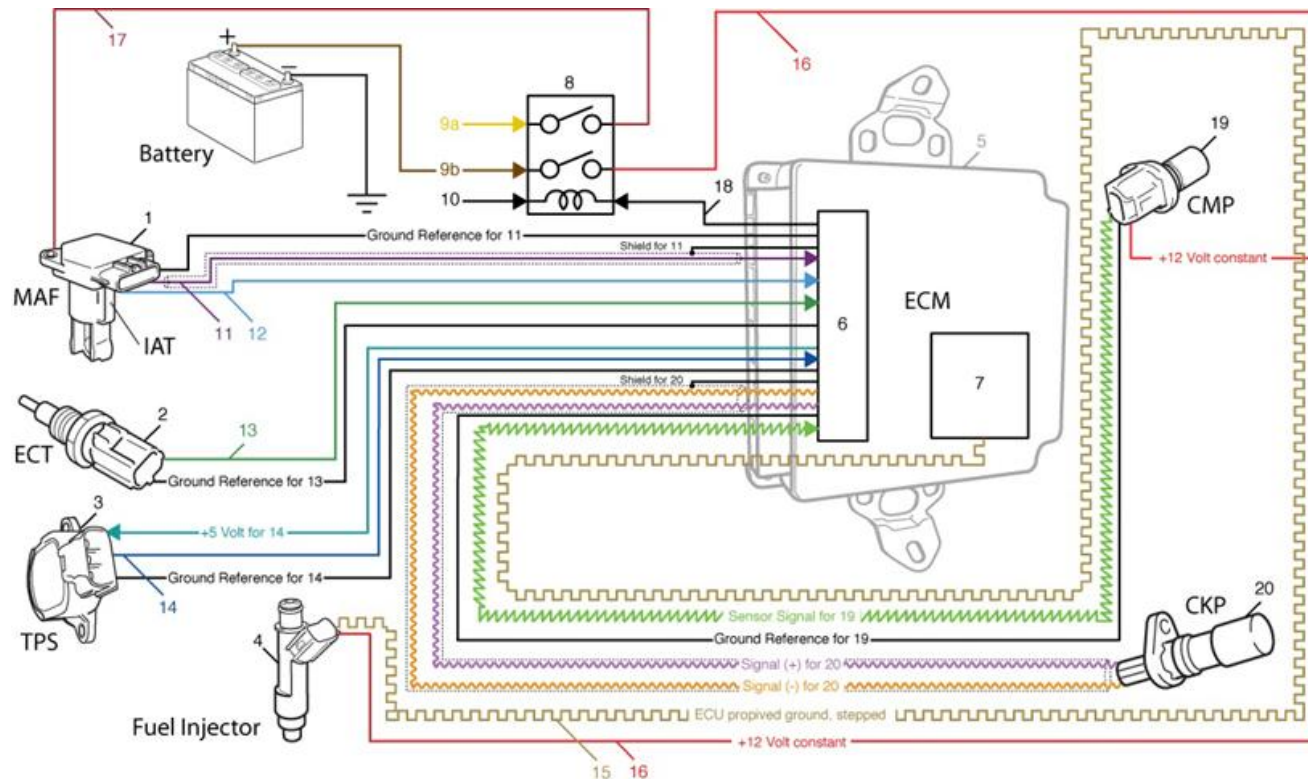


Building blocks - Systems

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1. Mass Air Flow w/ IAT Sensor (MAF)	8. Main or EFI relay	14. Throttle Position Sensor (TPS) sensor signal
2. Engine Coolant Temp (ECT) Sensor	9a. From "EFI or Main" fuse +5v	15. Drive Signals for fuel injector(s), ECU provides ground path
3. Throttle Position Sensor(s) (TPS)	9b. From "EFI or Main" SBF fuse +12v	16. Power Supply for fuel injector(s) and CMP sensor, +12v switched
4. Fuel Injector(s)	10. From Main relay	17. Power Supply MAF and IAT, +5v switched
5. Engine Control Module (ECM)	11. Mass Air Flow (MAF) sensor signal	18. Control signal for Main or EFI relay
6. Central Processing Unit (CPU)	12. Intake Air Temp (IAT) sensor signal	19. Camshaft Position (CMP) sensor
7. Drive circuit for Fuel Injectors	13. Engine Coolant Temp (ECT) sensor signal	20. Crankshaft Position (CKP) sensor

Engine Subsystems — Identify the signals to be measured or monitored.

1) Airflow & Intake Sensor

MAF Sensor — Hotwire

Analog
(0-5)V.

TPS — Potentiometer.

↳ variable resistance
(or) variable voltage (0-5)V.

2) Engine Condition sensor

• Engine Coolant Temperature Sensor (ECT)

— Thermistors
→ resistor → (pD) voltage

• Crankshaft position sensor

- Hall effect — varying magnetic field.
- Inductive — principle of inductor.

• Camshaft position sensor

- Hall effect

3) Exhaust & Emission

Oxygen sensor - lambda. $(0-1)V$.

4) Pressure
MAP - Piezoelectric - capacitive.

5) Knock sensor
Knock - Piezoelectric

What fundamentals?

1) Electrical terminology

→ voltage

→ Current

} sensor OPs are calibrated for different v & I levels

2) Technology

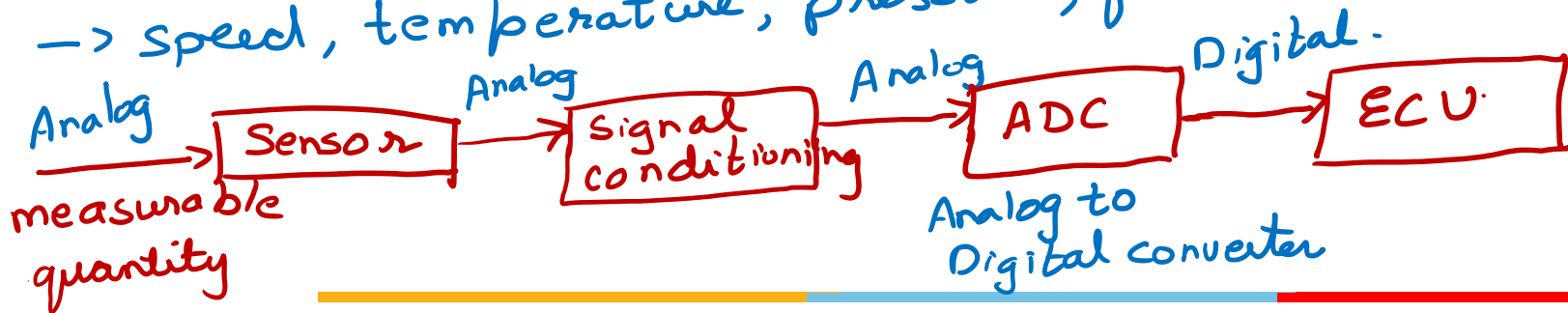
→ Resistive → potentiometers, voltage dividers
series & parallel

→ Inductive → magnetic field

→ Capacitive → voltage, distance betⁿ plates.

3) What are we measuring?

→ speed, temperature, pressure, flow, position.



Sensors in Electric Vehicles

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Thermal Management



Air Conditioning
Pressure & Temperature

Tire Monitoring



Valve Mounted
TPMS



Tire Mounted
TPMS



Tread Depth
Monitor

Electric Motor



Rotor
Position



Stator
Temperature



Oil Cooling Pressure
& Temperature

Brake System



Brake
Pressure



Pedal
Force



Drum
Force

Battery



Passive &
Active Fuses



Current Sensing
Module (IVT)



Insulation
Monitoring Device



Contactors

Sensors in Electric Vehicles

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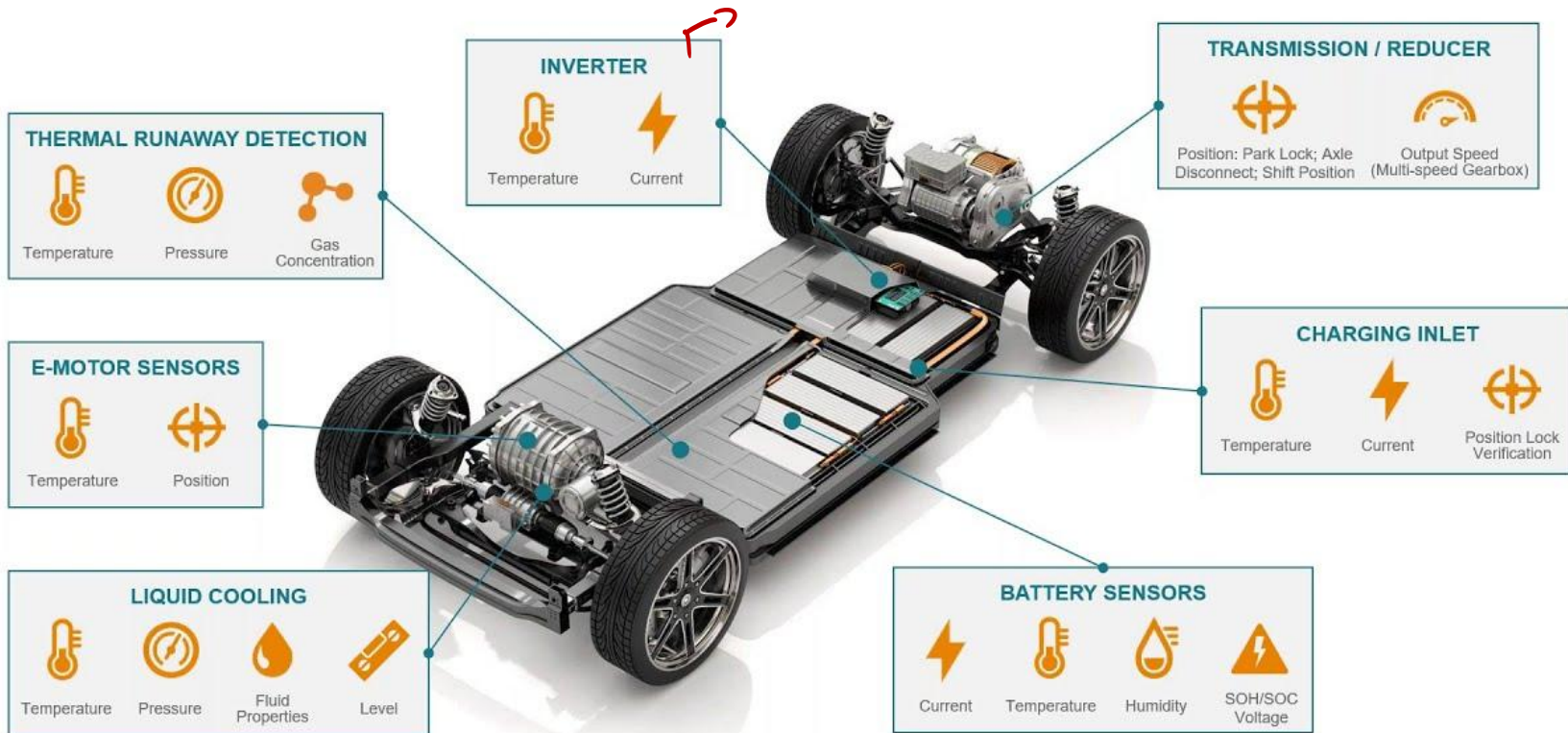
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V, I, temperature.

↳ Clamp meters



Electric Vehicle Powertrain Sensors



- Physical sensors
 - a. Temperature
 - b. Pressure
 - c. Position
 - d. level
- Electrical Sensors
 - a. Current
 - b. Voltage
- Chemical/environmental sensors
 - a. Gas concentration
 - b. Humidity
 - c. Fluid

- ✓ 1) Fundamentals of R, L, C, V, I
- ✓ 2) Fundamentals of magnetic field.
- 3) Signal conditioning
 - ↳ OPAMPs
 - ↳ Filters
- ✓ 4) Semiconductor switches
 - Protection
 - power conversion
- 5) Fundamentals of ADC & DAC
- ✓ 6) Digital fundamentals

Electrical fundamentals

Voltage & current

AC
Amplitude
&
frequency

DC
Amplitude

AC voltage $v(t) = V_m \sin \omega t$
 $\omega = 2\pi f$
 $\hookrightarrow$ frequency

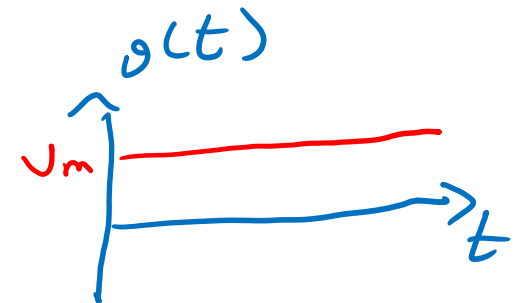
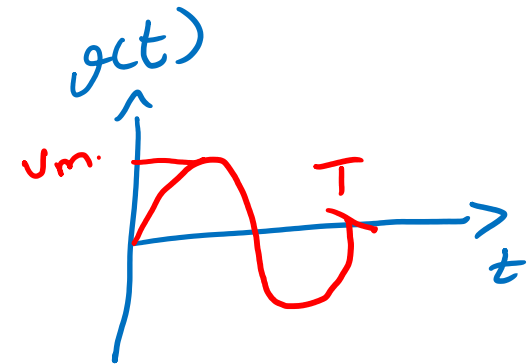
$$T = \frac{1}{f}$$

DC voltage $v(t) = V_m u(t)$
 $u(t) = 1$ for $t \geq 0$
 $= 0$ for $t < 0$
 unit step fn. $\leftarrow$

Ohm's law

$$V \propto I$$

$$\boxed{V = IR}$$



Passive components R, L, C

1) Resistor - dissipative element

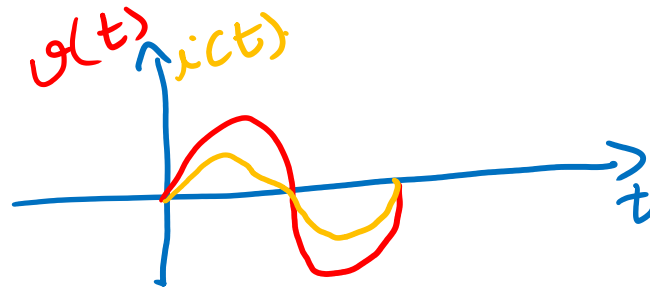

 $V = IR \rightarrow \text{ohm's law.}$

DC

$$V = IR$$

AC

$$V_m \sin \omega t = I_m \sin \omega t R.$$



Voltage & current are in phase

2) Inductor $\rightarrow$ any current carrying conductor produces a magnetic field.

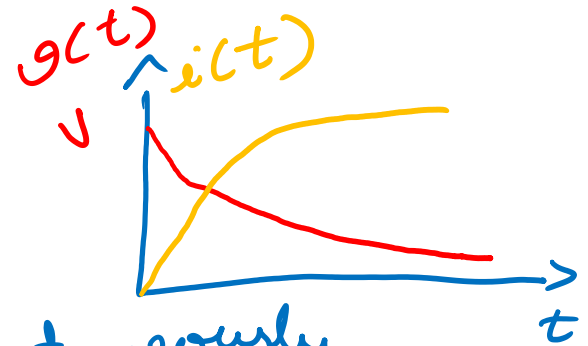
$\hookrightarrow$ Henry

$\rightarrow$ 'N' turn coil wound on a core.

electrical

magnetic

Ohm's law $V_L = L \frac{di}{dt}$

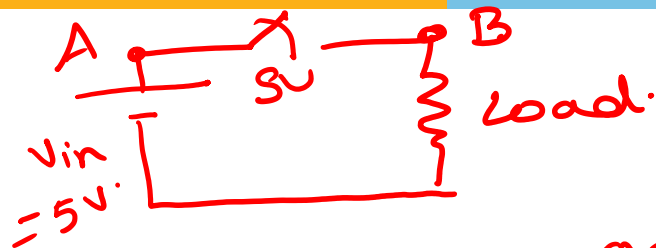


$\rightarrow$ current cannot change instantaneously.
 $\rightarrow$ energy storage element - fully charged inductor acts as a short circuit

$\rightarrow$ a charged inductor acts as a current source.

$$i(t) = I_0 e^{-R/L t}$$

$\frac{L}{R} \rightarrow$ time constant



$S \rightarrow$ open open circuit, infinite resistance

$$\rightarrow I = 0$$

$$V_{AB} = 5V = V_A - V_B$$

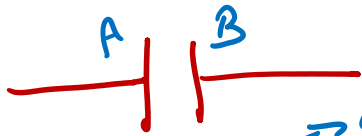
$$V_A = 5V \quad V_B = 0$$

$S \rightarrow$ closed short circuit $\rightarrow$ zero resistance.

$$I \neq 0$$

$$V_{AB} = 0V \quad V_A = V_B = 5V$$

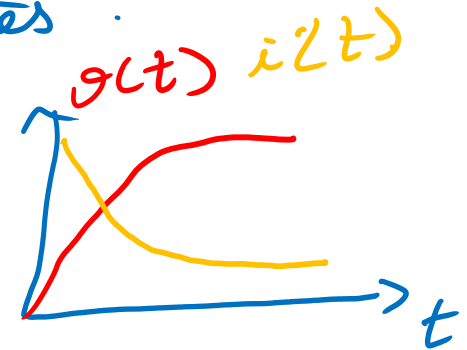
3) Capacitor - a parallel plate capacitor has stored electric field.





$$C = \frac{A \epsilon}{d}$$
 area betⁿ plates
 → permittivity of dielectric
 ← Farad
 → distance betⁿ the plates

→ Ohm's law $V_C = \frac{1}{C} \int i dt$
 $i = C \frac{dv}{dt}$



→ a fully charged capacitor acts as a voltage source instantaneously.

- voltage cannot change instantaneously.
- a fully charged capacitor acts as an open circuit.

- a fully charged capacitor acts as an open circuit.

AC Input



$$f = \frac{1}{T}$$

$$v(t) = V_m \sin \omega t$$

$$i(t) = ? \quad \text{Ohm's law for AC}$$

1) Resistor $V = I R$

2) Inductor $V_L = jI X_L \rightarrow \text{ohm}$

3) Capacitor $V_C = -I j X_C$

$X_L \rightarrow \text{inductive reactance} = \omega L \rightarrow \text{Henry}$

$X_C \rightarrow \text{capacitive reactance} = \frac{1}{\omega C} \rightarrow \text{Farad}$

RLC circuit $V = I Z$

$Z = R + jX$ $X = X_L \pm X_C$

$Z \rightarrow \text{impedance}$

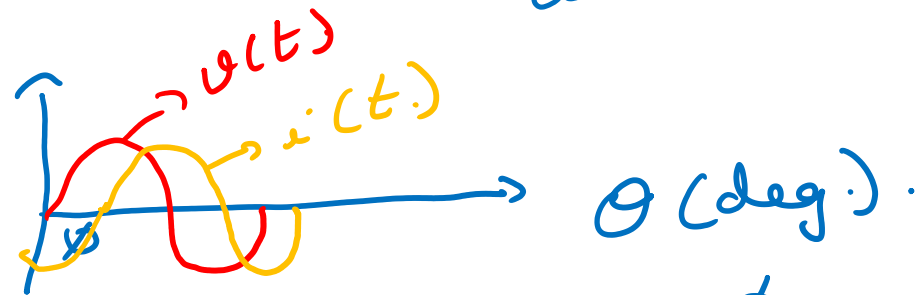
AC input $\rightarrow$ Phasor relationship

$$v(t) = V_m \sin \omega t$$

$$i(t) = I_m \sin \omega t$$

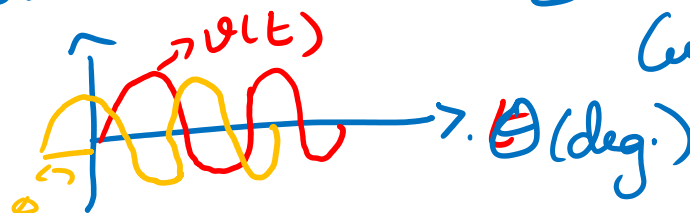
1) Resistor $\Rightarrow V = IR$

2) Inductor $v_L(t) = \frac{di}{dt} L$

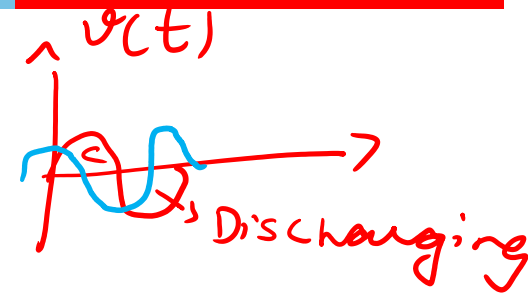
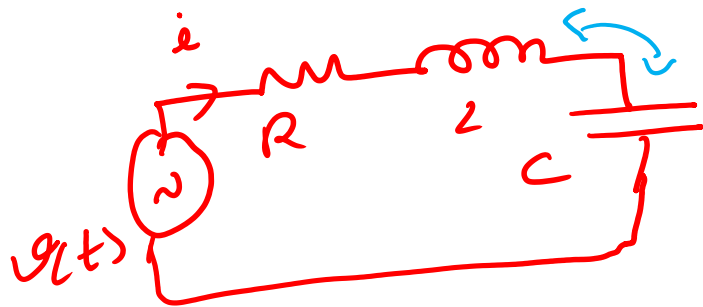


Current lags voltage by ϕ

3) Capacitor $v_C(t) = \frac{1}{C} \int i dt$



Current leads voltage by ϕ



$$v(t) = V_m \sin \omega t$$

$i(t) \rightarrow AC$

$C \rightarrow I$ leads V

$\rightarrow$ filter.

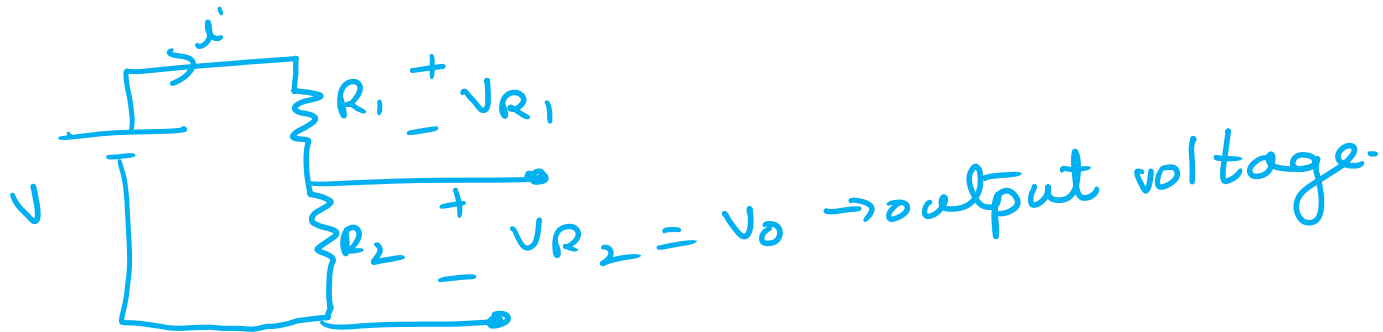
$\rightarrow$ L & C start charging

$$X_L < X_C$$

$$L < C \quad \forall$$

$\rightarrow$ exchange of energy betⁿ L & C.

Voltage divider.



$$V_0 = V \frac{R_2}{R_1 + R_2}$$

$$V_{R1} = V \frac{R_1}{R_1 + R_2}$$

Series circuit voltage divides, current is same.

$$V = V_{R1} + V_{R2}$$

$$V = I R_1 + I R_2$$

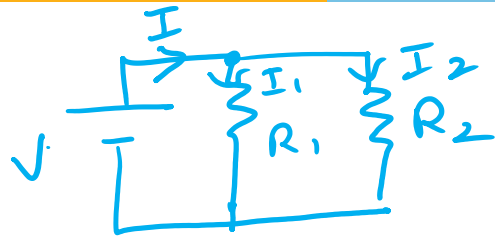
$$V = I (R_1 + R_2)$$

$$\boxed{\frac{V}{I} = R_1 + R_2}$$

ohm's law

$$R_{eq} = R_1 + R_2$$

Current division



$$V_{R_1} = V_{R_2} = V$$

$$\begin{aligned} I &= I_1 + I_2 \\ &= \frac{V}{R_1} + \frac{V}{R_2} \\ &= V \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \end{aligned}$$

$$\frac{V}{I} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} =$$

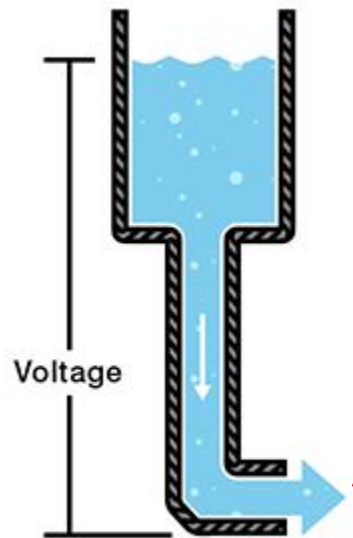
$$\frac{R_1 R_2}{R_1 + R_2}$$

$$\boxed{\frac{V}{I} = \frac{R_1 R_2}{R_1 + R_2}}$$

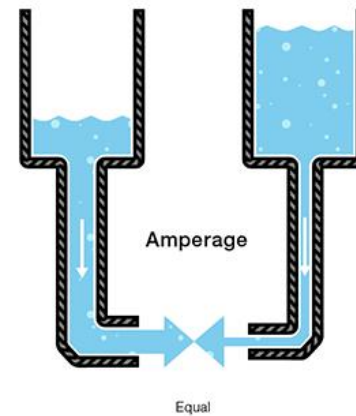
R in parallel

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Voltage / Current

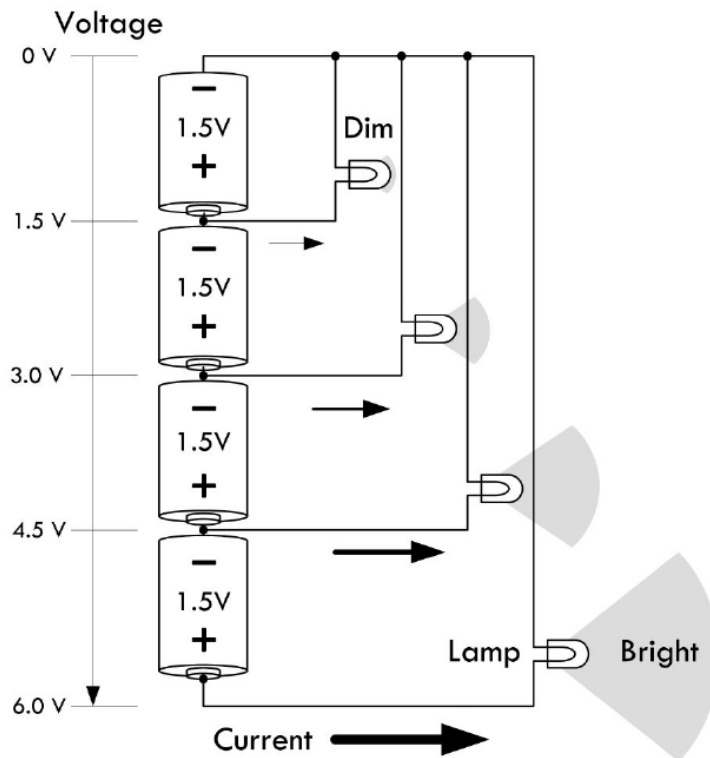


- Water = Charge
- Pressure = Voltage
- Flow = Current (Charge per unit time)

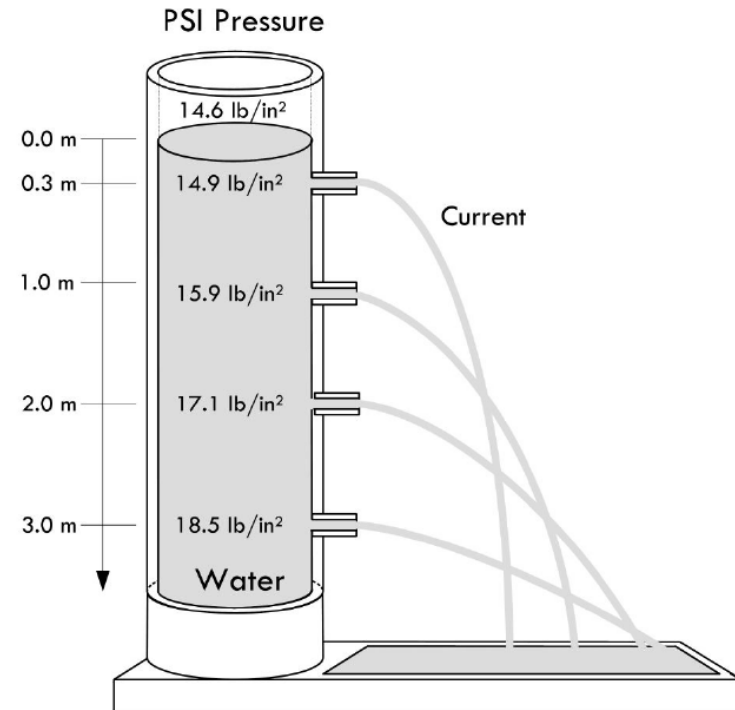


Voltage / Current

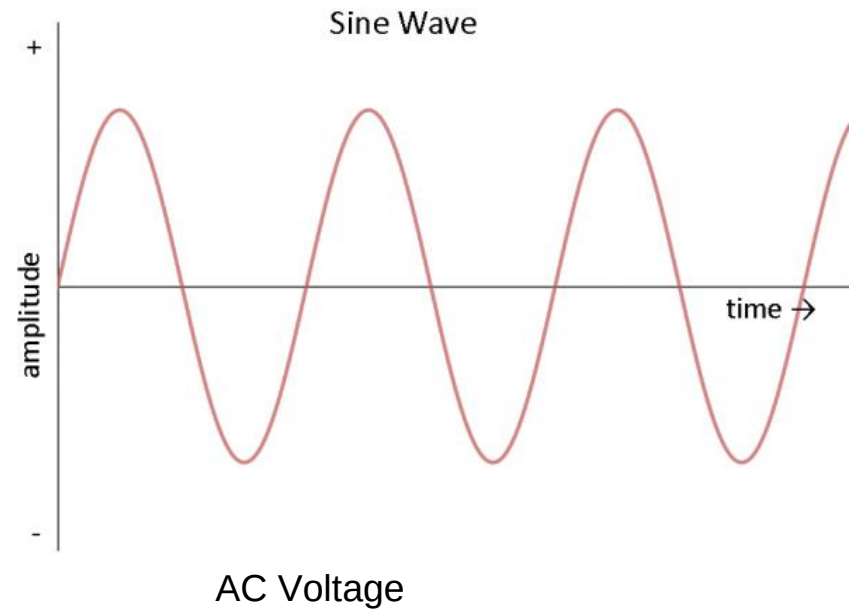
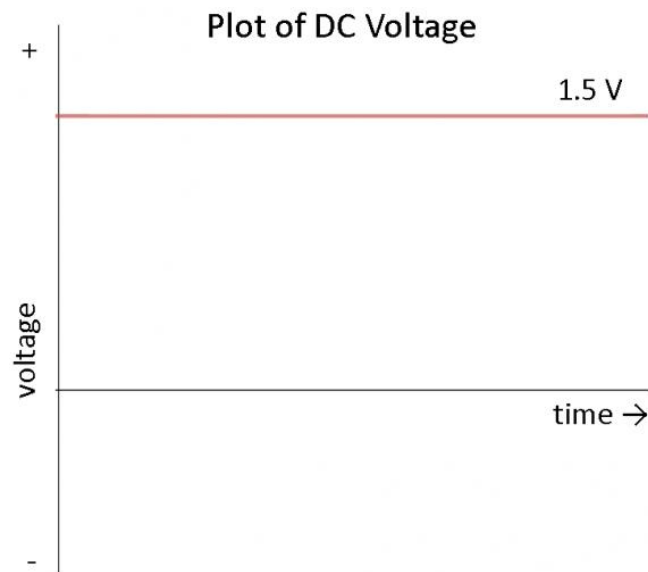
Electrical System



Water System



Voltage / Current



Resistance

A resistor is a dissipative element that converts electrical energy into heat. Ohm’s law defines the voltage-current characteristic of an ideal resistor:



$$V = IR$$

$$R = \frac{\rho L}{A}$$

4-Band-Code

2%, 5%, 10%

560k Ω ± 5%

5 6 x 10 = 560 Ω

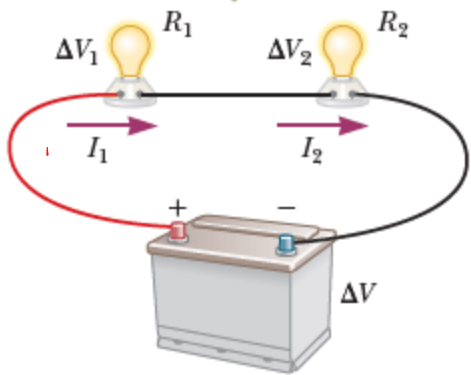
COLOR	1 ST BAND	2 ND BAND	3 RD BAND	MULTIPLIER	TOLERANCE
Black	0	0	0	1Ω	
Brown	1	1	1	10Ω	± 1% (F)
Red	2	2	2	100Ω	± 2% (G)
Orange	3	3	3	1KΩ	
Yellow	4	4	4	10KΩ	
Green	5	5	5	100KΩ	± 0.5% (D)
Blue	6	6	6	1MΩ	± 0.25% (C)
Violet	7	7	7	10MΩ	± 0.10% (B)
Grey	8	8	8		± 0.05%
White	9	9	9		
Gold				0.1Ω	± 5% (J)
Silver				0.01Ω	± 10% (K)

5-Band-Code

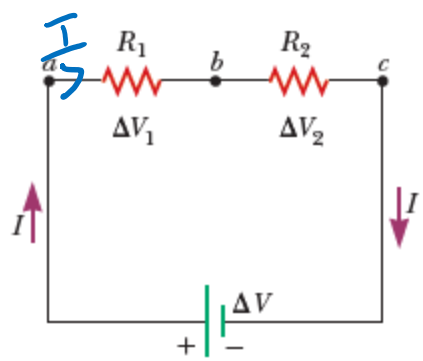
R???

Resistances in Series

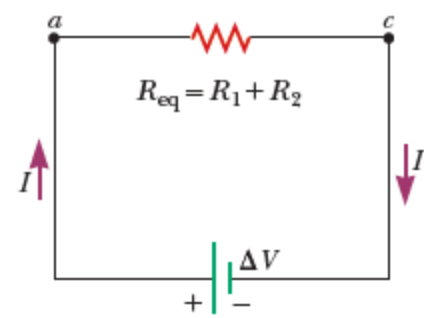
A pictorial representation of two resistors connected in series to a battery



A circuit diagram showing the two resistors connected in series to a battery

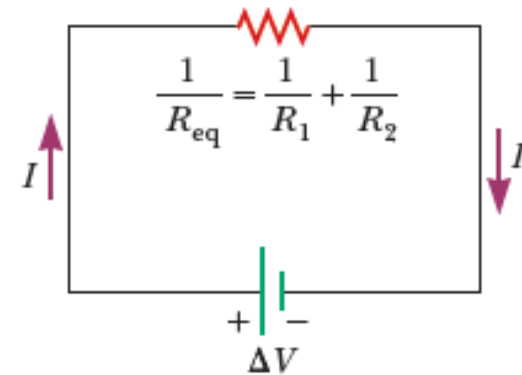
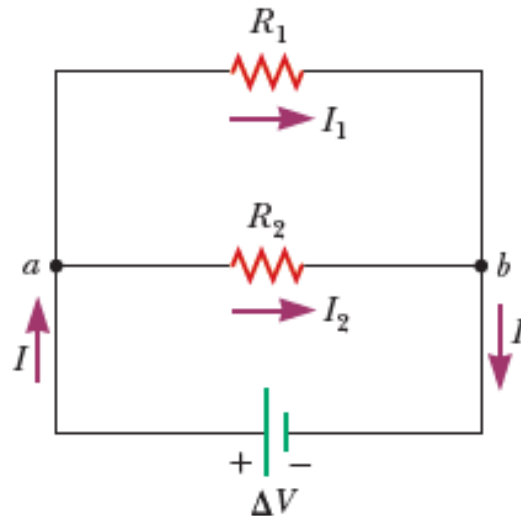
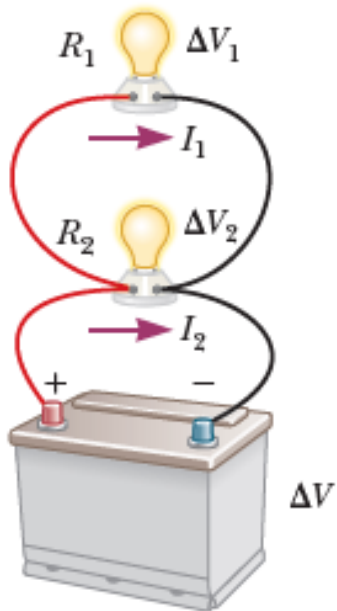


A circuit diagram showing the equivalent resistance of the resistors in series



$\Delta V = \Delta V_1 + \Delta V_2$
 $R_{eq} = R_1 + R_2$
 → divided voltage.
 voltage divider network

Resistances in Parallel

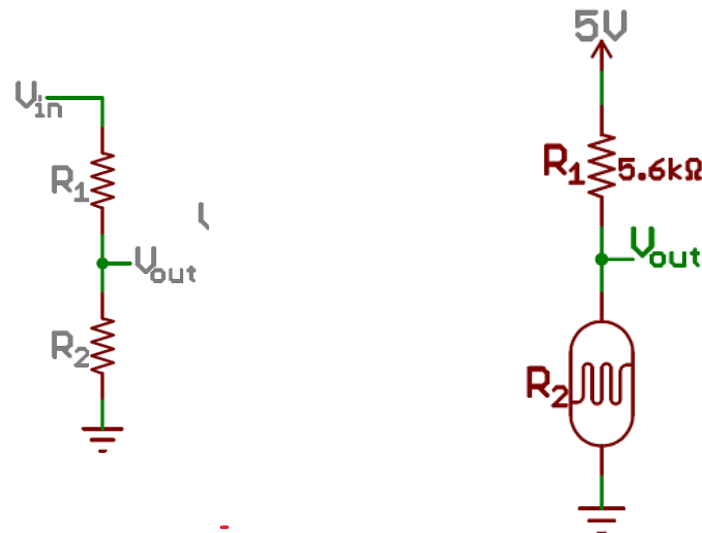


$$I = I_1 + I_2$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

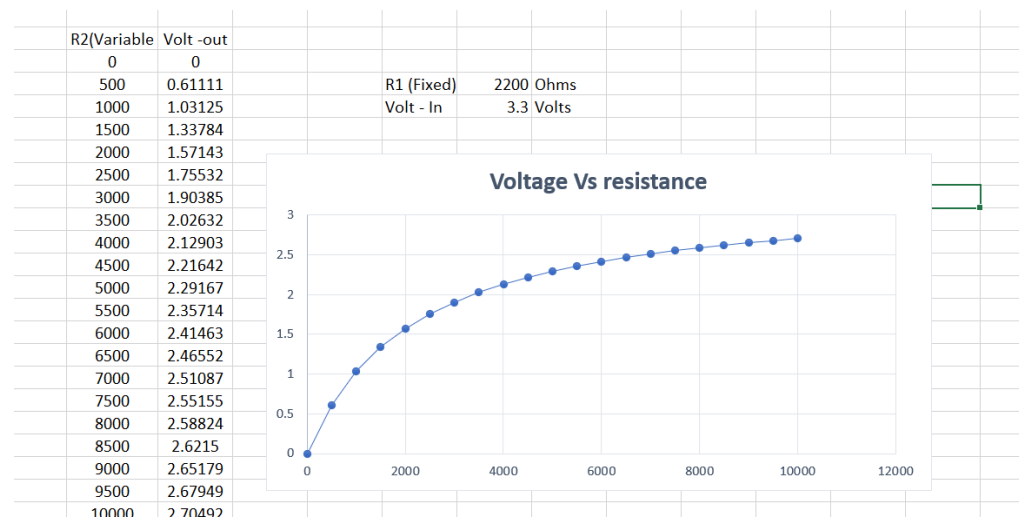
Current divider.

Voltage divider circuit



$$V_{out} = V_{in} \cdot \frac{R_2}{R_1 + R_2}$$

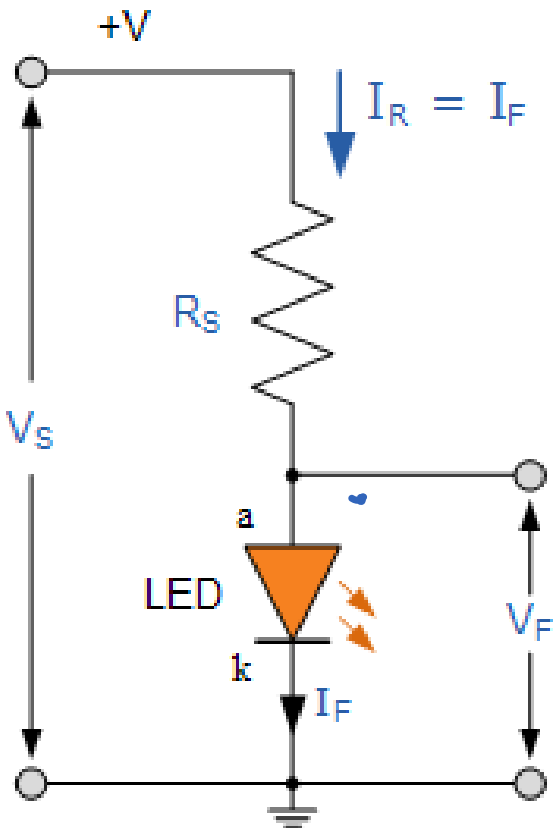
Light Level	R2 (Sensor)	R1 (Fixed)	Ratio R2/(R1+R2)	Vout
Light	1kΩ	5.6kΩ	0.15	0.76 V
Dim	7kΩ	5.6kΩ	0.56	2.78 V
Dark	10kΩ	5.6kΩ	0.67	3.21 V



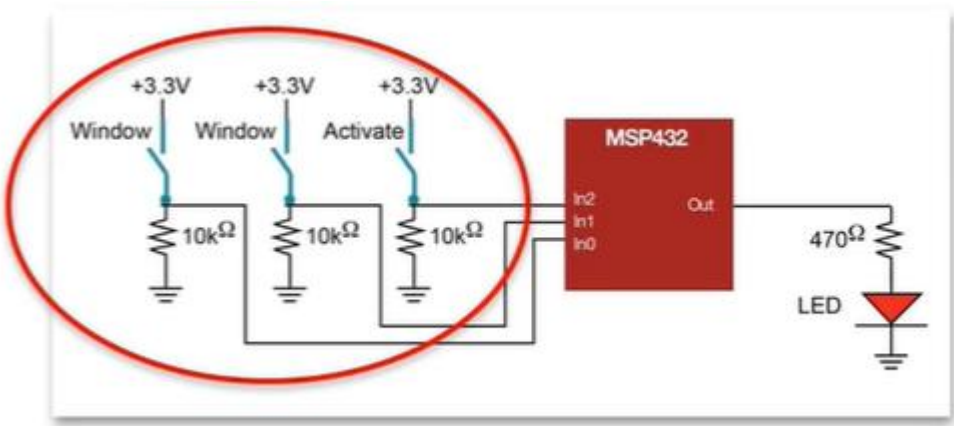
$V_o = 0.76V$
 $V_{in} = 3.3V$
 $V_o = V_{in} \frac{R_2}{R_1 + R_2}$
 $0.76 = 3.3 \times \frac{R_2}{5.6k + R_2}$

Voltage divider circuit

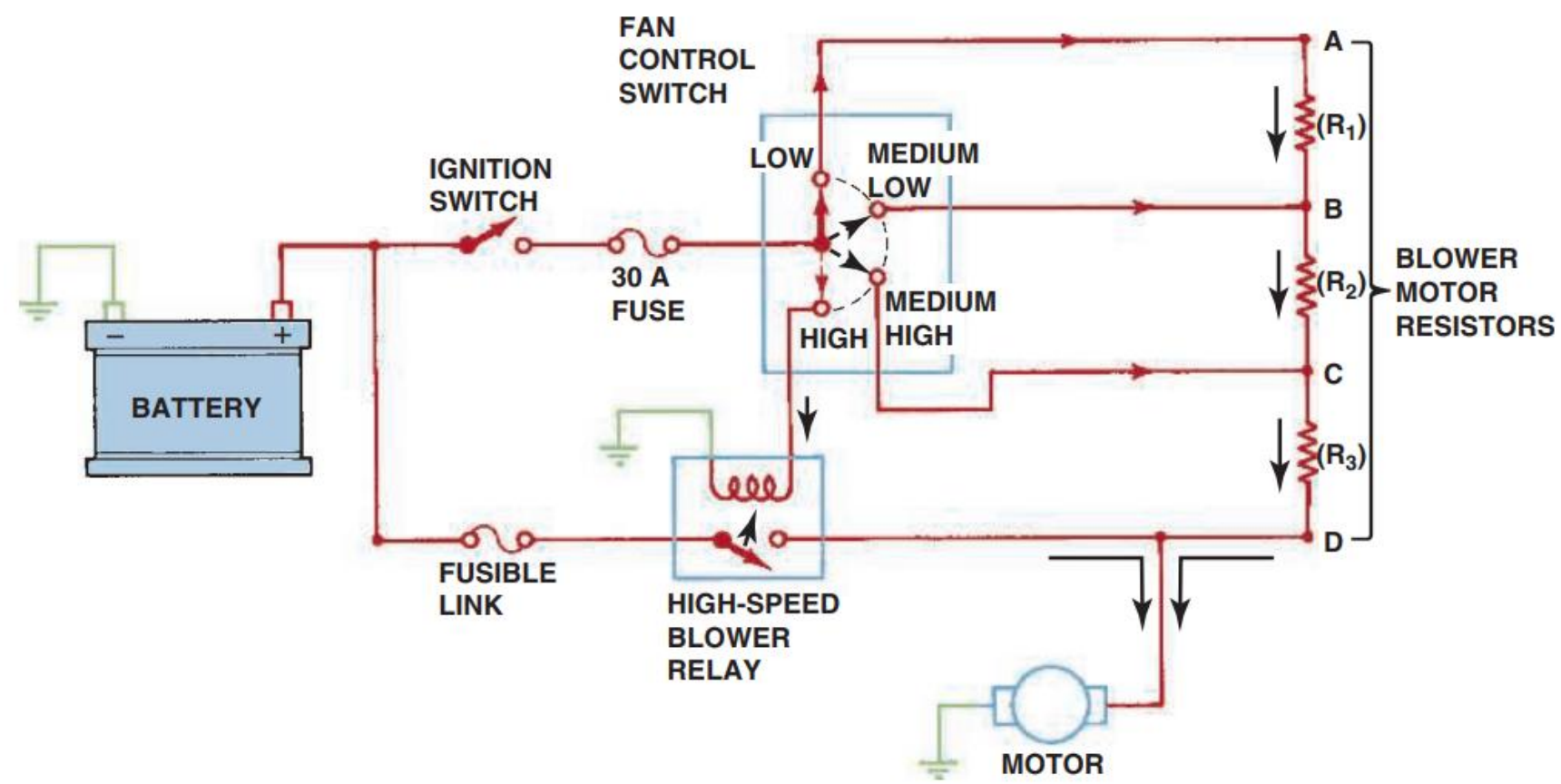
Hw



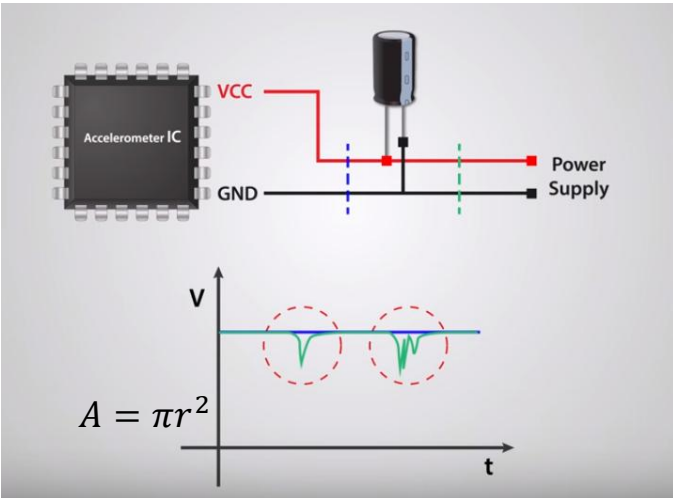
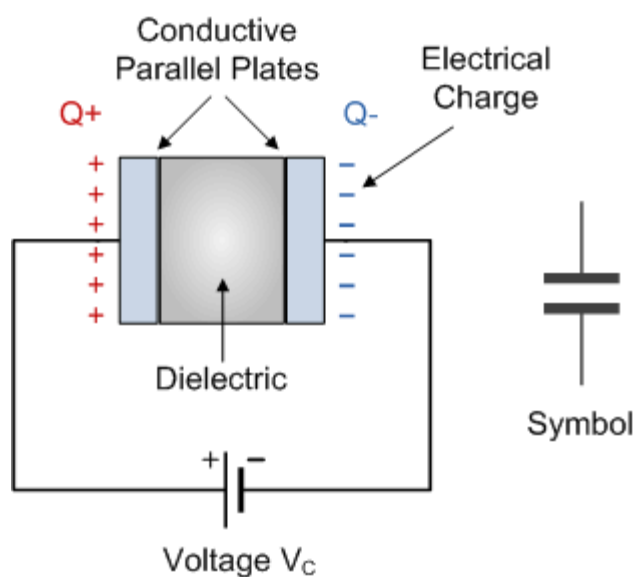
An amber coloured LED with a forward volt drop of 2 volts is to be connected to a 5.0v stabilised DC power supply. Using the circuit calculate the value of the series resistor required to limit the forward current to less than 10mA



Voltage divider circuit



Capacitor



$$C = Q/V$$

C is Capacitance in Farad
 Q is charge in Columb
 V - Volts

$$C = \frac{Q}{V}$$

$$Q = C \times V$$

$$\frac{dQ}{dt} = C \times \frac{dv}{dt}$$

$$I(t) = C \frac{dV}{dt}$$

$$W = \left(\frac{1}{2}\right) * C * V^2$$

Capacitor

$$W = \frac{1}{2} C V^2$$

Example capacitor:
 Capacitance : 470 μ F
 Voltage: 50V
 Dimensions: ~16x35.5mm

$$W = \frac{1}{2} \cdot 470 \cdot 10^{-6} \cdot 50^2$$

$$W = 0.58 \text{ [J]}$$



$$W = Q V$$

Example battery:
 Cell size: AAA
 Voltage: 1.225V
 Charge: 1.15Ah
 Dimensions: ~10.5x45.5mm

$$W = 1.225 \cdot 1.15 \cdot 3600$$

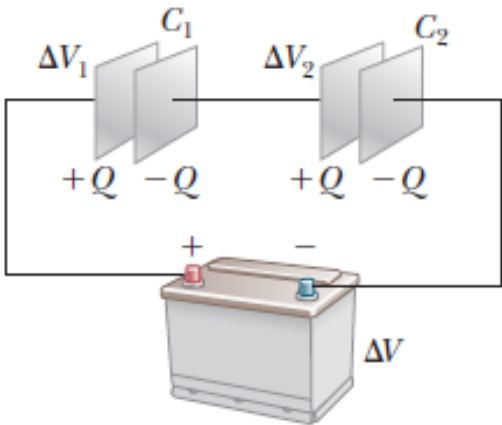
$$W = 5071 \text{ [J]}$$





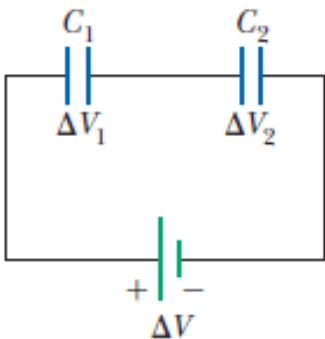
Capacitors in Series

A pictorial representation of two capacitors connected in series to a battery

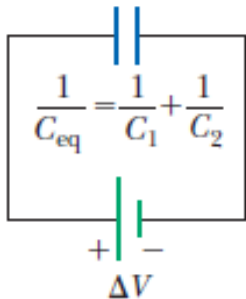


$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

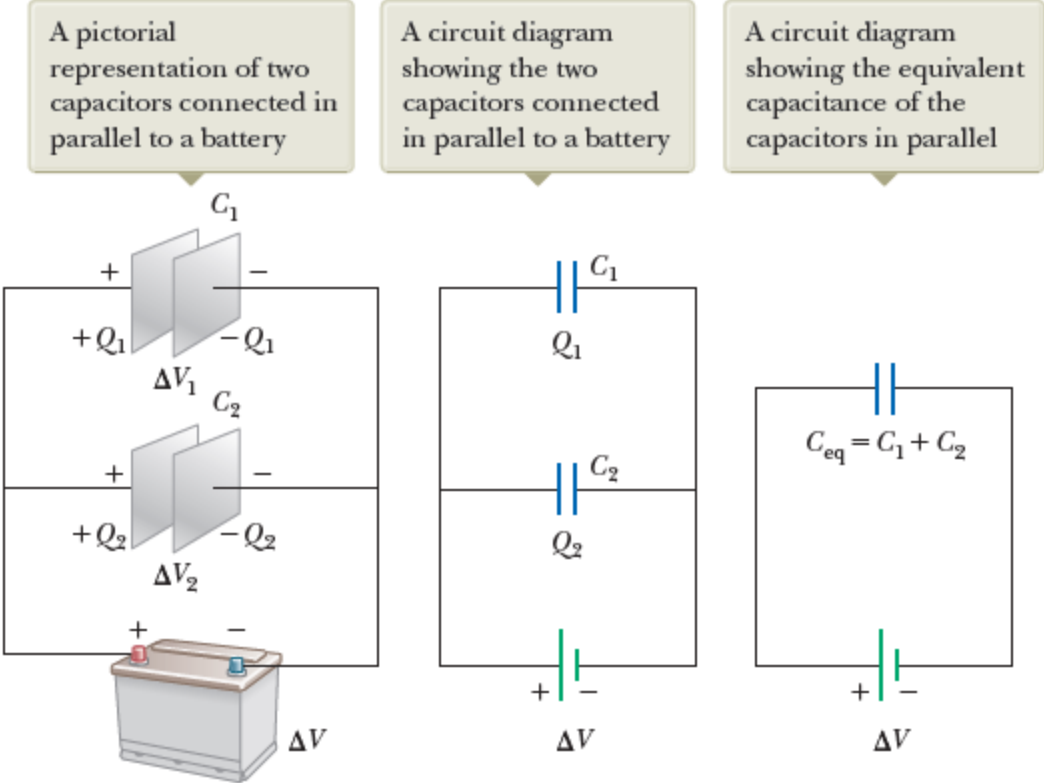
A circuit diagram showing the two capacitors connected in series to a battery



A circuit diagram showing the equivalent capacitance of the capacitors in series

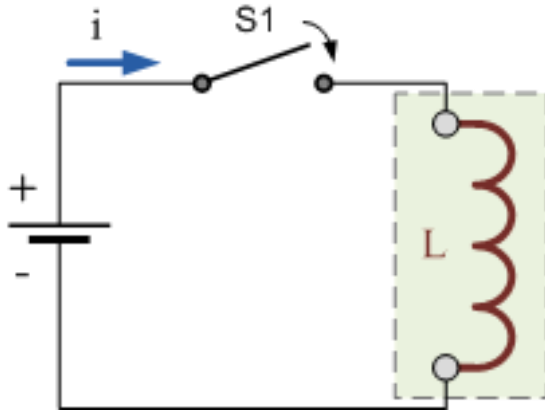


Capacitors in Parallel

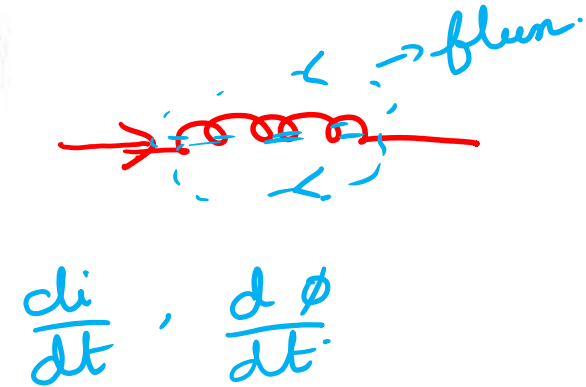


$C_{eq} = C_1 + C_2$

Inductors



Used as filters, motors, transformers etc,



An inductor opposes the rate of change of current flowing through it due to the build up of self-induced energy within its magnetic field.

$L =$

$$V_L(t) = \frac{d\phi}{dt} = \frac{dLi}{dt} = -L \frac{di}{dt}$$

rate of change of flux

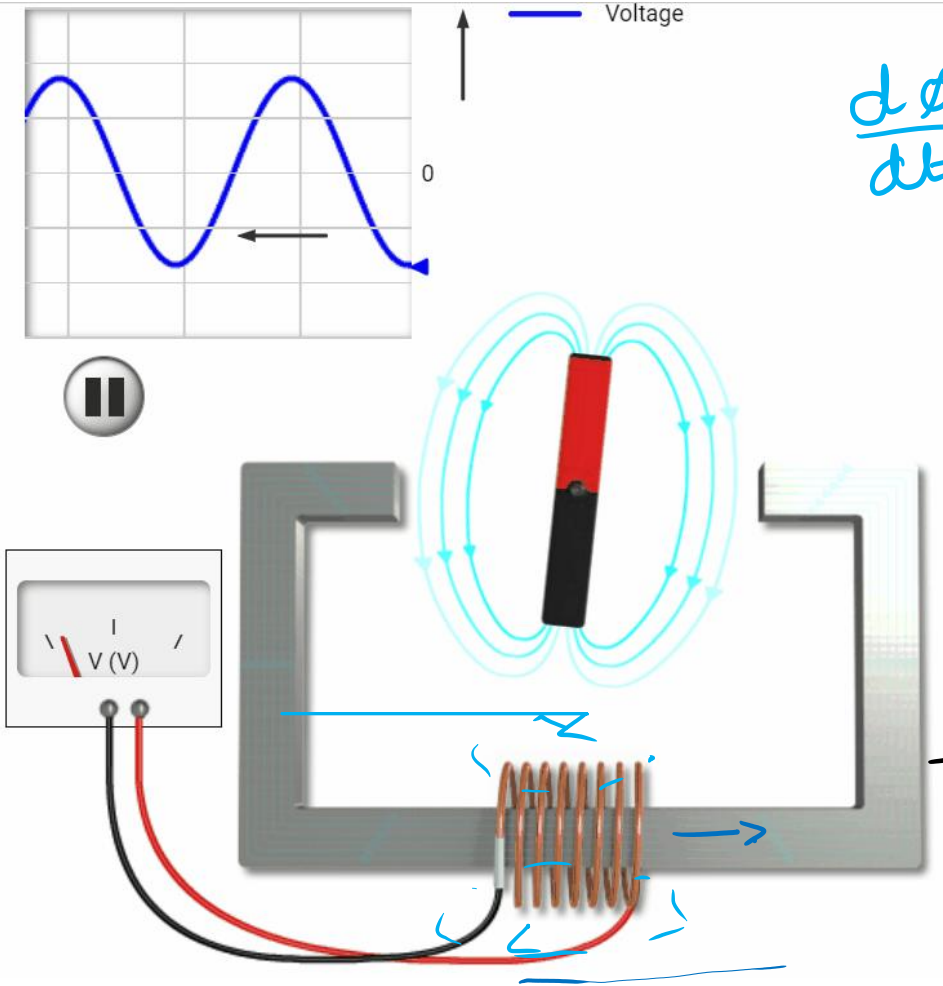
Φ - Magnetic flux (Weber)

L – Inductance (Henry)

i - Ampere

V - Voltage

Inductors → sensors



$\frac{d\phi}{dt}$ → voltage
↓
induced emf.

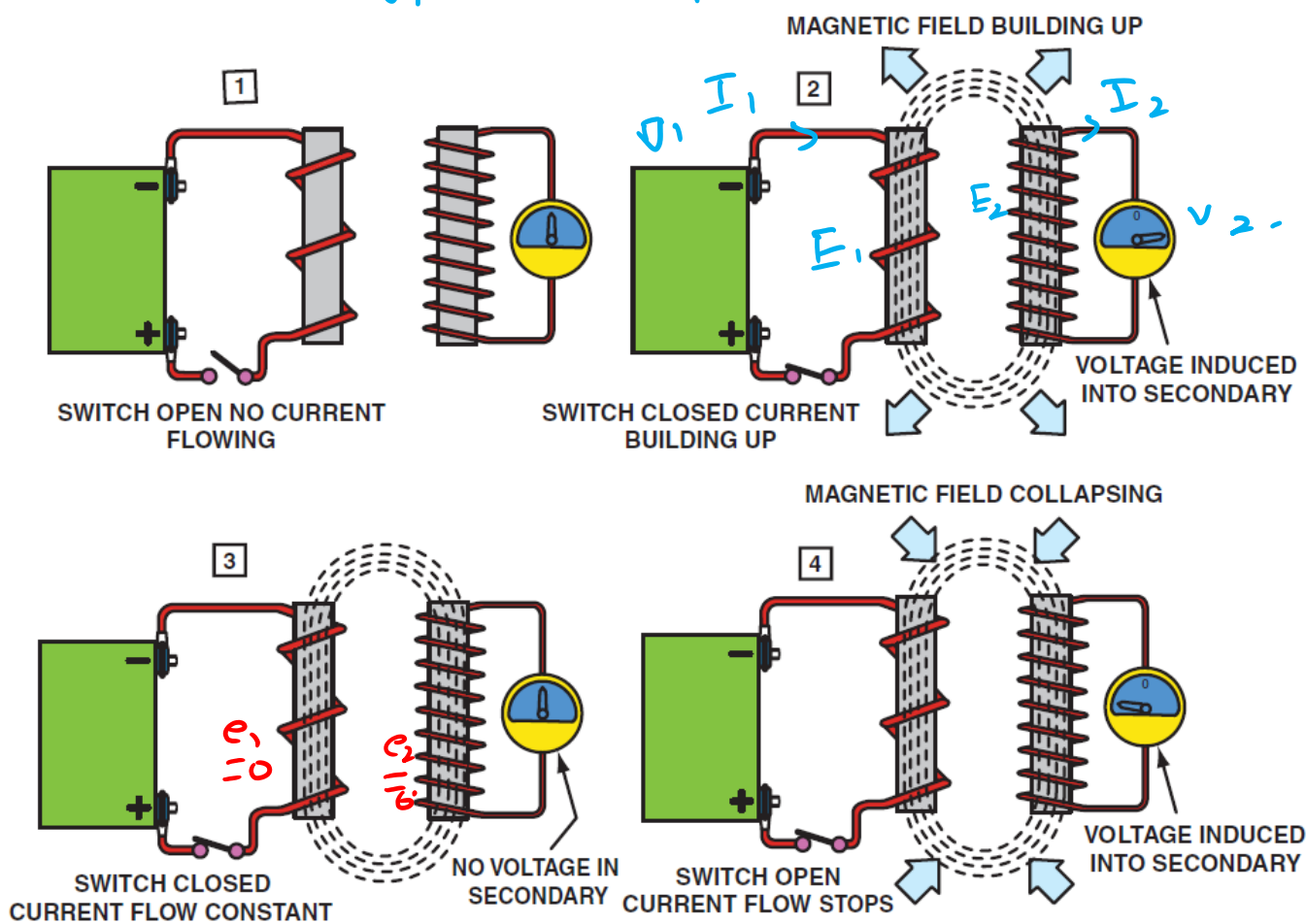
→ Core.

Magnetic field

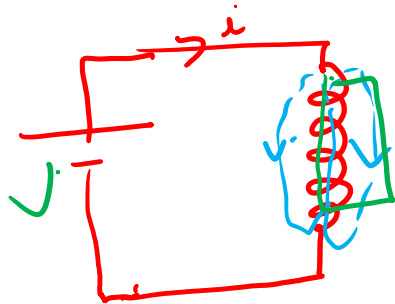
induced emf in coil
 $V_L = L \frac{di}{dt} = E_1$

$V_1 \rightarrow I_1 \rightarrow E_1 \rightarrow E_2 \rightarrow I_2 \rightarrow V_2$

only for AC
 no DC.



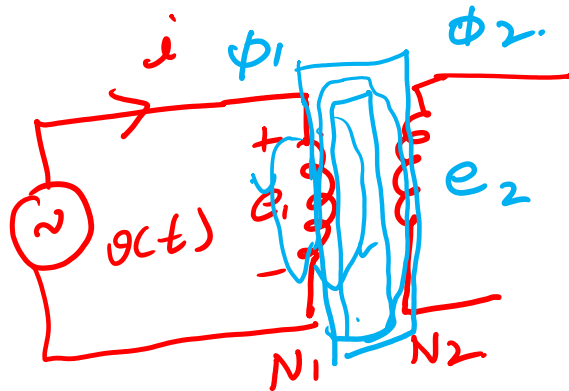
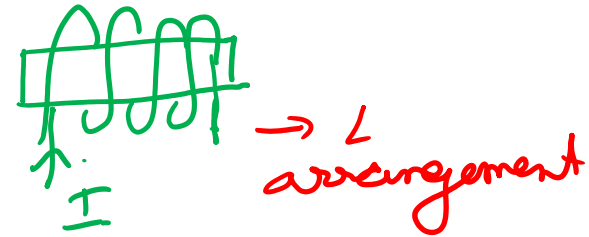
L is wound core.



$$V_L = L \frac{di}{dt}$$

$L \rightarrow$ fully charged.

$\rightarrow L$ is short circuit

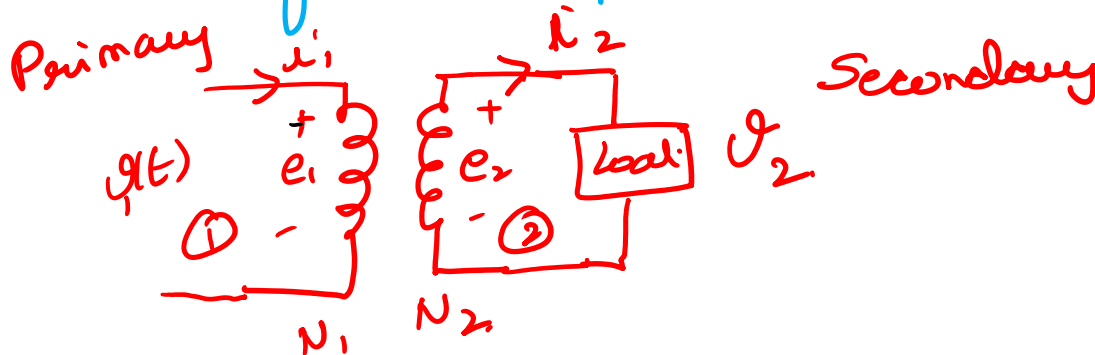


N_1 & $N_2 \rightarrow$ no. of turns.

Faraday's law - emf induced in N_1 $e_1 = -N_1 \frac{d\phi}{dt}$
 In N_2 e_2 is generated due to flux produced in N_1 .

h

Transformer principle

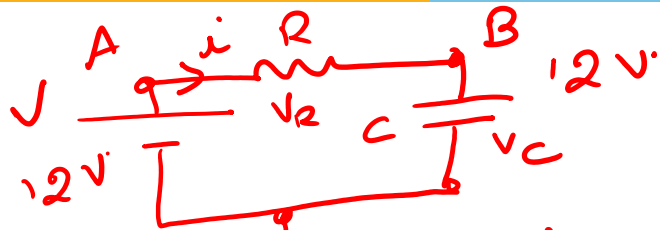


$$v_1 \rightarrow i_1 \rightarrow \phi_1 \xrightarrow{e_1} e_2 \rightarrow i_2 \rightarrow v_2$$

$$e_1 = -N_1 \frac{d\phi}{dt}$$

$$e_2 = -N_2 \frac{d\phi}{dt}$$

if $N_1 = N_2$.
 $\hookrightarrow$ isolation $\sim$
 $\hookrightarrow$ protection.



Kirchoff's voltage law

$$V = V_R + V_C$$

$$= iR + \frac{1}{C} \int i dt$$

- current 'i' starts flowing
- C starts charging

$$V_A = 12V$$

$$V_B = 12V$$

$$V_{AB} = V_A - V_B$$

$$= 0$$

$V_C = 12V \rightarrow$ capacitor is fully charged.

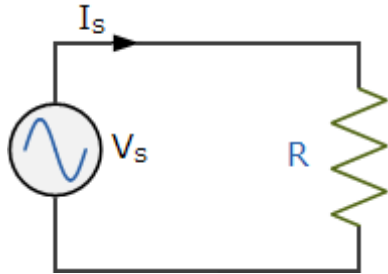
→ a fully charged capacitor is open circuit

Today's class

- ↳ Engine sub system
 - ↳ Sensors → types
- ↳ Fundamentals of electrical technology
 - ↳ V, I, R, L, C
(AC & DC)

Circuits

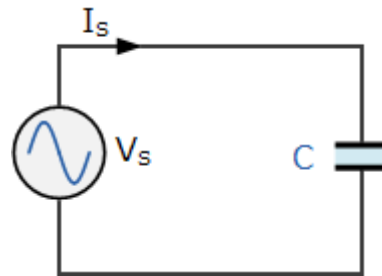
Pure Resistance Circuit



$$Z = \frac{V_R}{I_R} = R$$

$$I_s = \frac{V_s}{R}$$

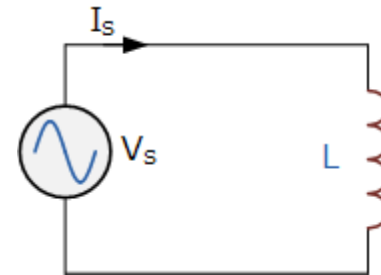
Pure Capacitance Circuit



$$X_C = \frac{V_C}{I_C} = \frac{1}{2\pi fC}$$

$$I_s = \frac{V_s}{X_C}$$

Pure Inductive Circuit

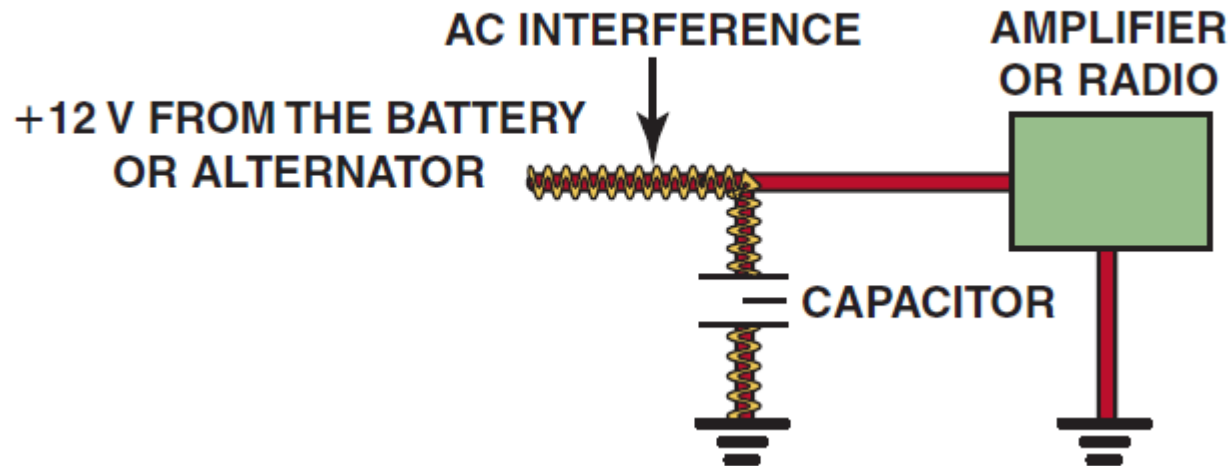


$$X_L = \frac{V_L}{I_L} = 2\pi fL$$

$$I_s = \frac{V_s}{X_L}$$

Unlike Resistor which just impedes the current (Constant Ohms), Capacitor and inductor impede currents (Variable ohms) based on frequency of power supply

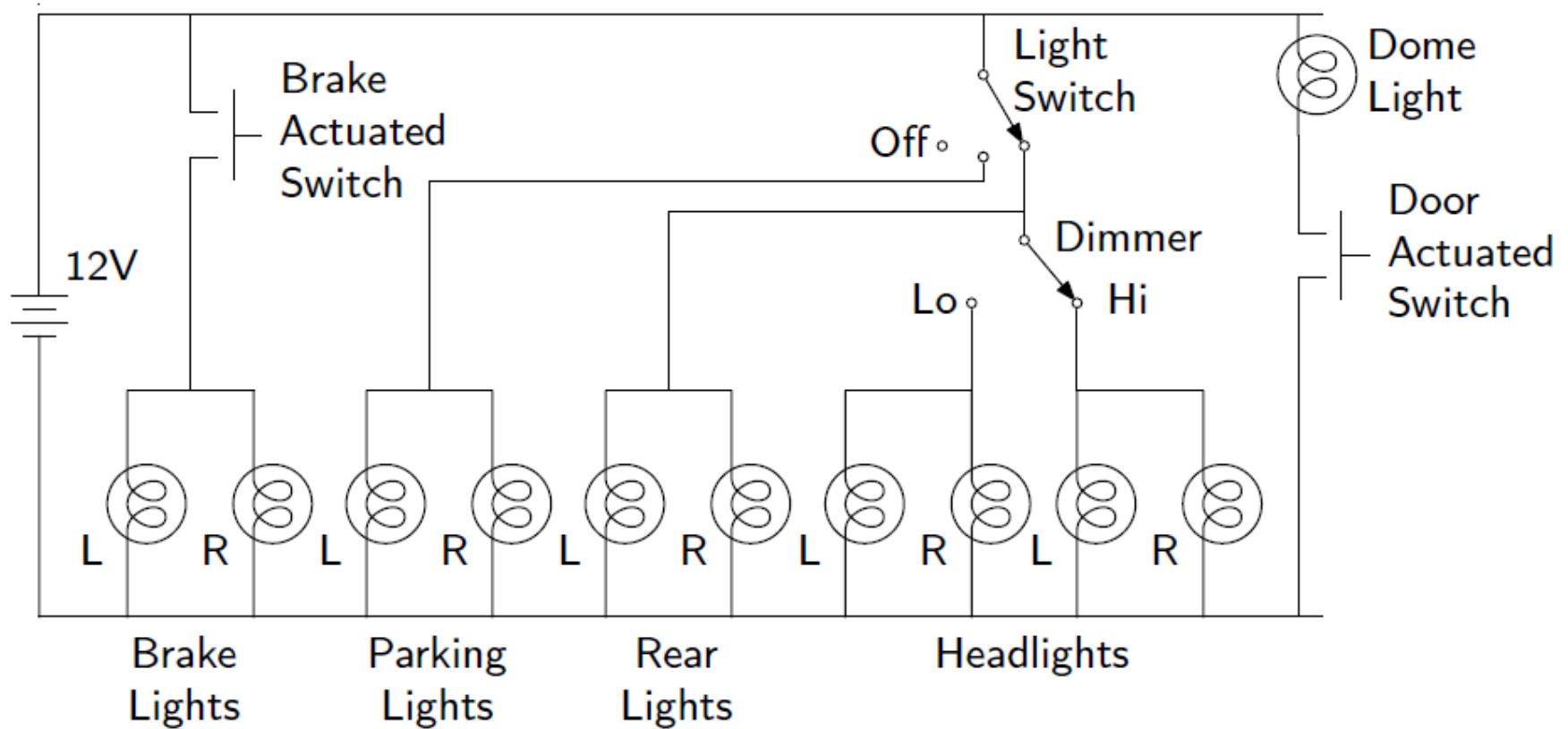
Circuits



- ✘ A capacitor blocks direct current (DC) but passes alternating current (AC).

A capacitor makes a very good noise suppressor because most of the interference is AC and the capacitor will conduct this AC to ground before it can reach the radio or amplifier.

Circuit analysis

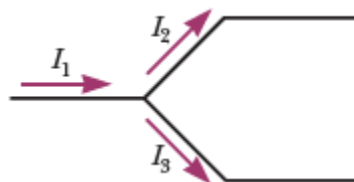


Kirchoff's Law

1. **Junction rule.** At any junction, the sum of the currents must equal zero:

KCL

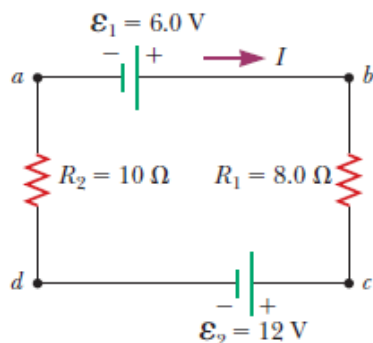
$$\sum_{\text{junction}} I = 0 \quad (28.9)$$



2. **Loop rule.** The sum of the potential differences across all elements around any closed circuit loop must be zero:

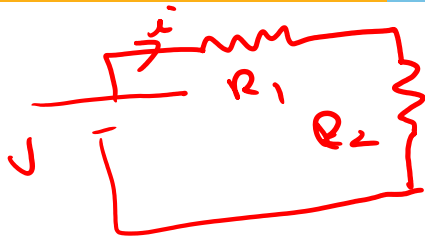
KVL

$$\sum_{\text{closed loop}} \Delta V = 0 \quad (28.10)$$

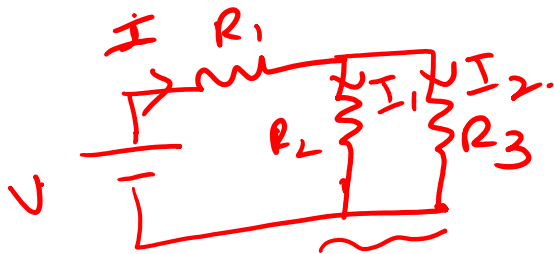


$$\sum \Delta V = 0 \rightarrow \mathcal{E}_1 - IR_1 - \mathcal{E}_2 - IR_2 = 0$$

$$(1) \quad I = \frac{\mathcal{E}_1 - \mathcal{E}_2}{R_1 + R_2} = \frac{6.0 \text{ V} - 12 \text{ V}}{8.0 \, \Omega + 10 \, \Omega} = -0.33 \text{ A}$$



$$V = V_{R_1} + V_{R_2} \rightarrow KVL.$$



$$V = V_{R_1} + V_{R_2} \quad (V_{R_2} = V_{R_3})$$

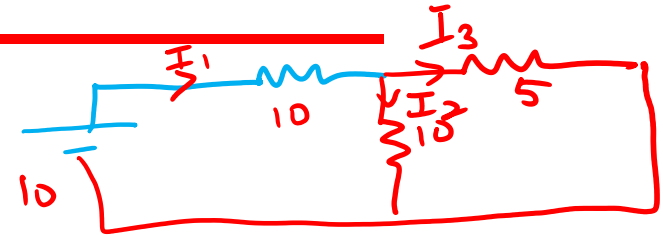
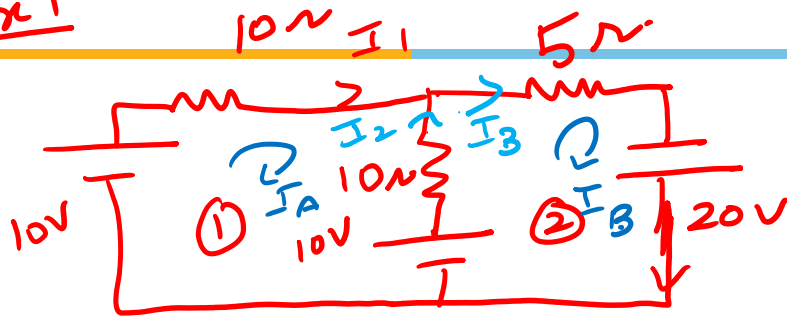
$$= IR_1 + I_1 R_2 \quad \text{X}$$

$$R_2 \parallel R_3$$

$$R_{eq} = \frac{R_2 R_3}{R_2 + R_3}$$

$$V = IR_1 + \frac{I R_2 R_3}{R_2 + R_3}$$

Ex 1



$$I_1 = I_2 + I_3$$

$$I_3 = I_1 + I_2$$

Mesh analysis
Method 1

→ assume mesh currents

equation in ①,

$$10I_A + 10(I_A - I_B) + 10 - 10 = 0$$

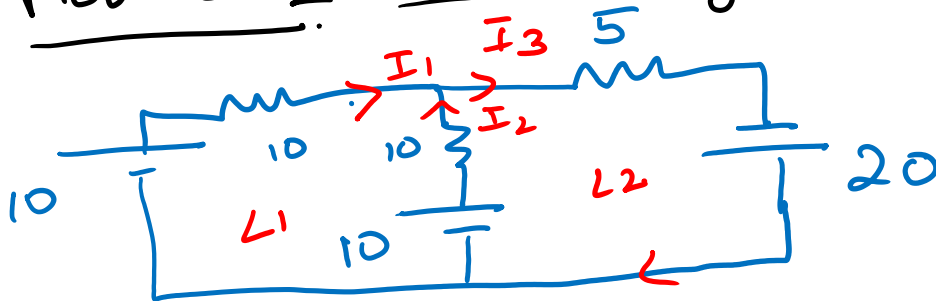
equation in ②

$$5I_B - 20 - 10 + 10(I_B - I_A) = 0$$

HW → Solving set of simultaneous eqns.

avoid.)

Method 2 Nodal analysis.



in L1

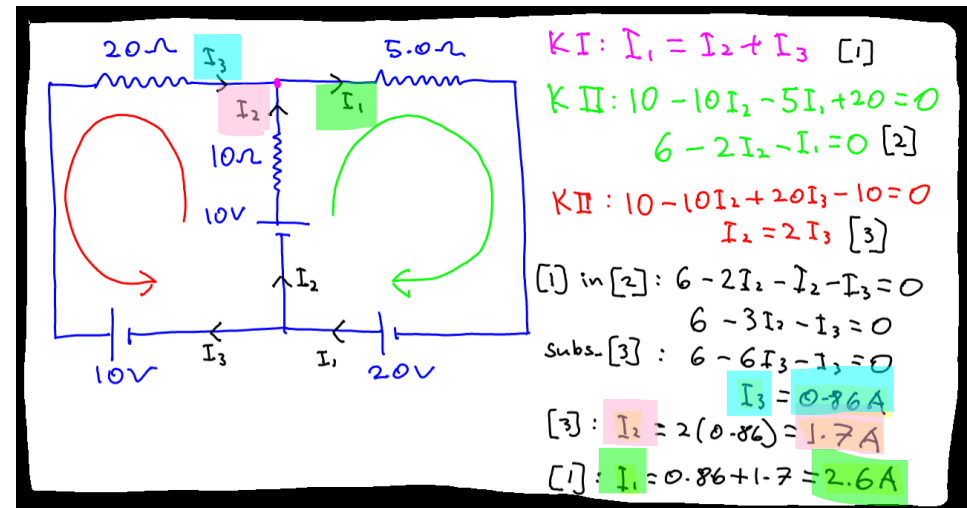
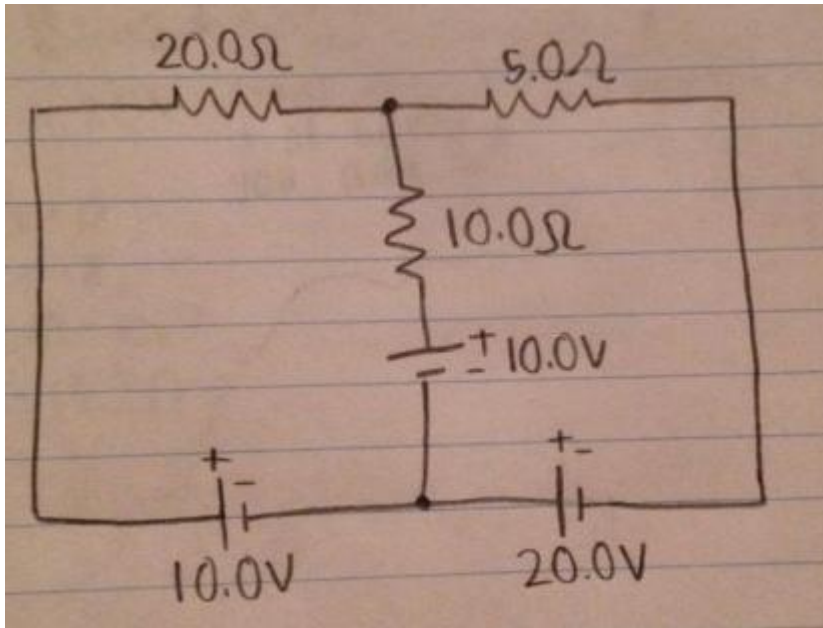
$$10I_1 - 10I_2 + 10 = 10 \quad \dots \textcircled{1}$$

in L2

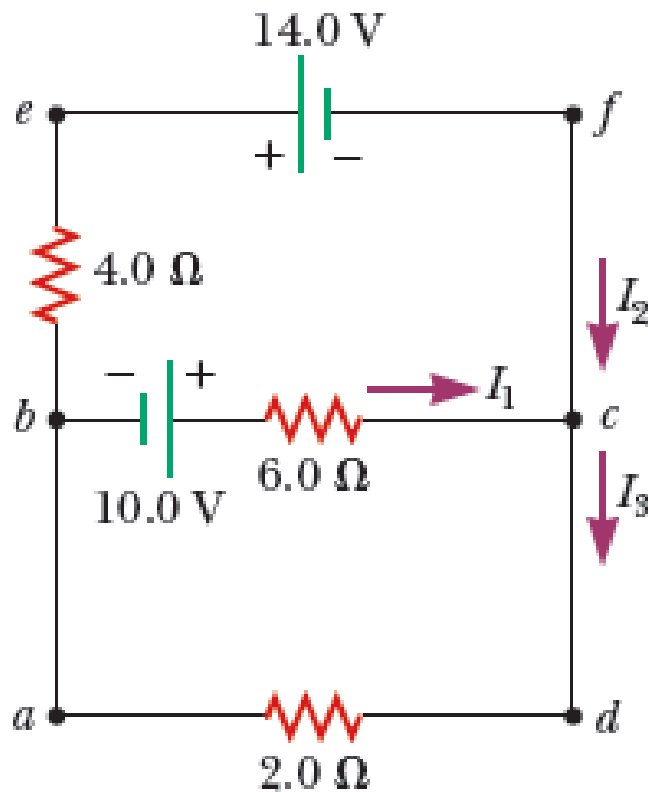
$$5I_3 - 20 - 10 + 10I_2 = 0 \quad \dots \textcircled{2}$$

$$I_1 + I_2 = I_3 \quad \dots \textcircled{3}$$

Kirchoff's Law



Kirchoff's Law (Home work)



Kirchoff's Law (HW)

$$V_1 = 20\text{V}, V_2 = 12\text{V}, V_3 = 40\text{V}$$

$$R_1 = 18\Omega, R_2 = 10\Omega, R_3 = 16\Omega$$

$$R_4 = 6\Omega, R_5 = 15\Omega, R_6 = 8\Omega$$

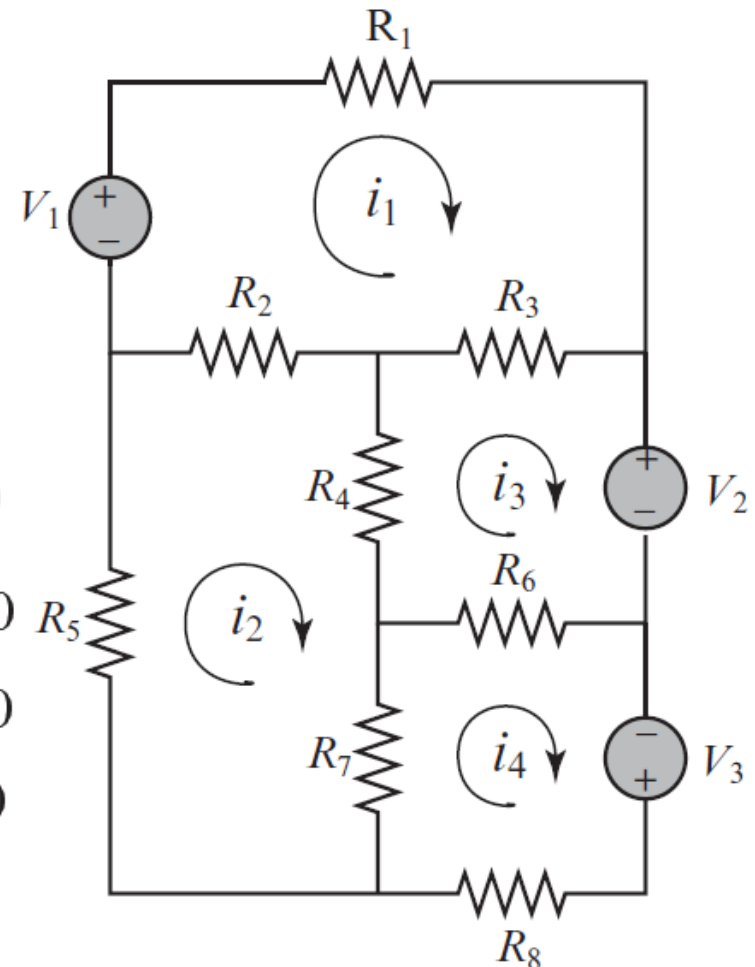
$$R_7 = 12\Omega, R_8 = 14\Omega$$

$$V_1 - R_1 i_1 - R_3(i_1 - i_3) - R_2(i_1 - i_2) = 0$$

$$-R_5 i_2 - R_2(i_2 - i_1) - R_4(i_2 - i_3) - R_7(i_2 - i_4) = 0$$

$$-V_2 - R_6(i_3 - i_4) - R_4(i_3 - i_2) - R_3(i_3 - i_1) = 0$$

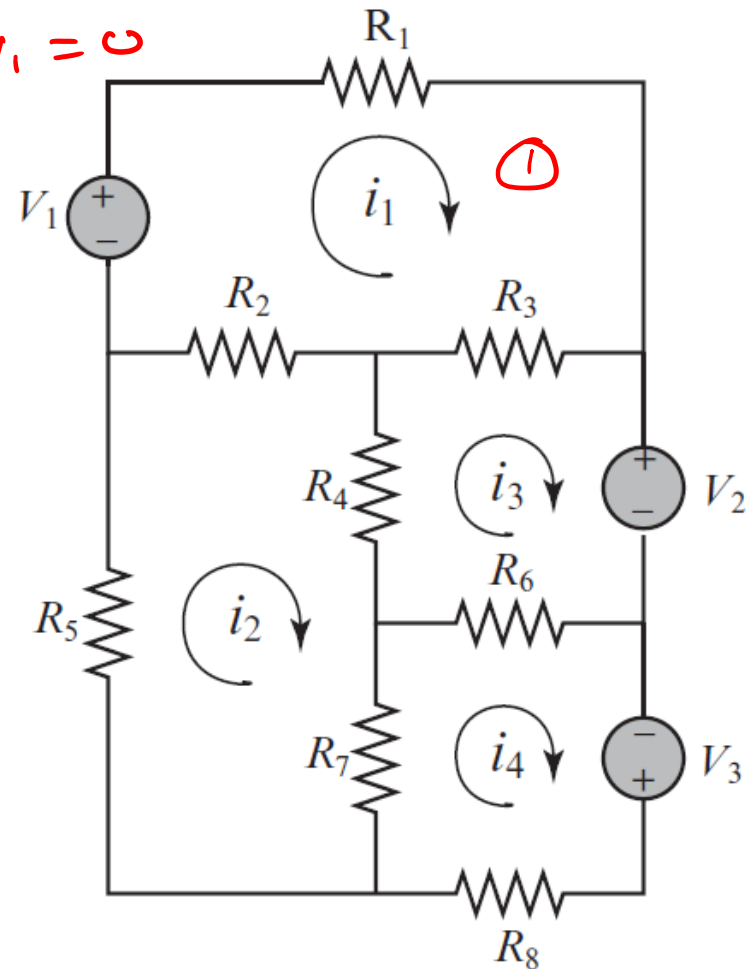
$$V_3 - R_8 i_4 - R_7(i_4 - i_2) - R_6(i_4 - i_3) = 0$$



HW

For L1

$$I_1 R_1 + (I_1 - I_3) R_3 + (I_1 - I_2) R_2 - V_1 = 0$$



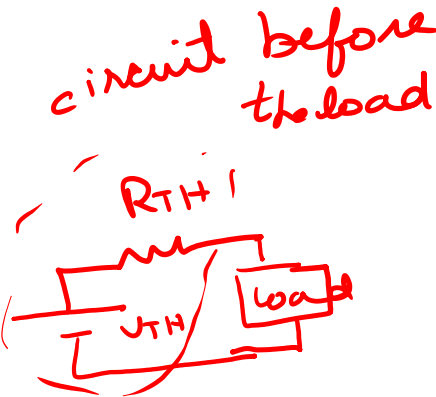
Thevenin's Theorem

$$I_L = \frac{V_{TH}}{R_{TH} + R_L}$$

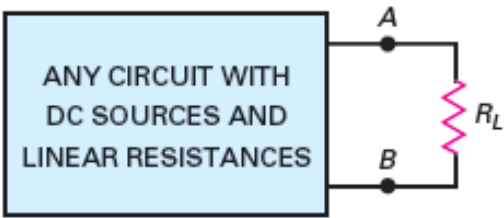
Thevenin voltage V_{TH} is defined as the voltage across the load terminals when the load resistor is open

R_{TH}

The **Thevenin resistance** is defined as the resistance that an ohmmeter measures across the load terminals

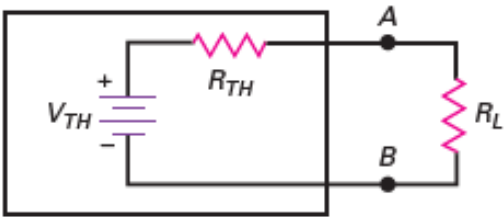


- ↳ Load 1 – TV
- Load 2 – WM
- Load 3 – Microw wave



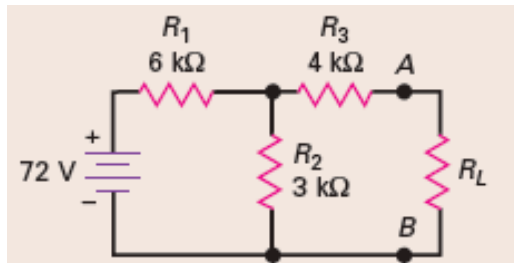
(a)

→ TV
↳ Load

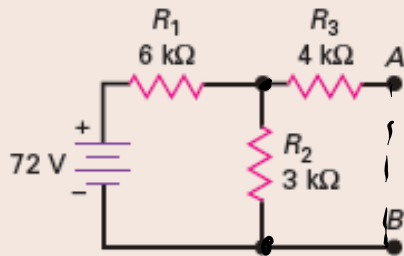


(b)

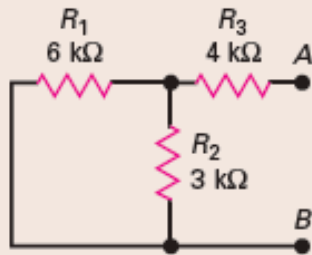
Thevenin's Theorem



(a)

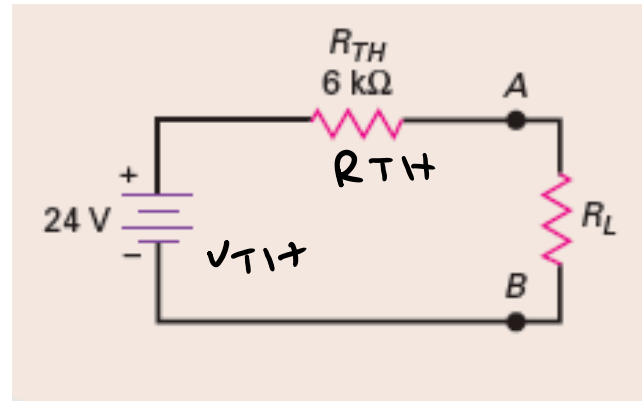


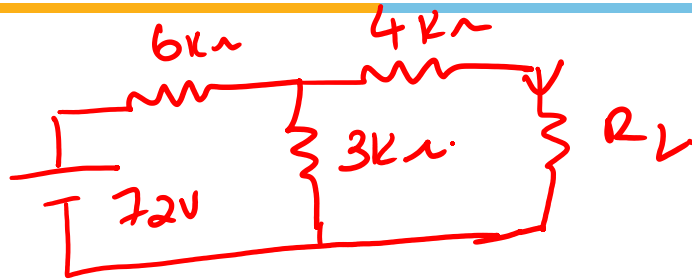
(b)



(c)

d) After c, find R_{eq} .
 e) $V_{TH} \rightarrow SC R_L$ in (a)



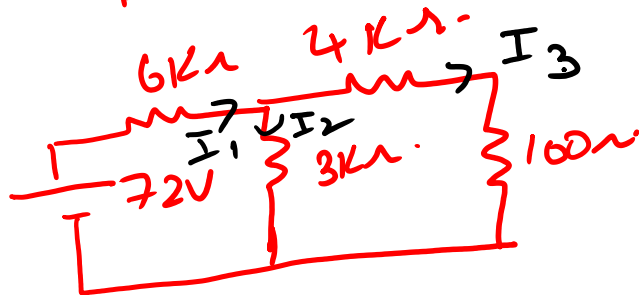


$$R_{L1} = 100\Omega$$

$$R_{L2} = 1000\Omega$$

$$R_{L3} = 5k\Omega$$

Find current thro' R_L .



calculate R_{eq} .

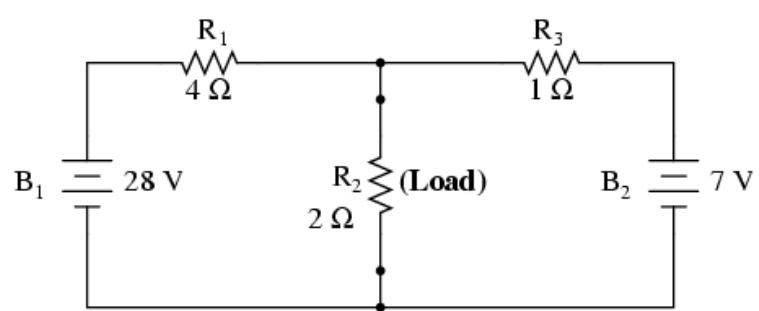
i) 100Ω in series with $4k\Omega = 4100\Omega$

ii) 4100Ω in parallel to $3k\Omega = \frac{4100 \times 3000}{7100}$

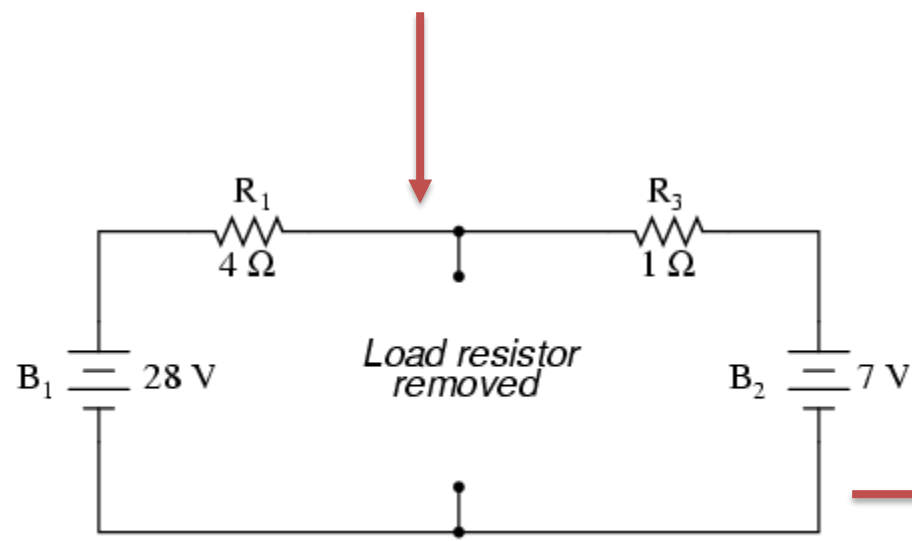
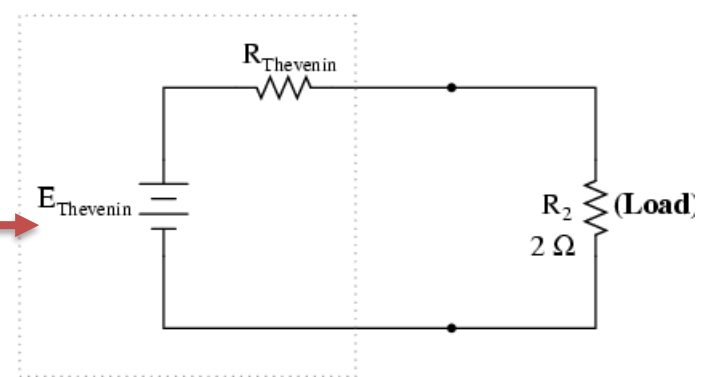
iii) answer in (ii) in series with $6k\Omega$ R_{eq} .
calculate I_1, I_2, I_3

$$I = \frac{72}{R_{eq}}$$

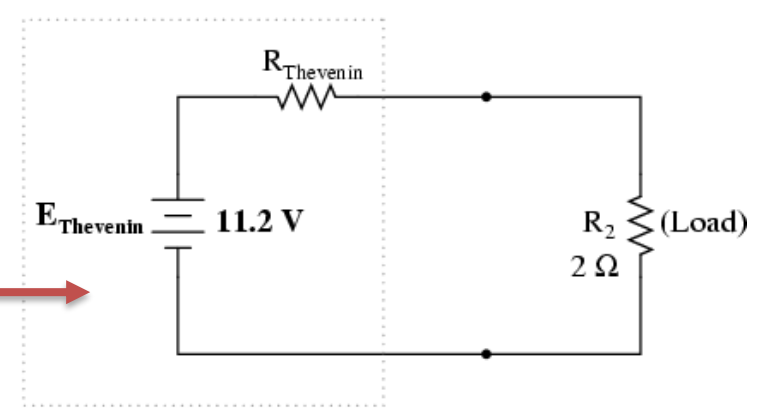
Thevenin's Theorem



Thevenin Equivalent Circuit



Thevenin Equivalent Circuit



Class on 9 Aug 26

Electrical fundamentals

- ↳ R, L, C, V, I
- ↳ Ohm's law for AC & DC input
 - ↳ Impedance
 - ↳ Reactance
- ↳ Series parallel connection of R & C
- ↳ Solving circuits using KCL, KVL, mesh analysis
- ↳ Magnetic circuit - transformer.
- ↳ Thevenin Theorem

Subsystem

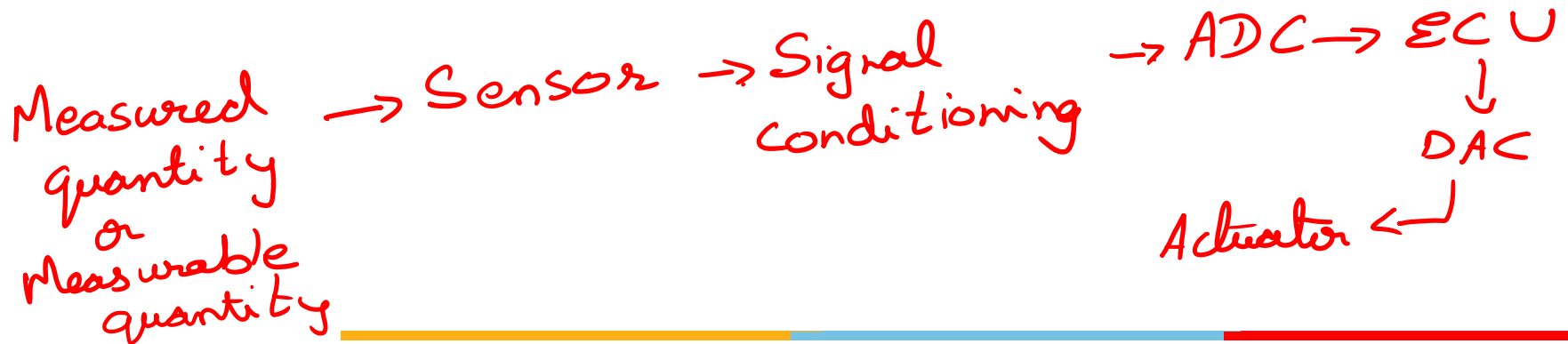
↳ Engine subsystem - class on 1st Aug.

↳ Created a sequence

↳ Identified the sensors

↳ Type of sensors

↳ What is the OP?
V, I, analog,
digital



Today's class 9 Aug 26

↳ Sensor characteristics

↳ Static

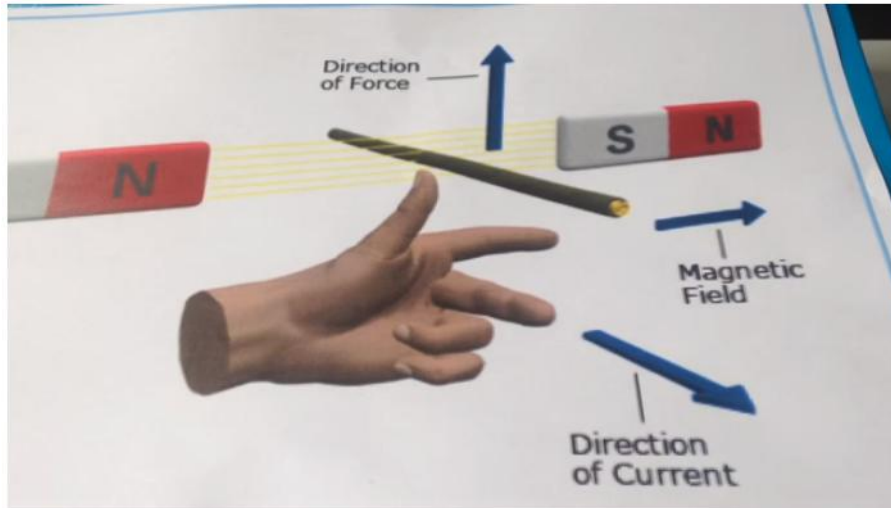
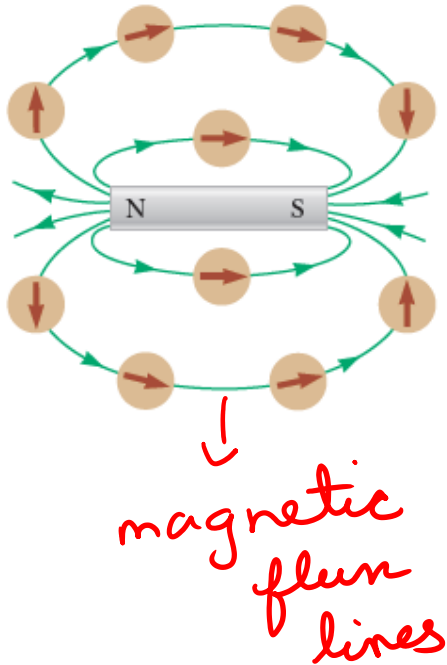
↳ Dynamic

↳ Completing the electronics fundamentals.

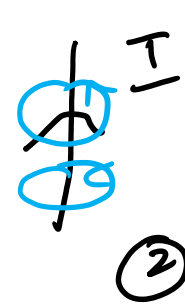
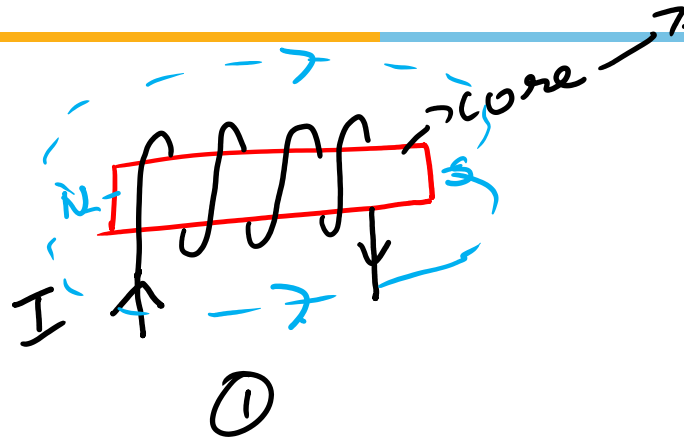
Magnetic field

1. A current-carrying wire produces a magnetic field in the area around it.
2. A time-changing magnetic field induces a voltage in a coil of wire if it passes through that coil. (This is the basis of *transformer action*.)
3. A current-carrying wire in the presence of a magnetic field has a force induced on it. (This is the basis of *motor action*.)
4. A moving wire in the presence of a magnetic field has a voltage induced in it. (This is the basis of *generator action*.)

Magnetic field



CO1



$\rightarrow R, L$

Right hand rule

$\hookrightarrow$ Current carrying conductor - Inductor
 $\downarrow$
 magnetic field.

Faraday's laws

$$e = -N \frac{d\phi}{dt}$$

$\nwarrow$ induced emf

$\nearrow$ no. of turns

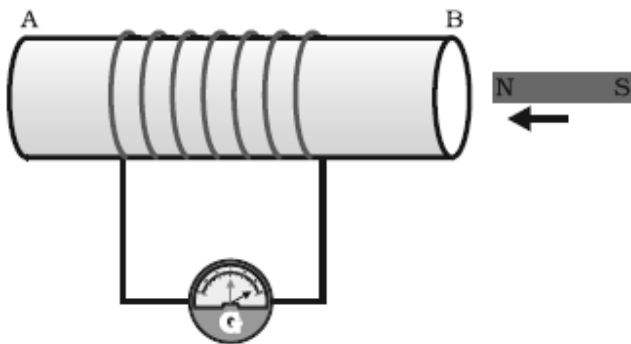
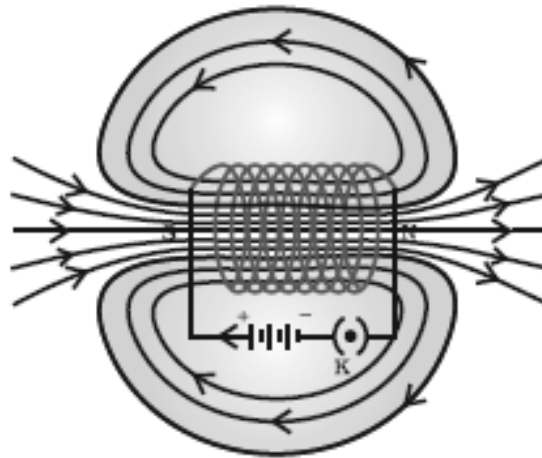
$\rightarrow$ flux.

Magnetic field

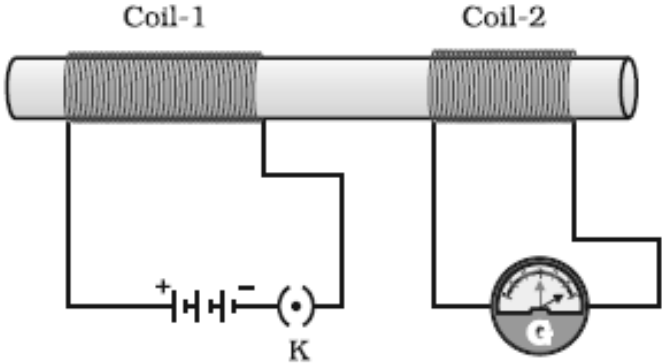
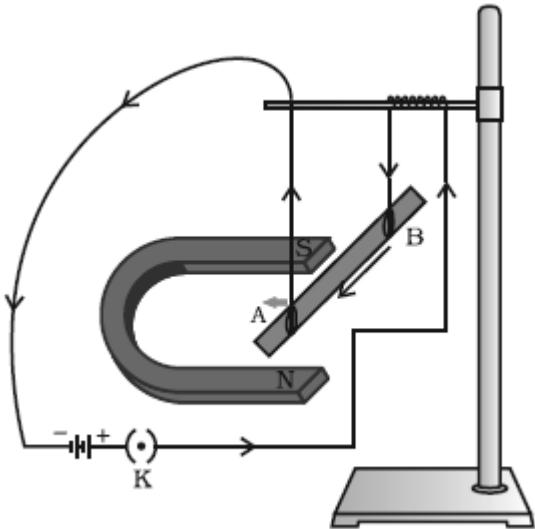
innovate

achieve

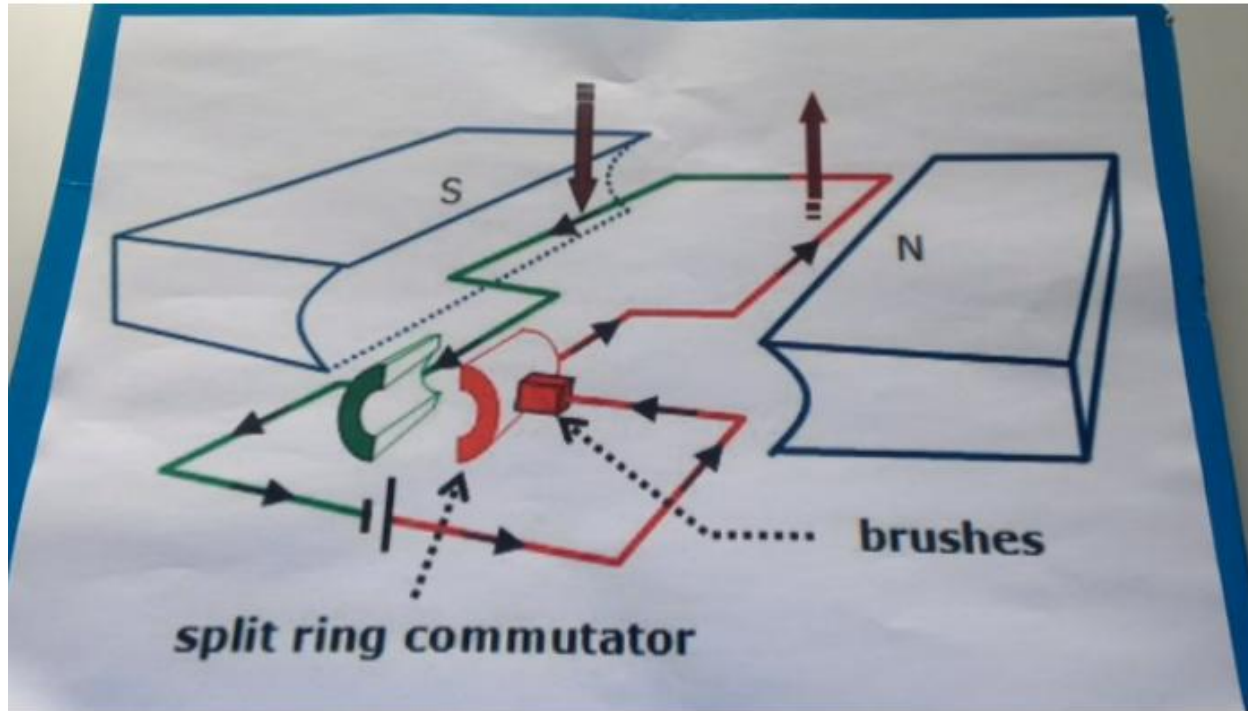
lead

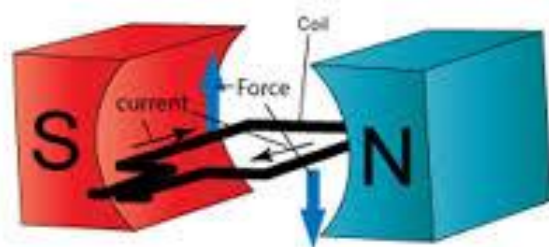


Magnetic field

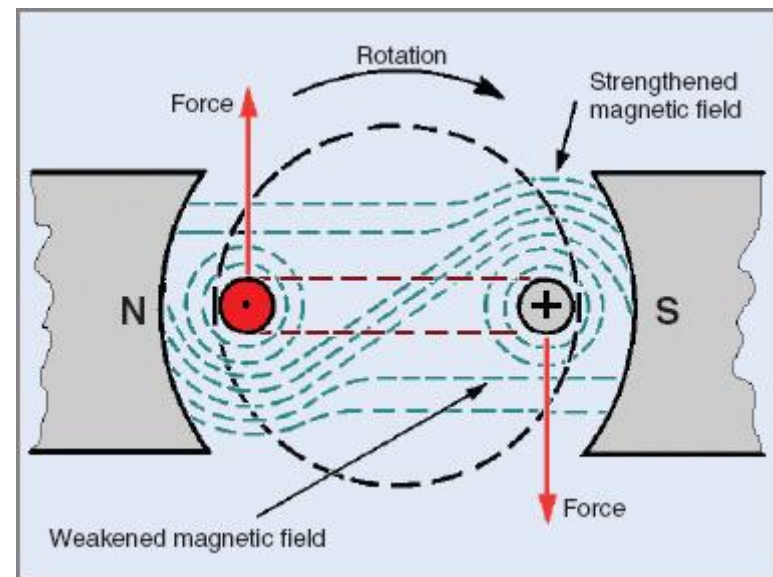
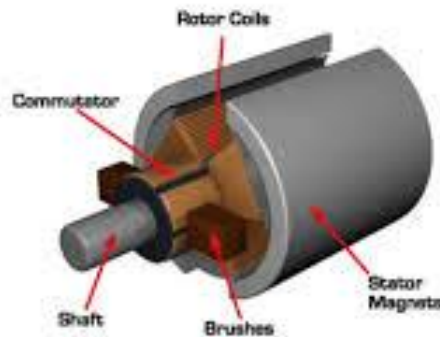


DC motor (Actuators)





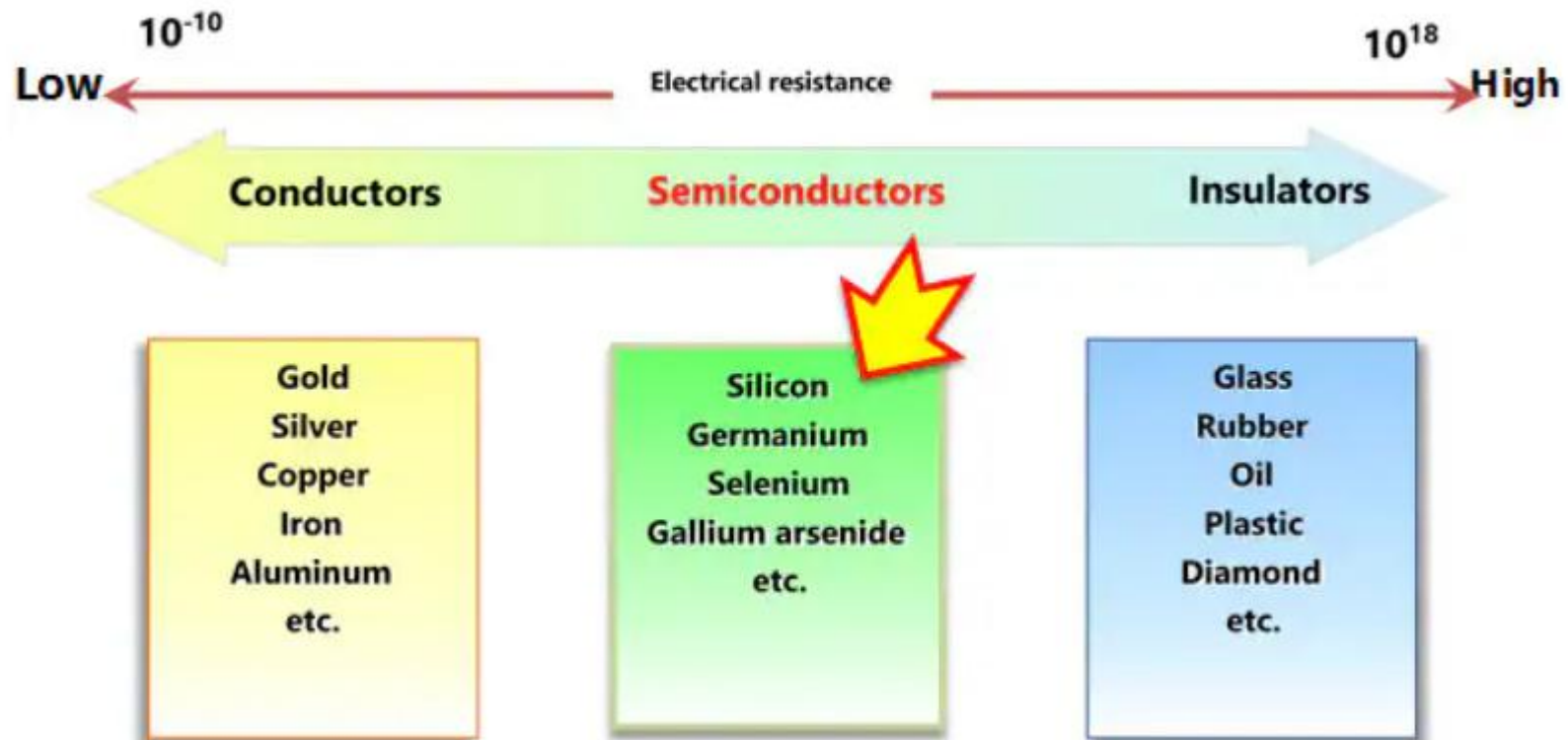
Turning Effect of a Coil in a Magnetic Field



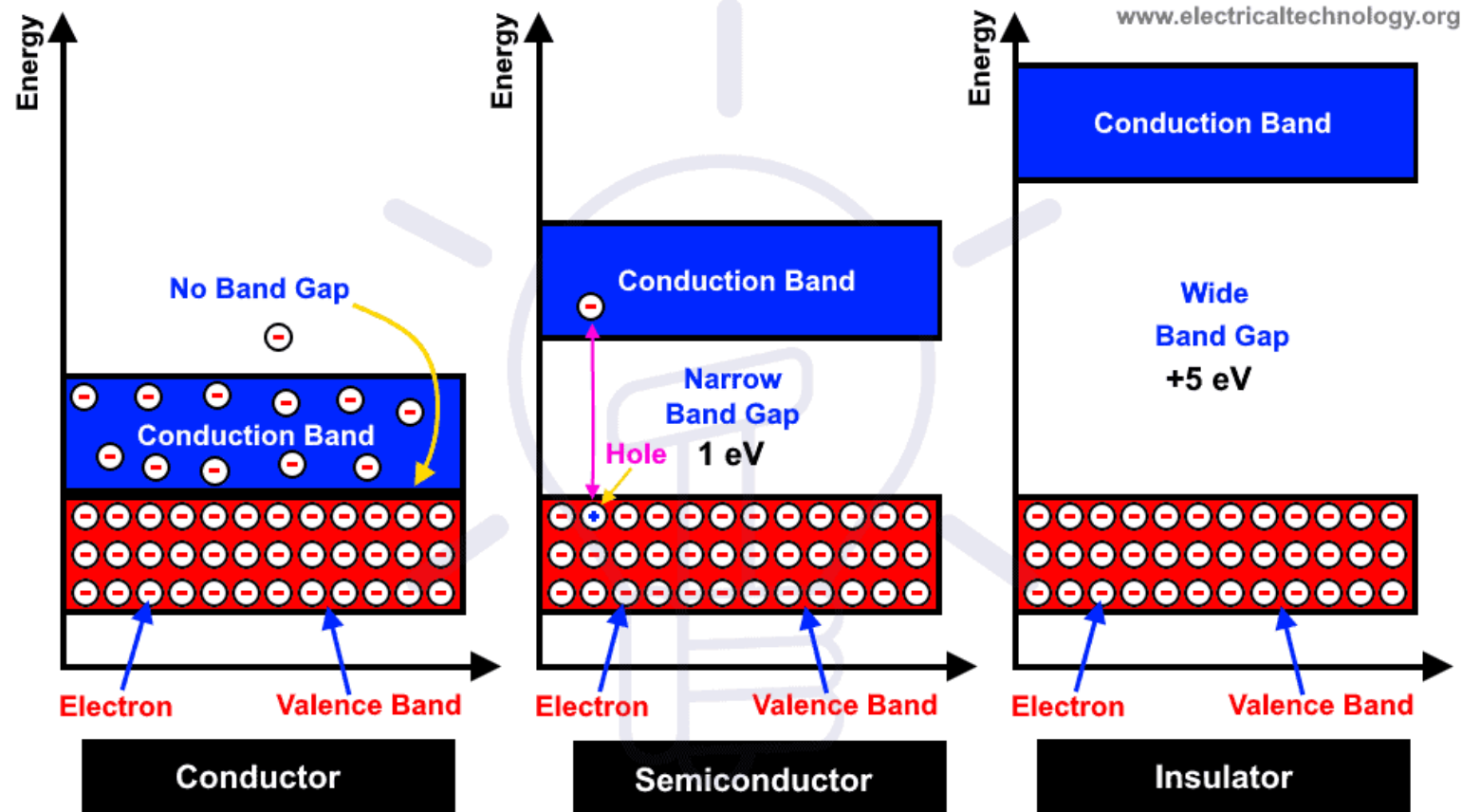
Semiconductor switches

electrons $\rightarrow$ responsible for flow of
(-) current
 $\rightarrow$ free moving electrons.

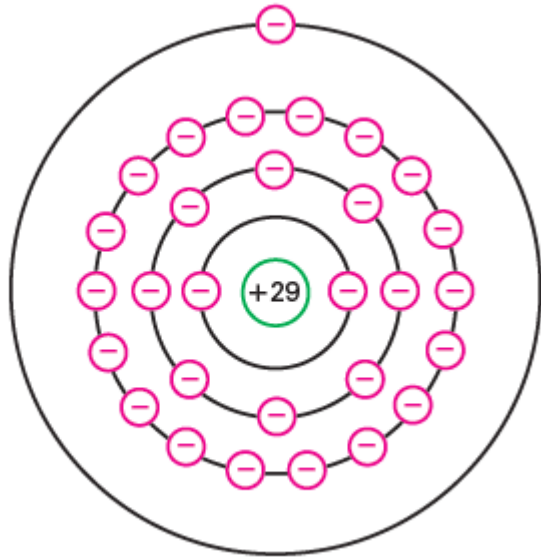
Classification of materials



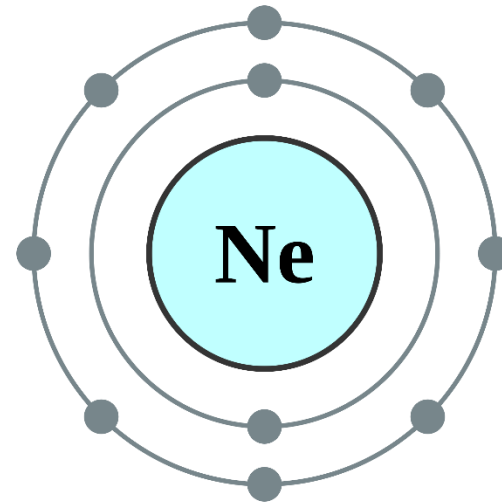
Classification of materials



Conductor / Insulator



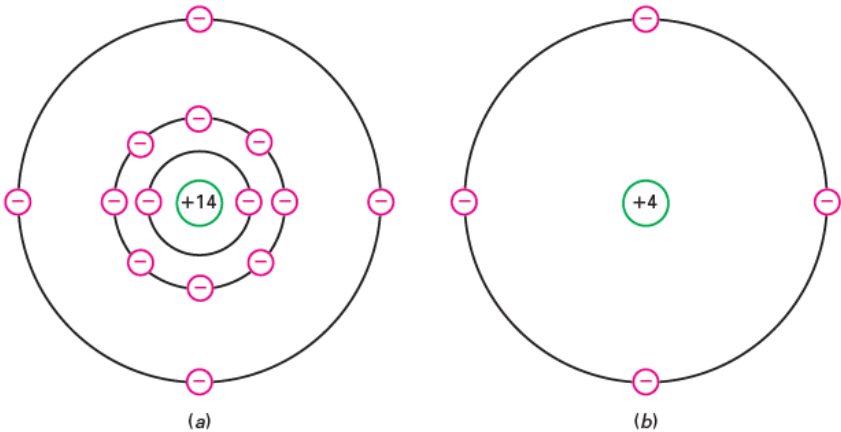
Copper atom, Good conductor – One electron in outer most orbit



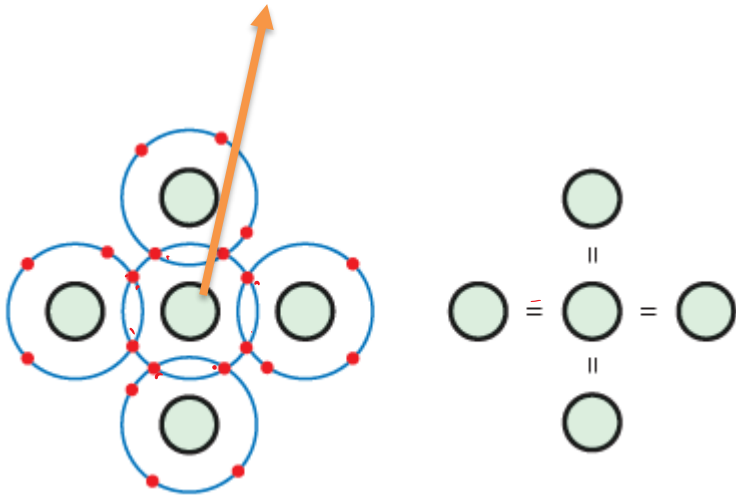
Neon atom, Poor conductor – Eight electrons in outer most orbit

Semi conductor

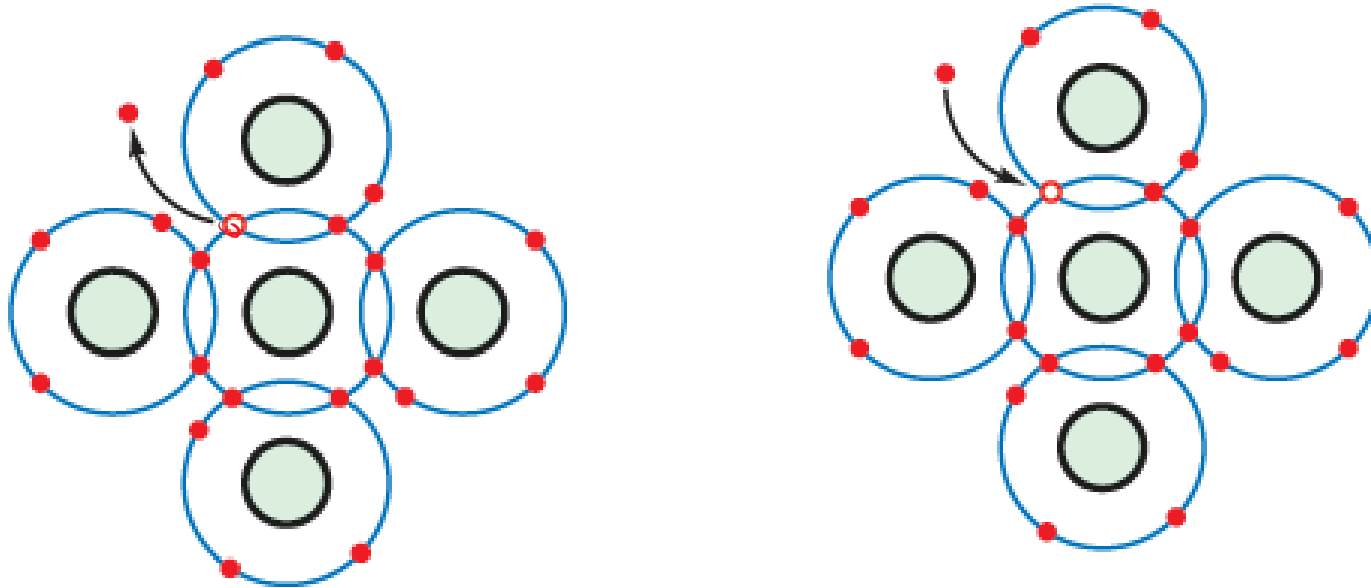
Figure 2-3 (a) Silicon atom; (b) core diagram.



Silicon Atom in the center surrounded by other 4 silicon atoms sharing one electron with center atom



Semi conductor

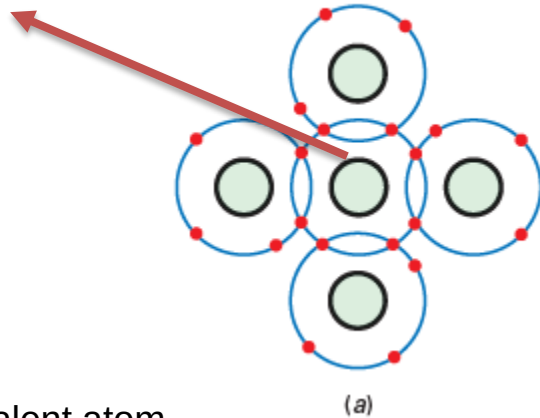


At room temperatures, valence electrons leave the orbit and their absence is marked by “holes” (Positive charge)

Doping

Pentavalent atom

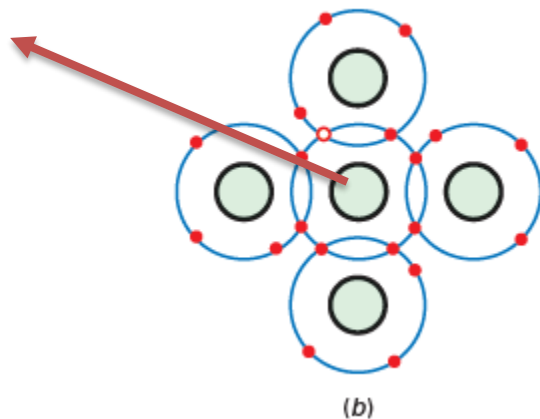
• FREE ELECTRON



Pentavalent atoms have five electrons in the valence orbit. Examples of pentavalent atoms include arsenic, antimony, and phosphorus.

Because these materials *will donate an extra electron* to the silicon crystal, they are often referred to as *donor impurities*

Trivalent atom



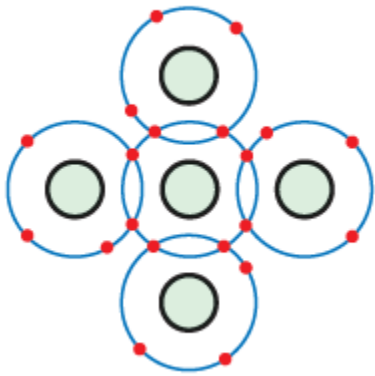
How can we dope a pure silicon crystal to get an **excess of holes**?

By using a *trivalent impurity*, one whose atoms have only three valence electrons. Examples include aluminum, boron, and gallium.

Doping

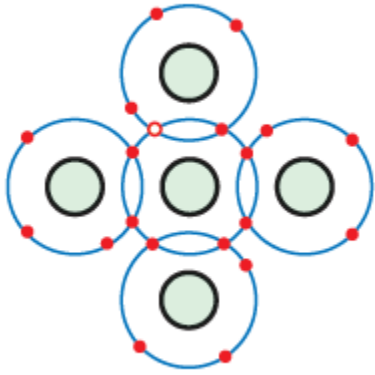
N type Semi conductor

• FREE ELECTRON



(a)

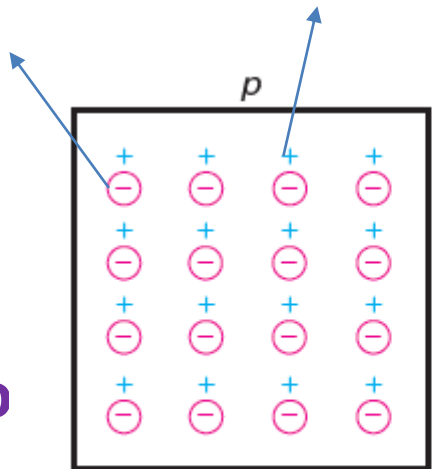
P type Semi conducto



(b)

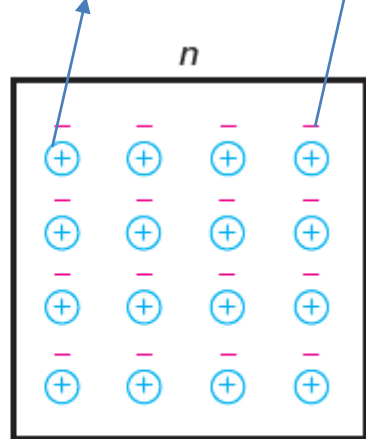
Trivalent impurities

holes



Pentavalent impurities

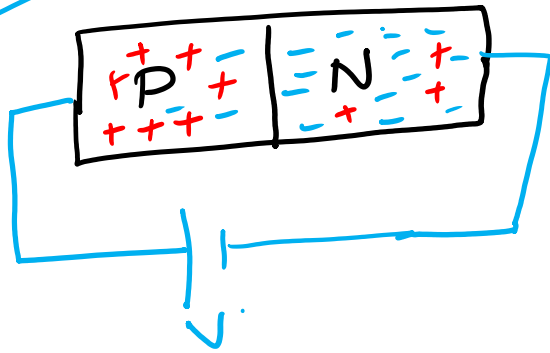
Electrons



$N \rightarrow e^-$
 $p \rightarrow \text{holes}$
 e^- have more mobility than holes

Forward bias

Switch is ON



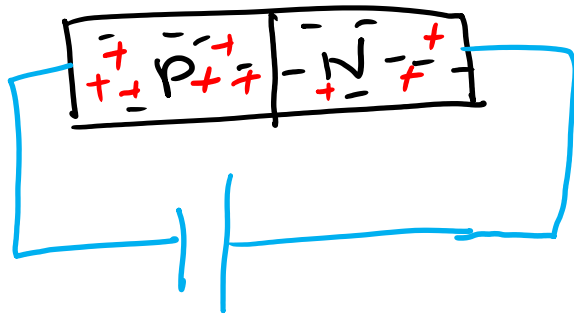
P $\rightarrow$ majority charge carriers are holes

N $\rightarrow$ majority charge carriers are electrons.

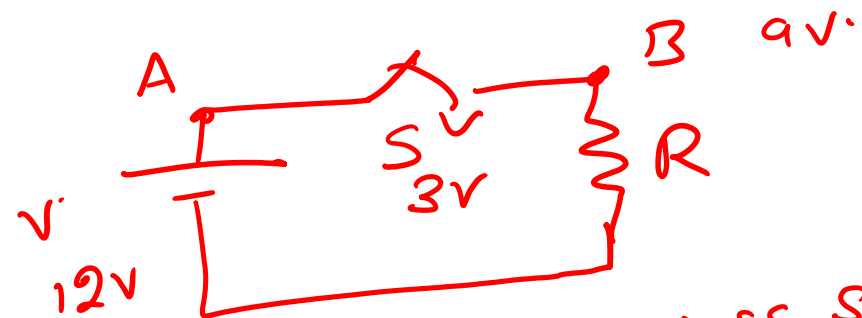
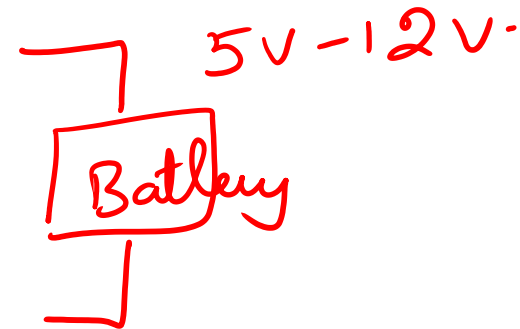
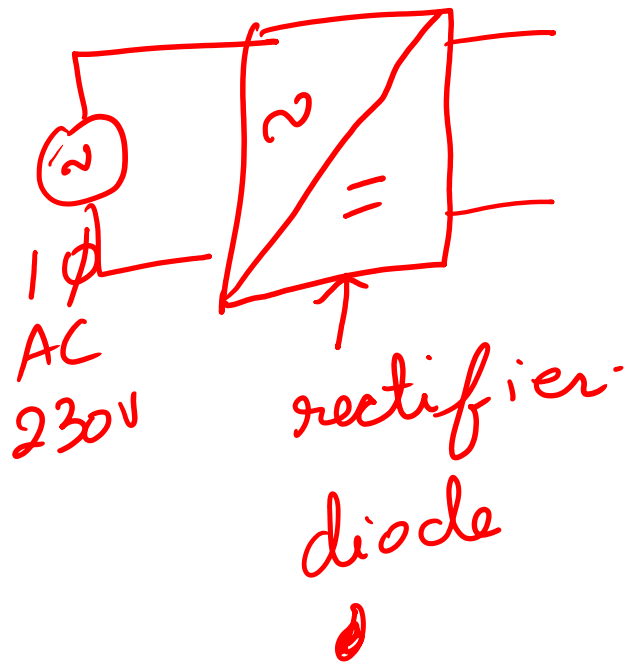
$\hookrightarrow$ holes & e^- recombine @ the j_n & current flows

Reverse bias

OFF



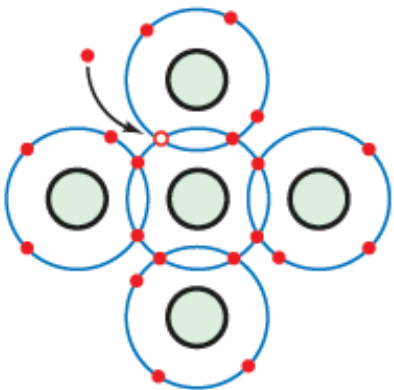
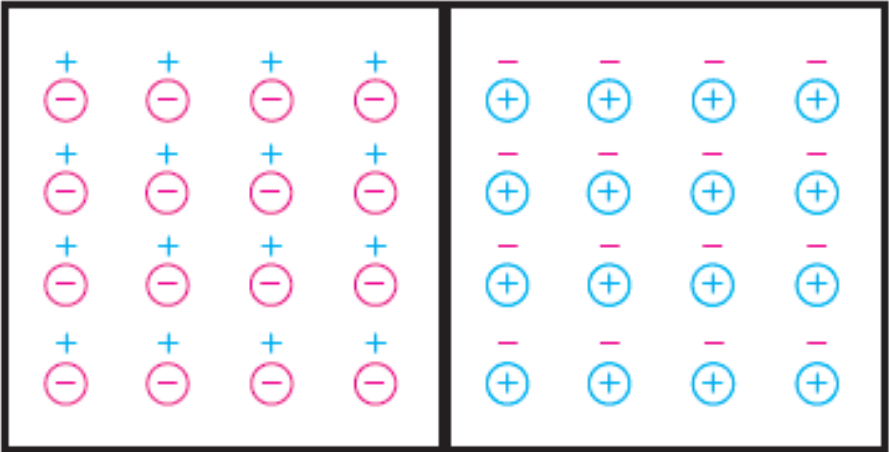
$\rightarrow j_n$ does not have majority charge carriers so no current
 $\rightarrow$ leakage current



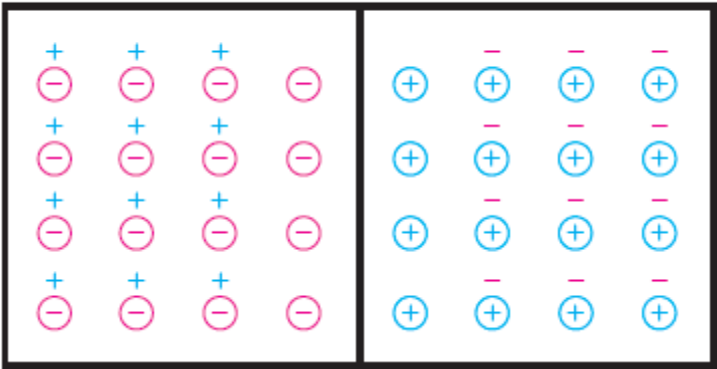
$S_{ON} \rightarrow \text{drop across } S = 0$
 $S_{OFF} \rightarrow \text{block } 12V$

P-N Junction

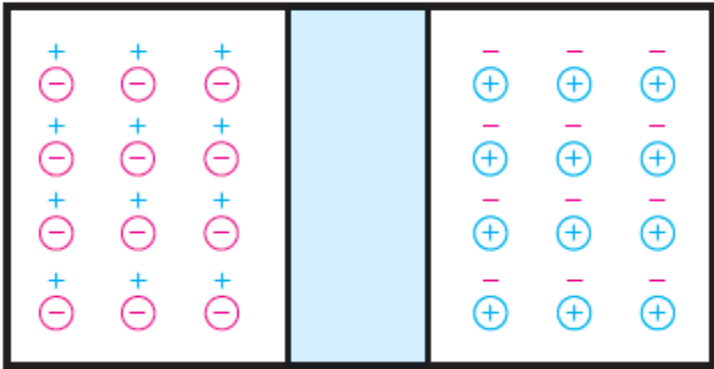
The *pn* junction.



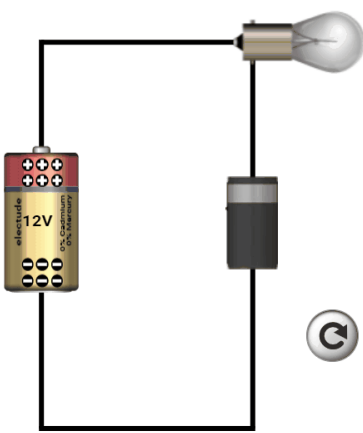
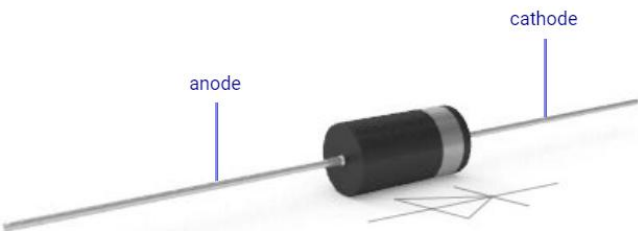
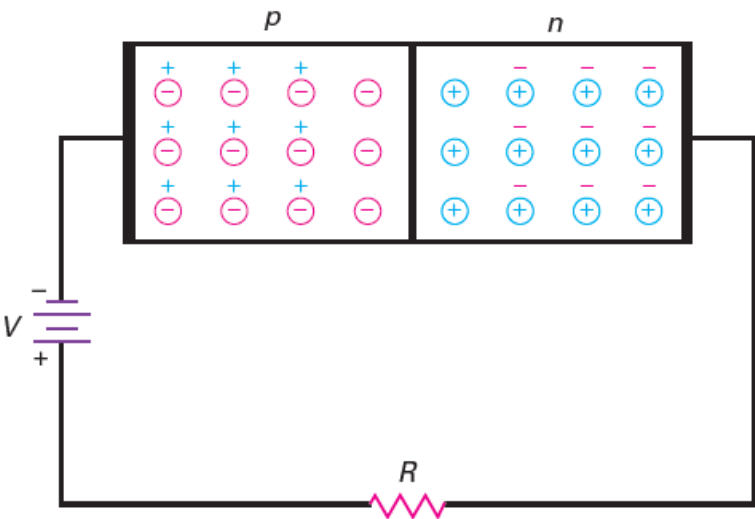
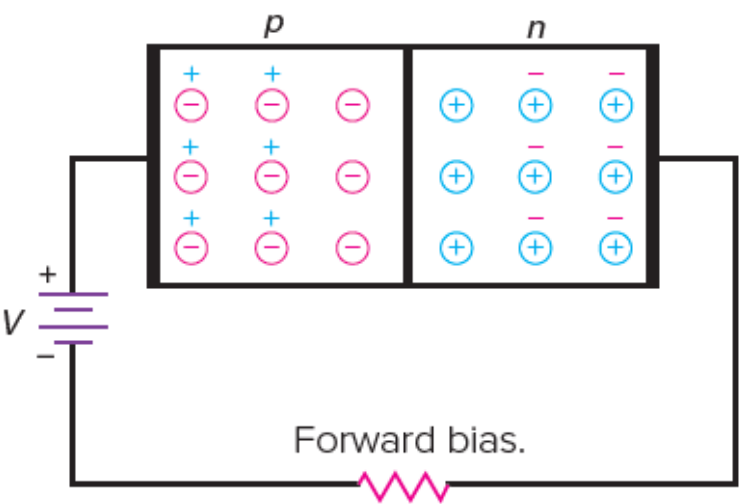
IONS



DEPLETION LAYER

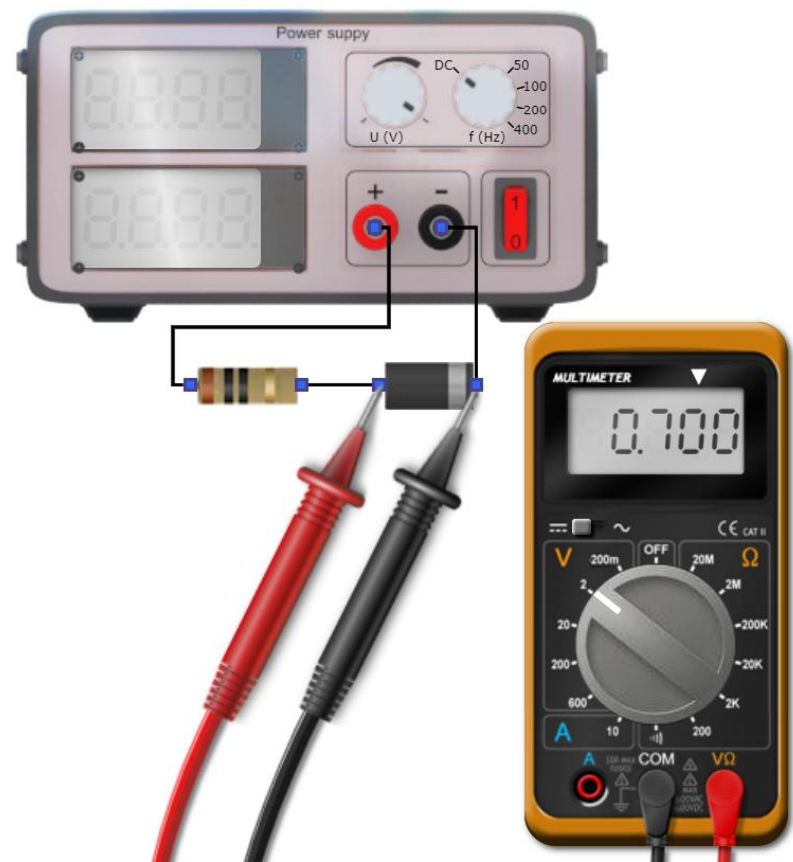


P-N Junction

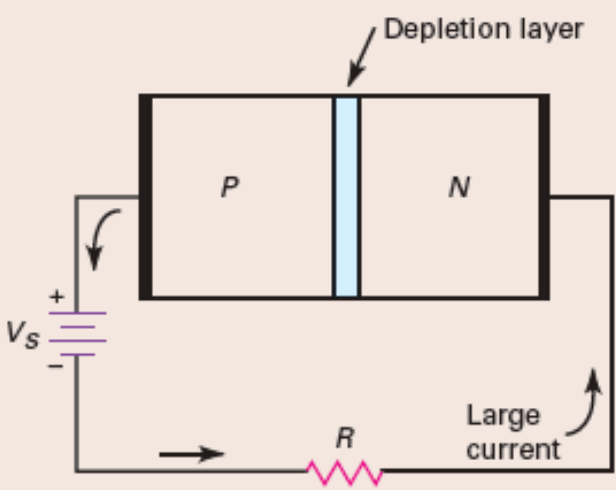
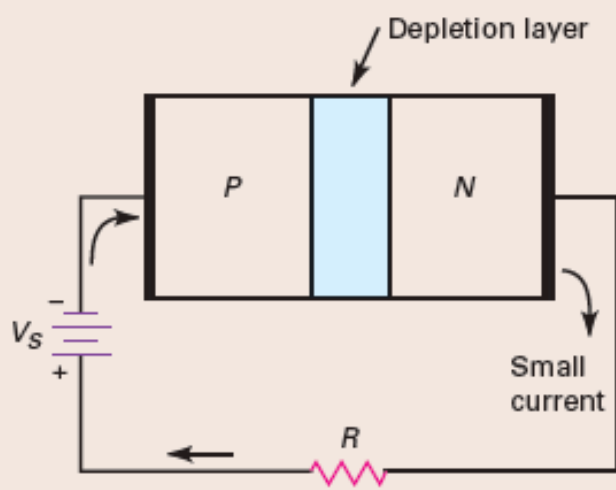


Reverse bias.

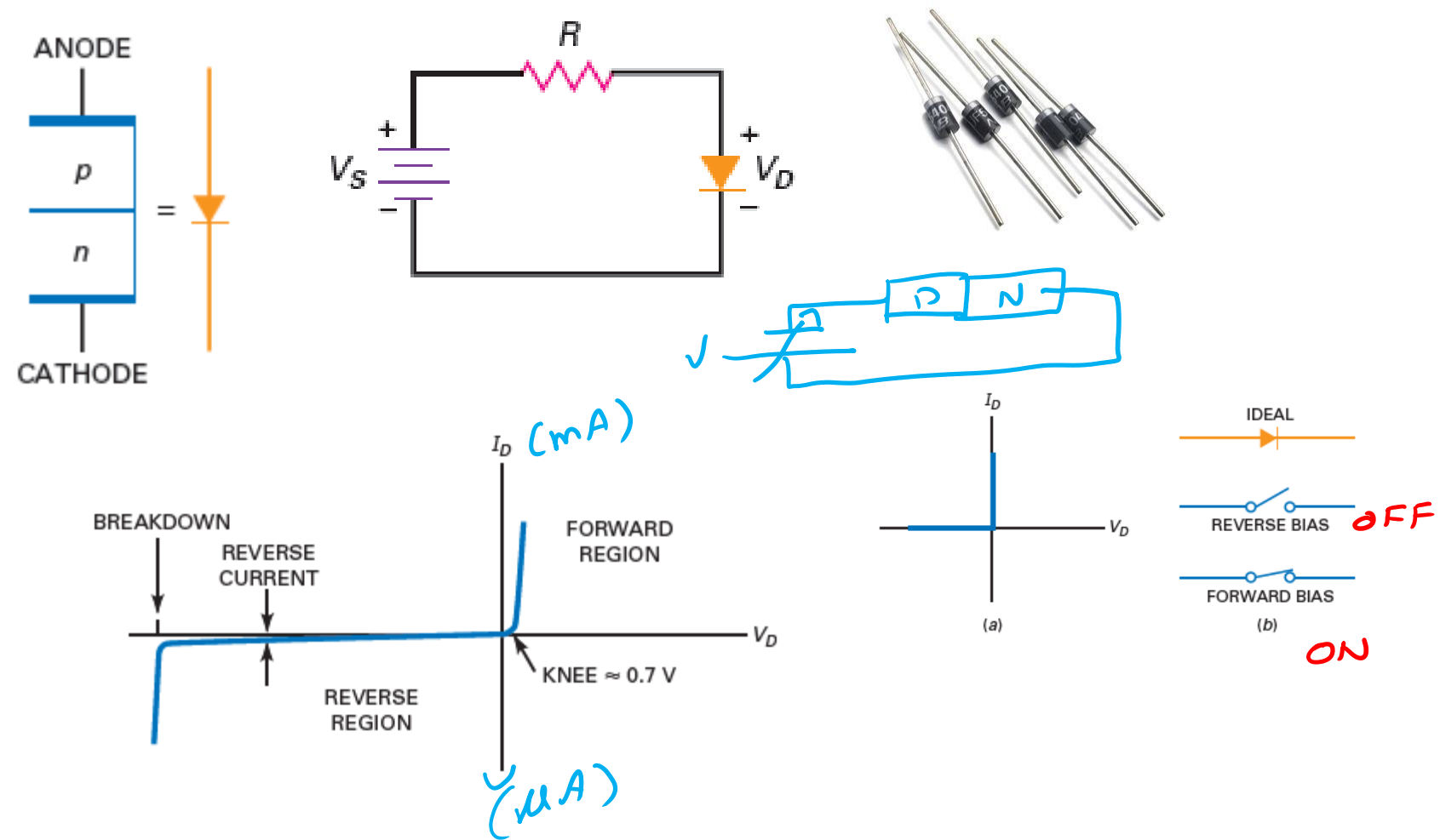
P-N Junction

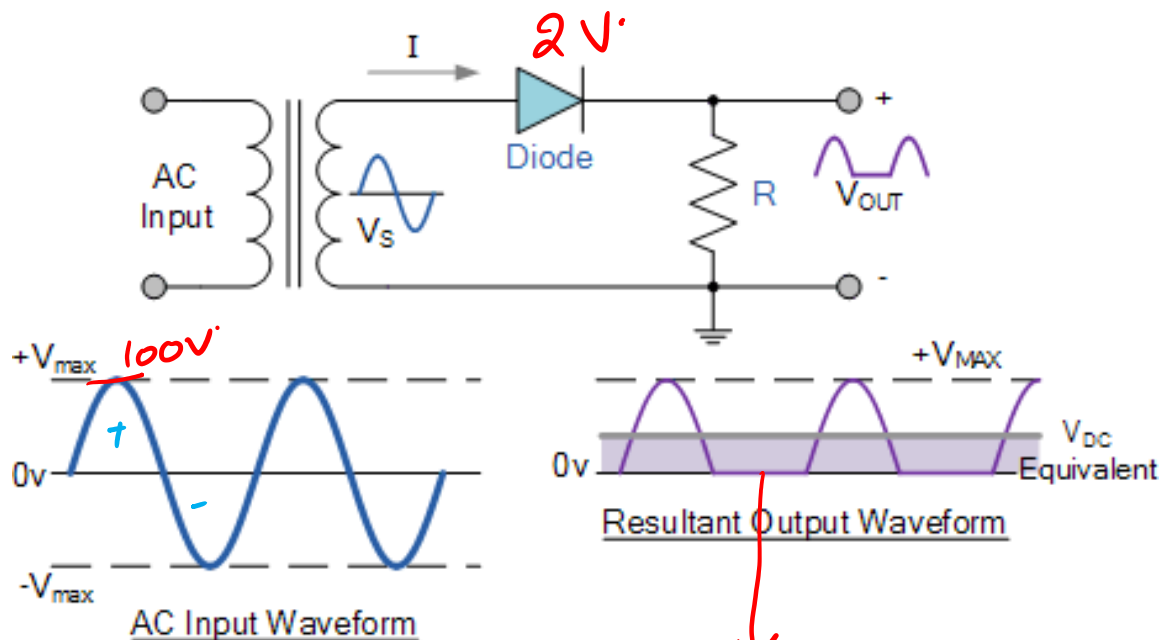


P-N Junction

Forward bias 	Reverse bias 
V_s polarity (+) to <i>P</i> material (-) to <i>N</i> material	(-) to <i>P</i> materials (+) to <i>N</i> material
Current flow Large forward current if $V_s > 0.7\text{ V}$	Small reverse current (saturation current and surface leakage current) if $V_s < \text{breakdown voltage}$
Depletion layer Narrow	Wide

Diodes



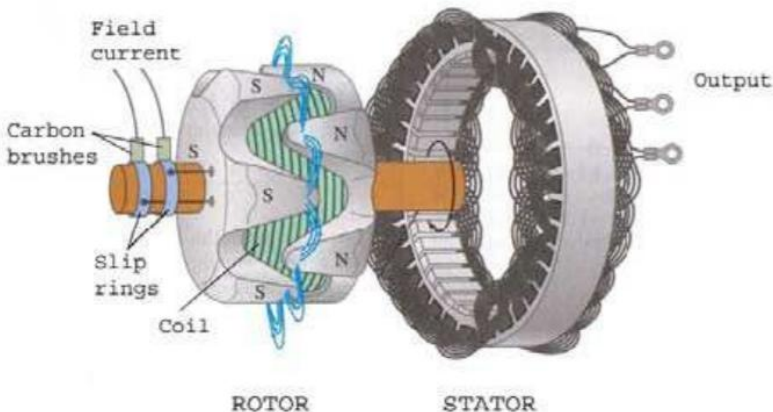
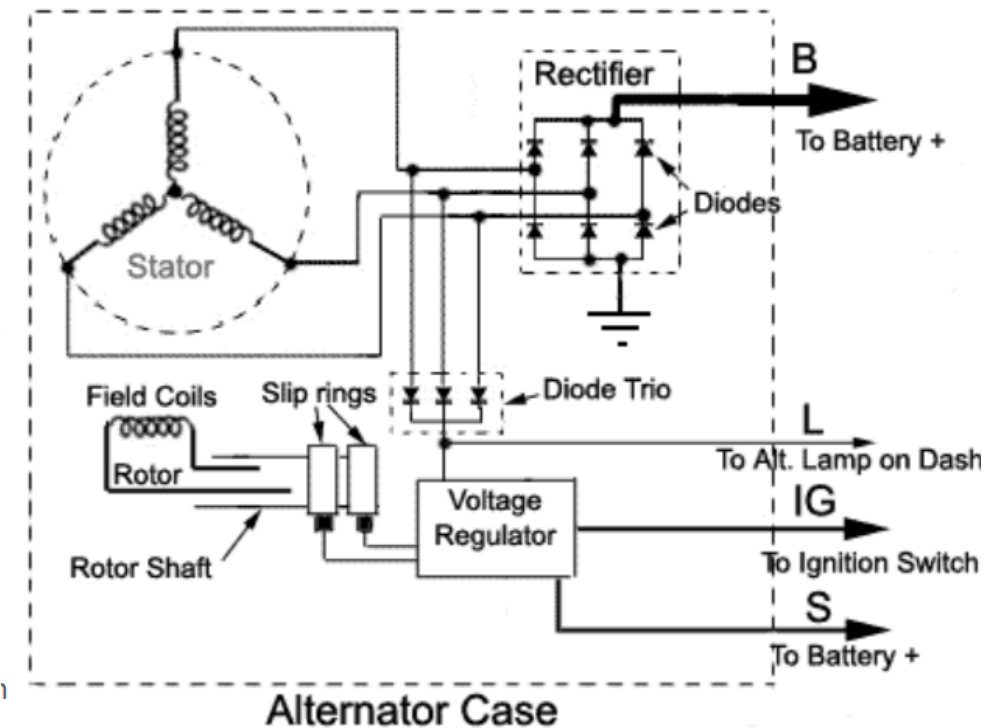


pulsated DC



Alternator Charging

Typical Alternator Circuit

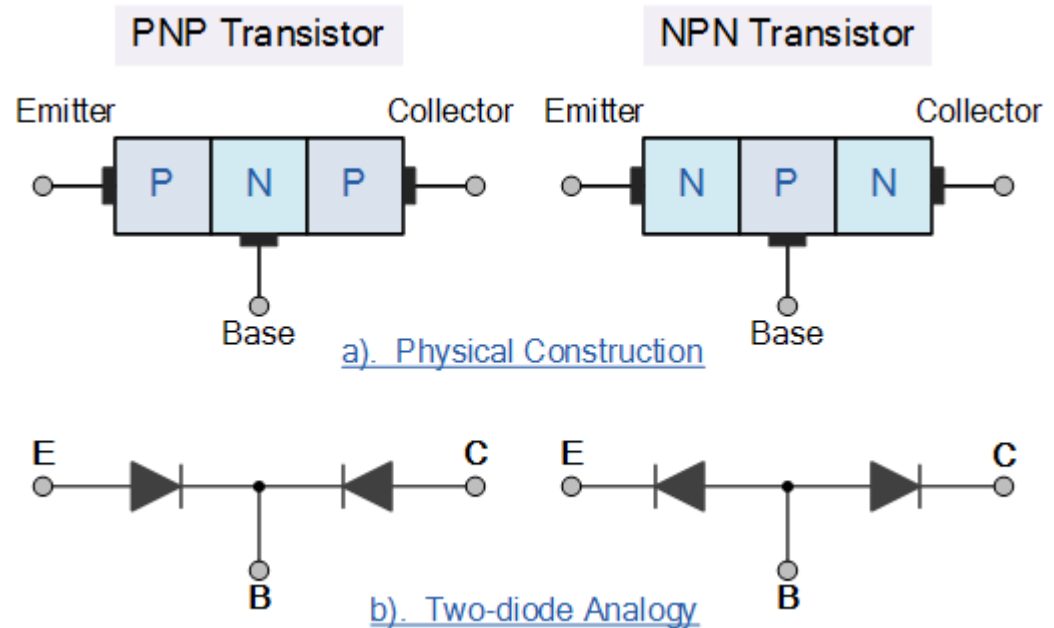


Transistors (BJTs) – Bipolar Junction transistors

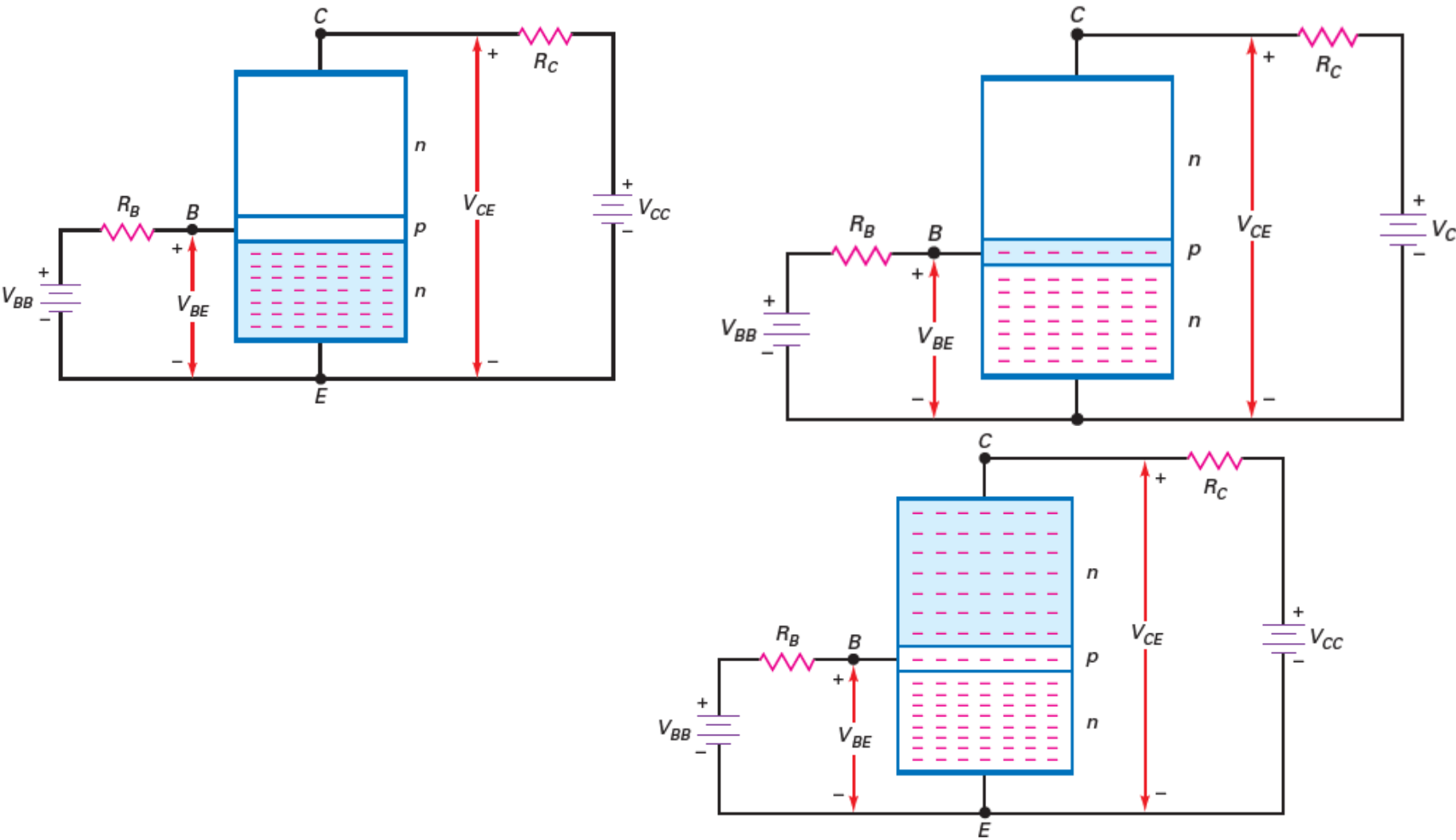
- The emitter is heavily doped.
- The base is lightly doped.
- The doping level of the collector is intermediate, between the heavy doping of the emitter and the light doping of the base.

Why called as Bipolar?

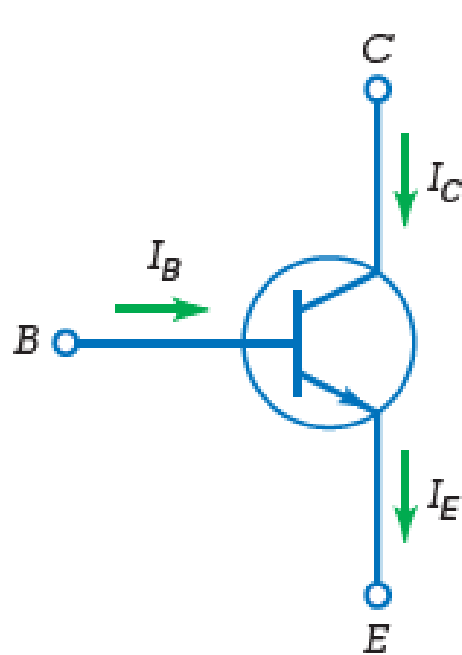
Operation depends on Free Electrons and Holes



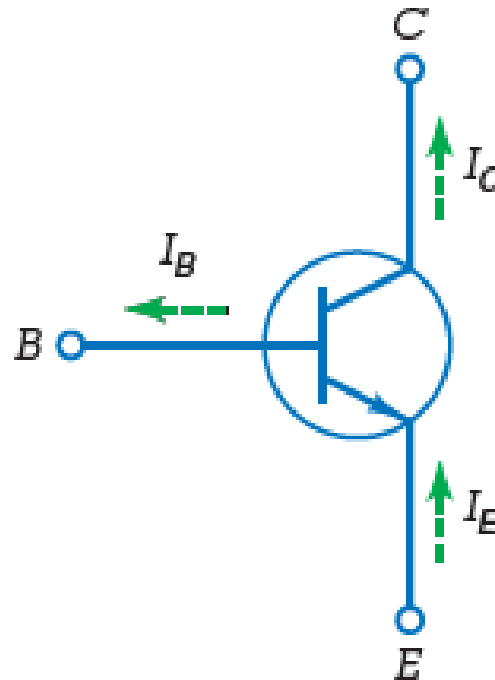
Transistors



Transistors



Conventional Current



Electron Flow

$$I_E = I_C + I_B$$

$$I_C \approx I_E$$

$$\alpha_{dc} = \frac{I_C}{I_E}$$

β_{dc} is the current gain of the transistor – A small base current controls much larger collector current

$$\beta_{dc} = \frac{I_C}{I_B}$$

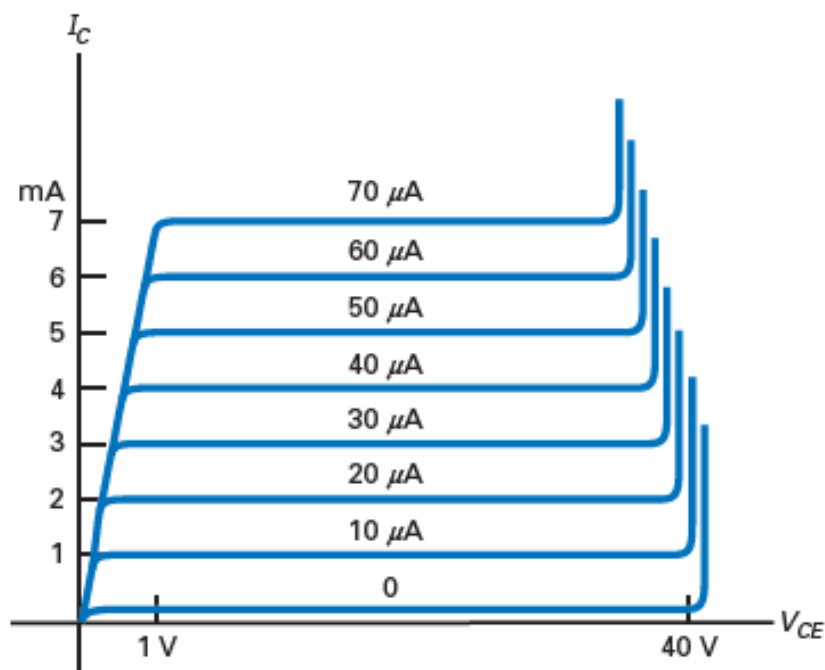
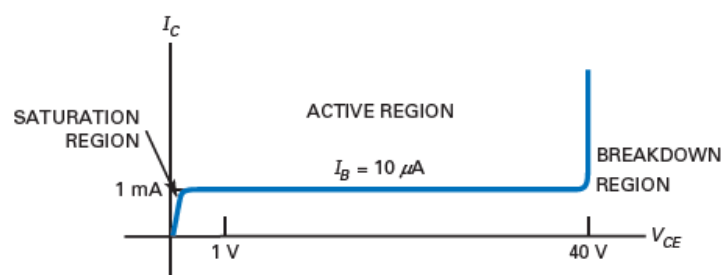
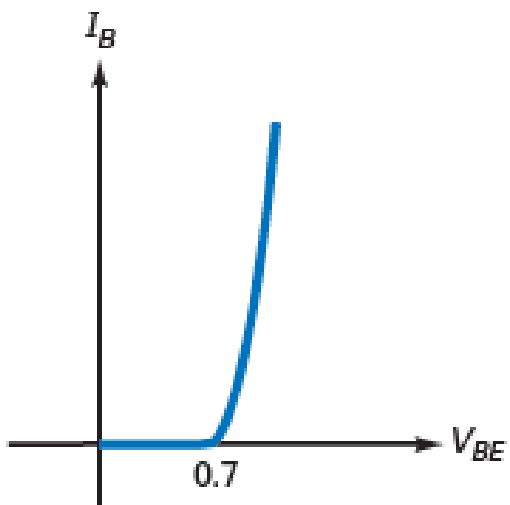
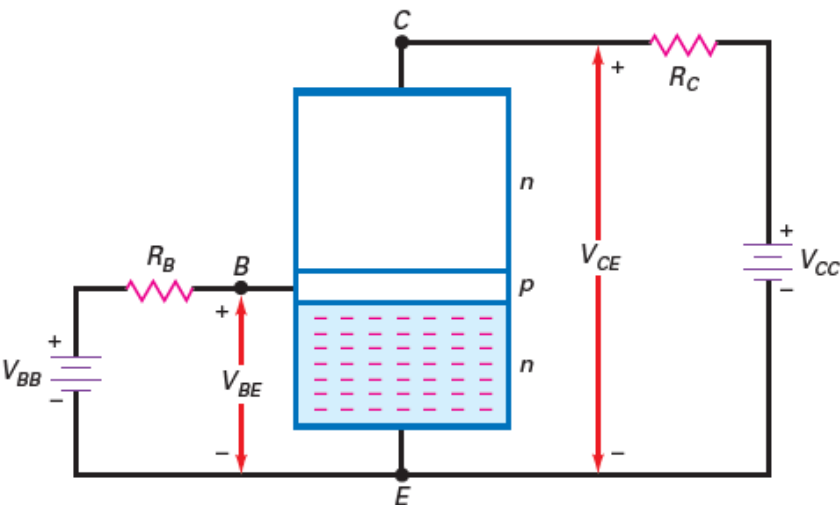
Transistors

A transistor has a collector current of 10 mA and a base current of 40 μA . What is the current gain of the transistor?

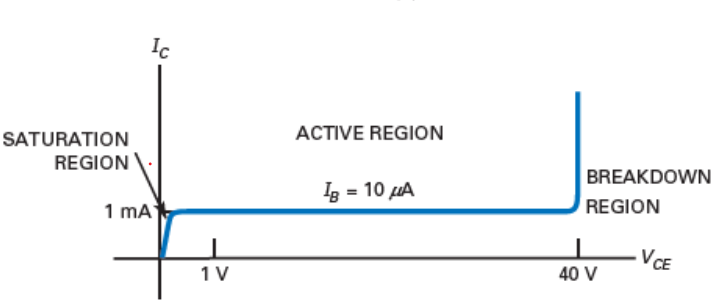
SOLUTION Divide the collector current by the base current to get:

$$\beta_{\text{dc}} = \frac{10 \text{ mA}}{40 \mu\text{A}} = 250$$

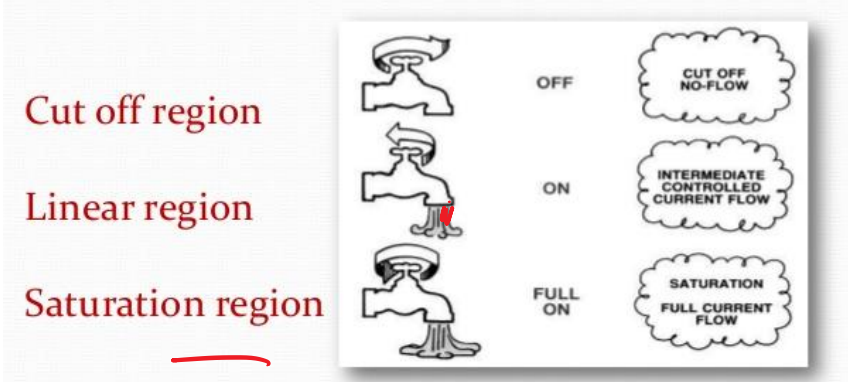
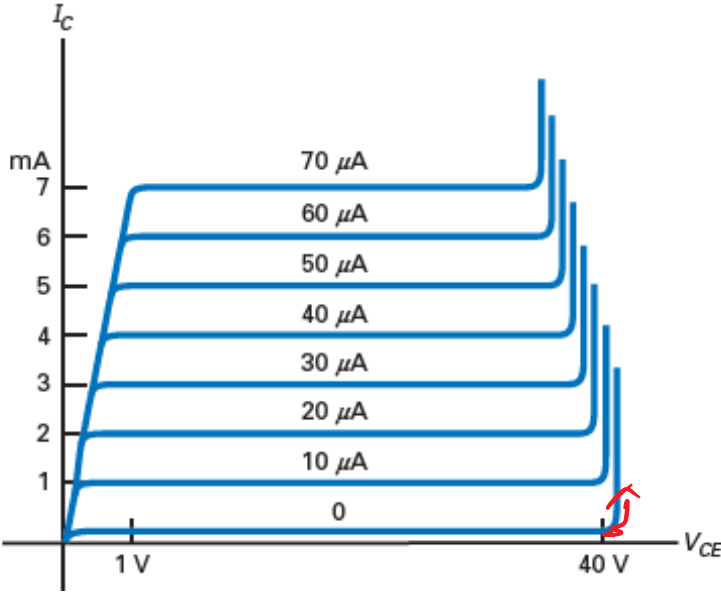
Transistors



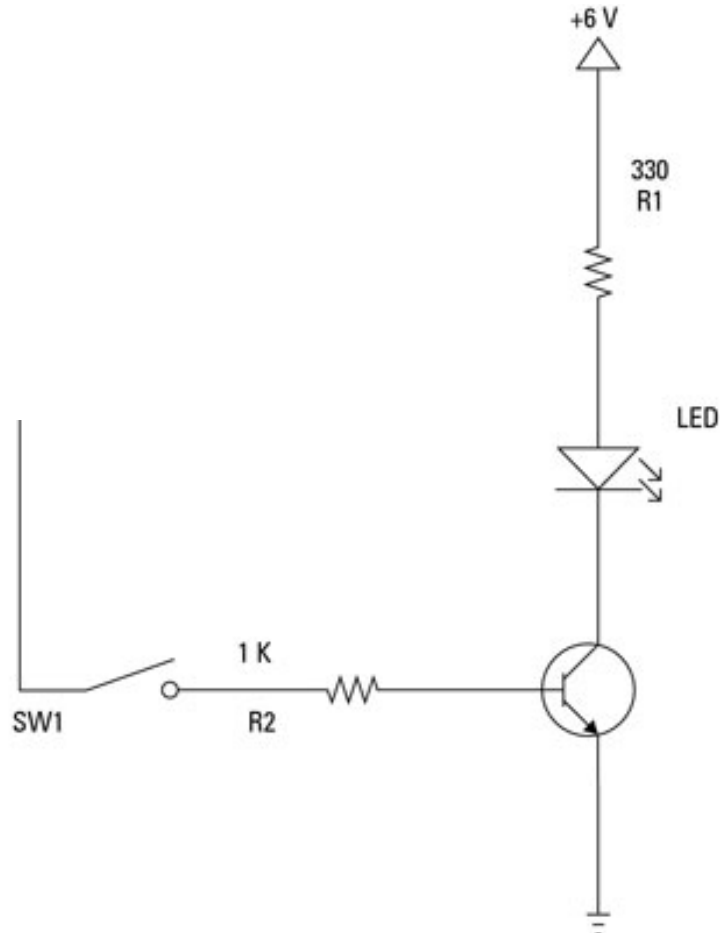
Transistors



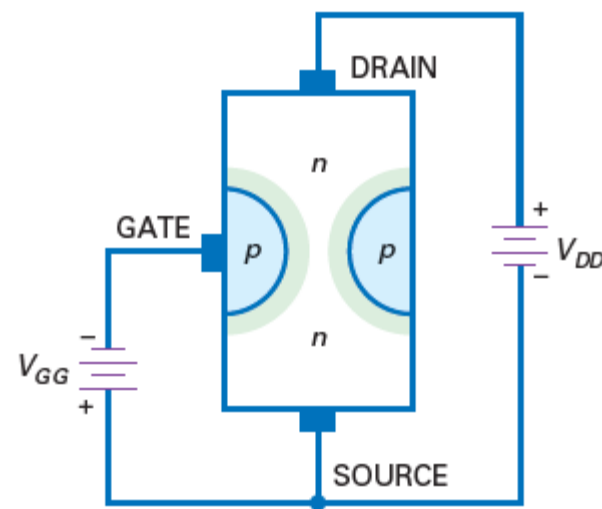
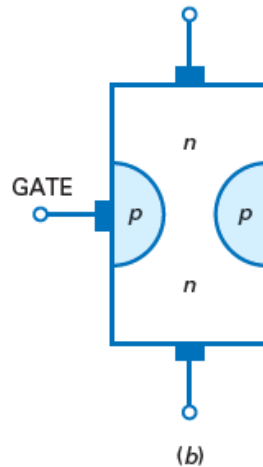
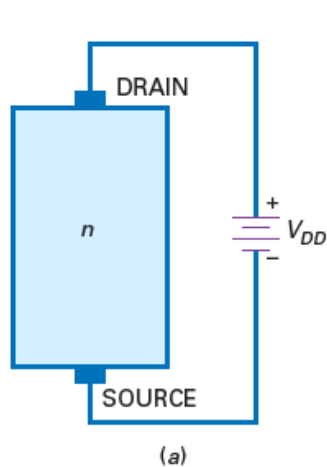
- Saturation** – The transistor acts like a **short circuit**. Current freely flows from collector to emitter.
- Cut-off** – The transistor acts like an **open circuit**. No current flows from collector to emitter.
- Active** – The current from collector to emitter is **proportional** to the current flowing into the base.



Transistor as a switch



Field Effect transistors (unipolar)



Gate Voltage Controls Drain Current

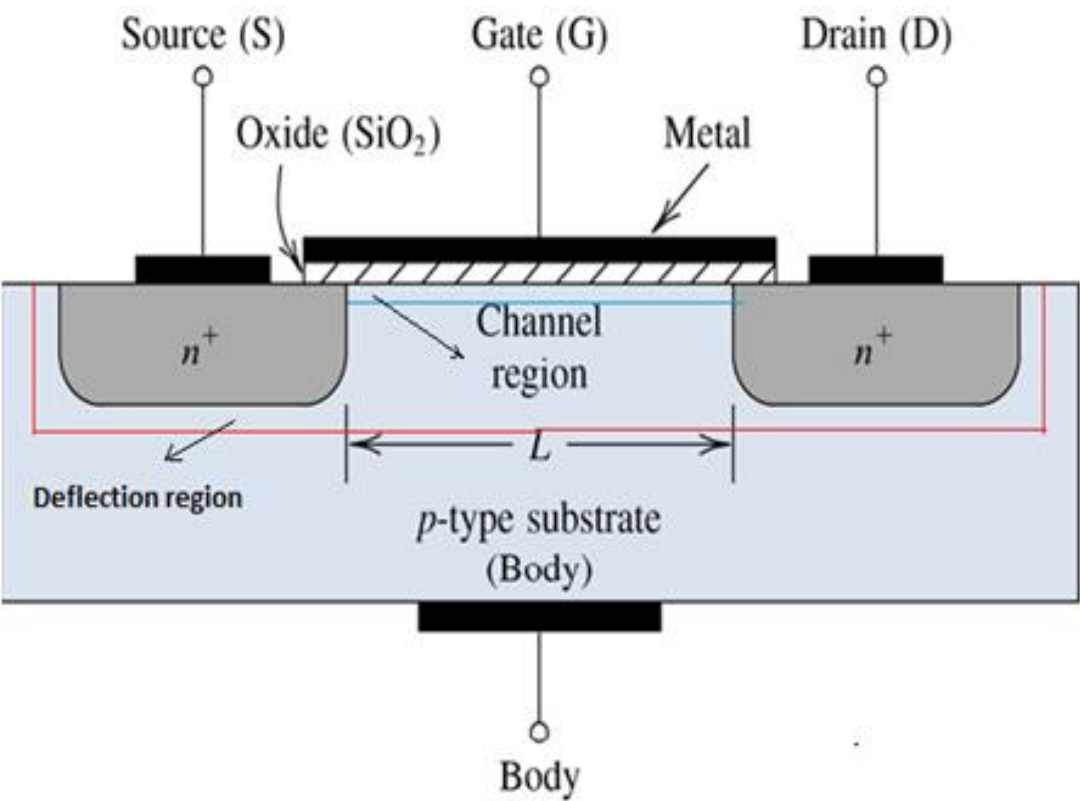
Why called as Unipolar?

Operation depends on either Free Electrons or Holes

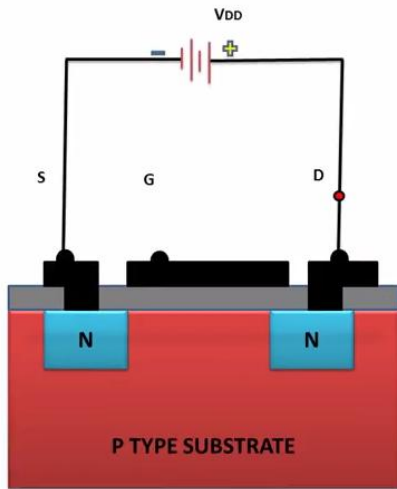
Much faster switching circuit than BJTs.



MOSFET



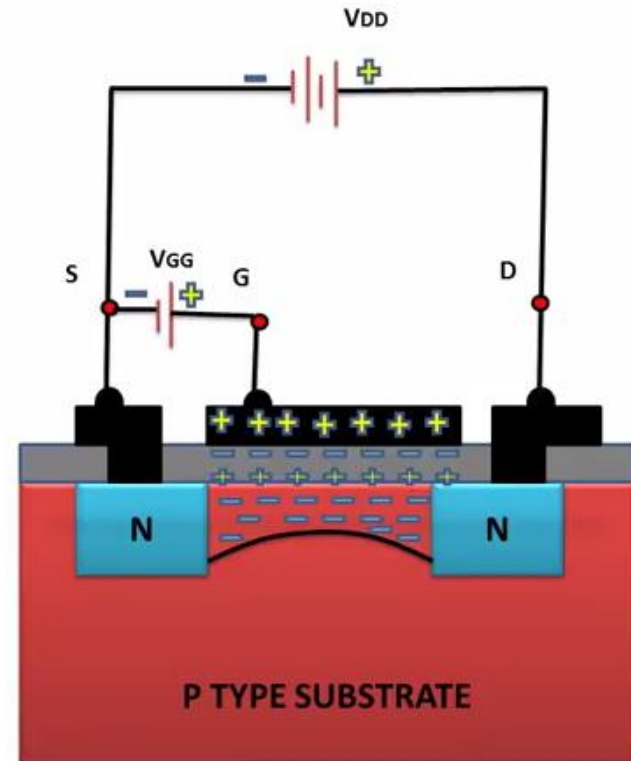
MOSFET



INPUT IMPEDANCE- VERY HIGH

SIZE – VERY SMALL

LESS NOISY THAN BJT



Frequency	Percentage
Never	10%
Sometimes	40%
Often	50%

NO THERMAL RUNAWAYS

Digital electronics
(for reference.)

AND Gate *(multiplication)*

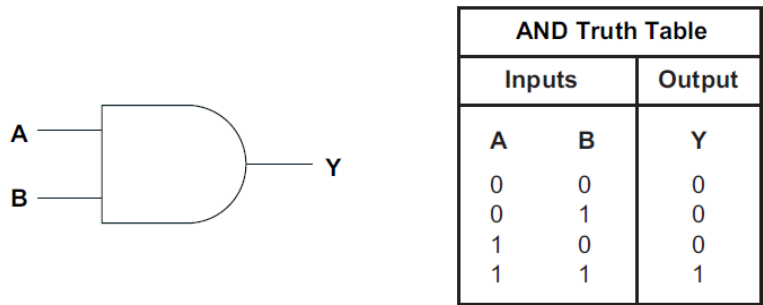
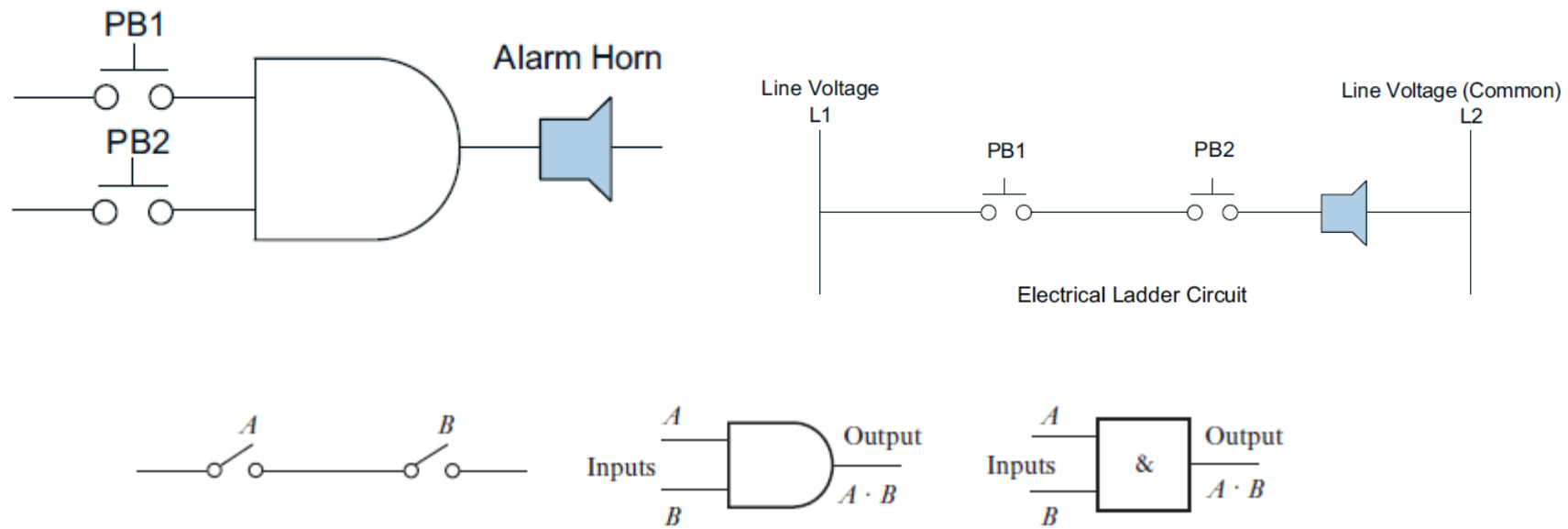
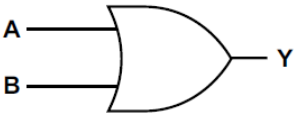


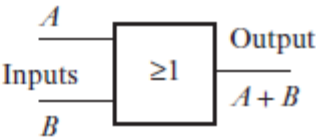
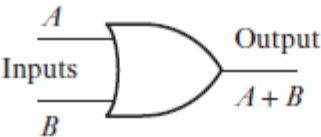
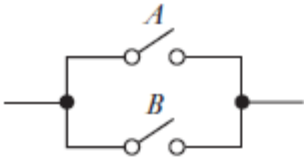
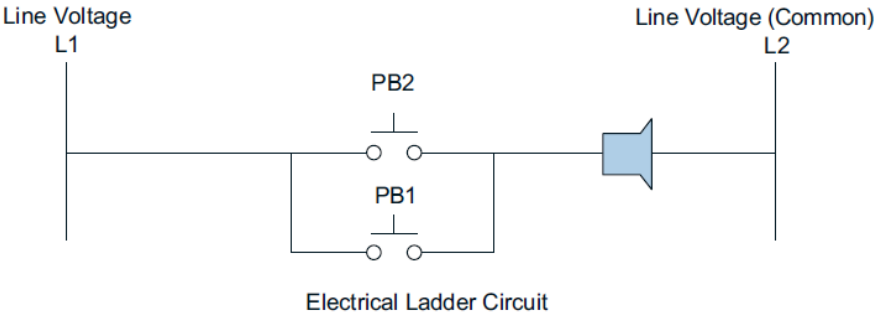
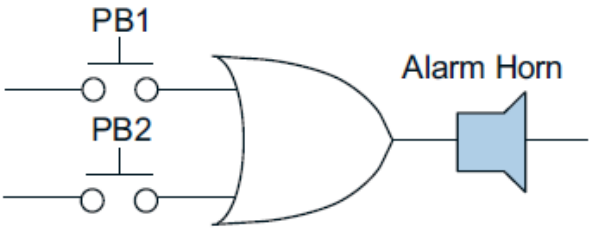
Figure 3-2. Two-input AND gate and its truth table.

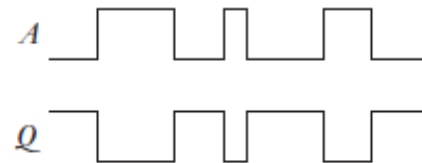


OR Gate (addition)



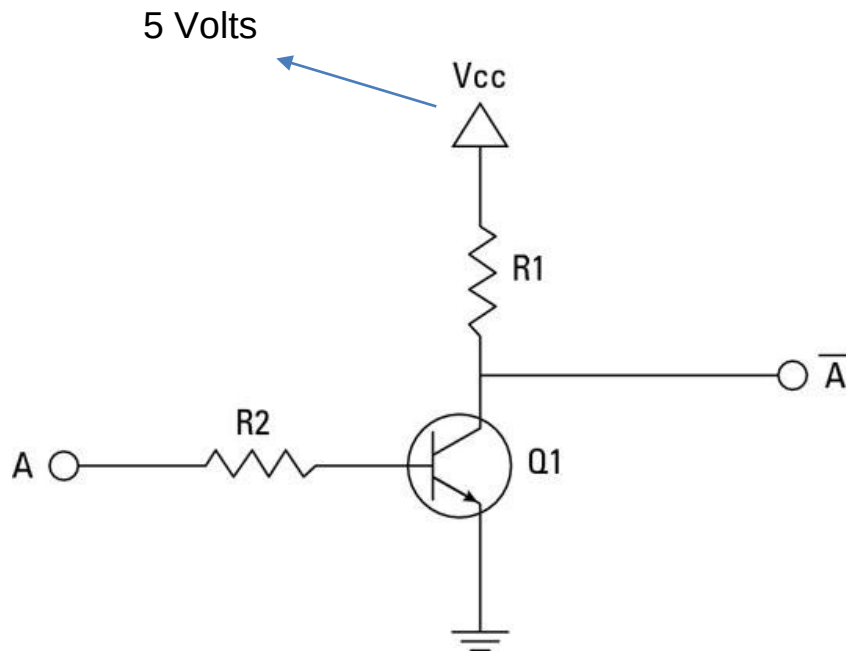
OR Truth Table		
Inputs		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1





Input A	Output Q
0	1
1	0

NOT Gate Hardware

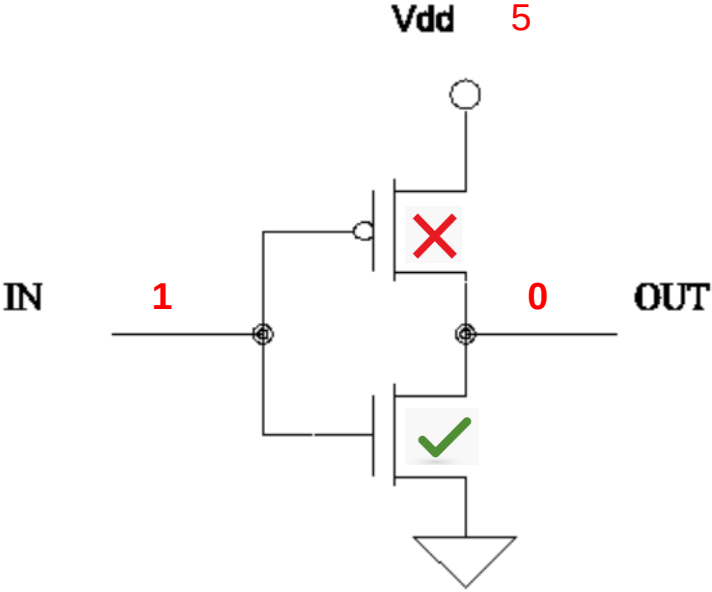
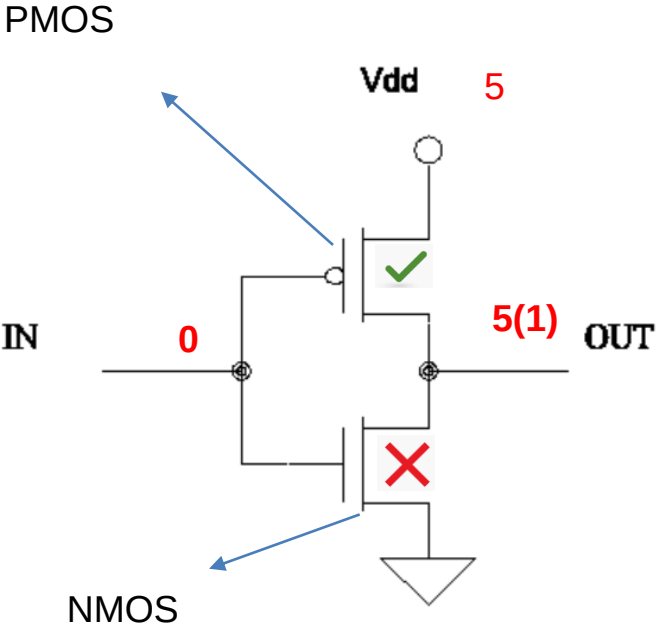


When the transistor is off, no current flows through the collector-emitter path. In this way, the circuit's output is HIGH when its input is LOW.

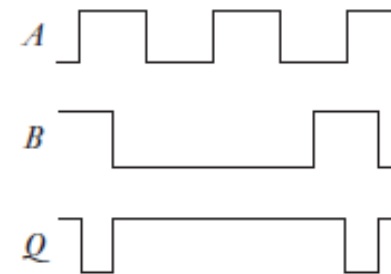
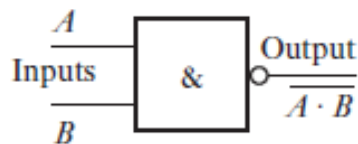
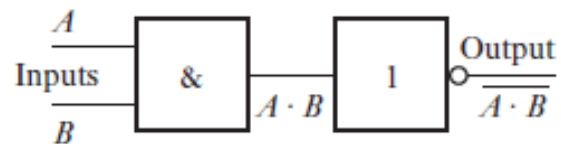
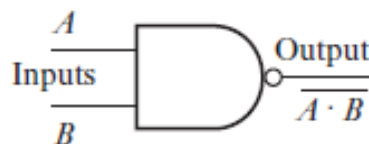
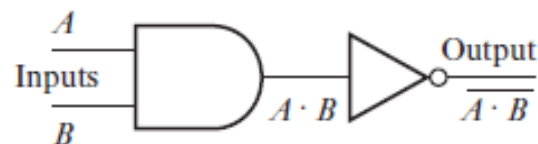
When voltage is present at the input, the transistor turns on, allowing current to flow through the collector-emitter circuit directly to ground. This ground path creates a shortcut that bypasses the output, which causes the output to go LOW.

In this way, the output is HIGH when the input is LOW and LOW when the input is HIGH.

NOT Gate (CMOS)

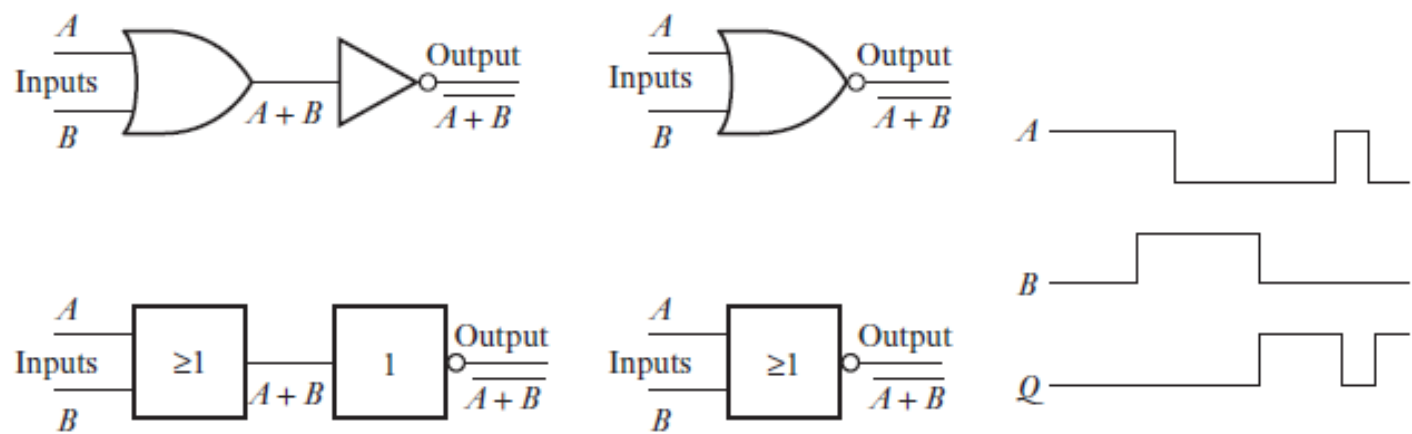


NAND Gate (AND + NOT)



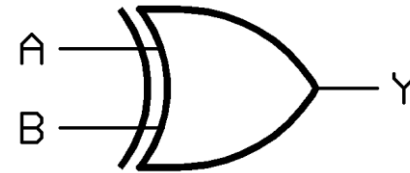
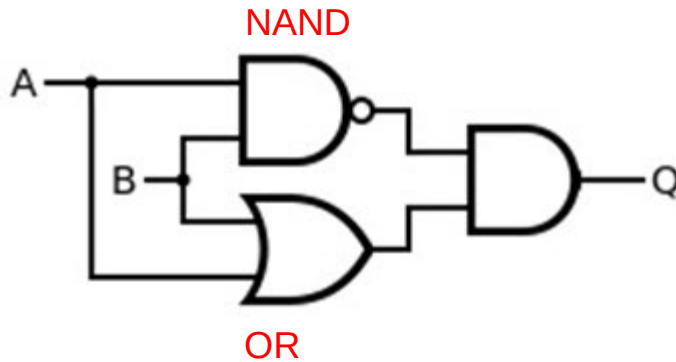
Inputs		Output Q
A	B	
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate (OR + NOT)



Inputs		Output Q
A	B	
0	0	1
0	1	0
1	0	0
1	1	0

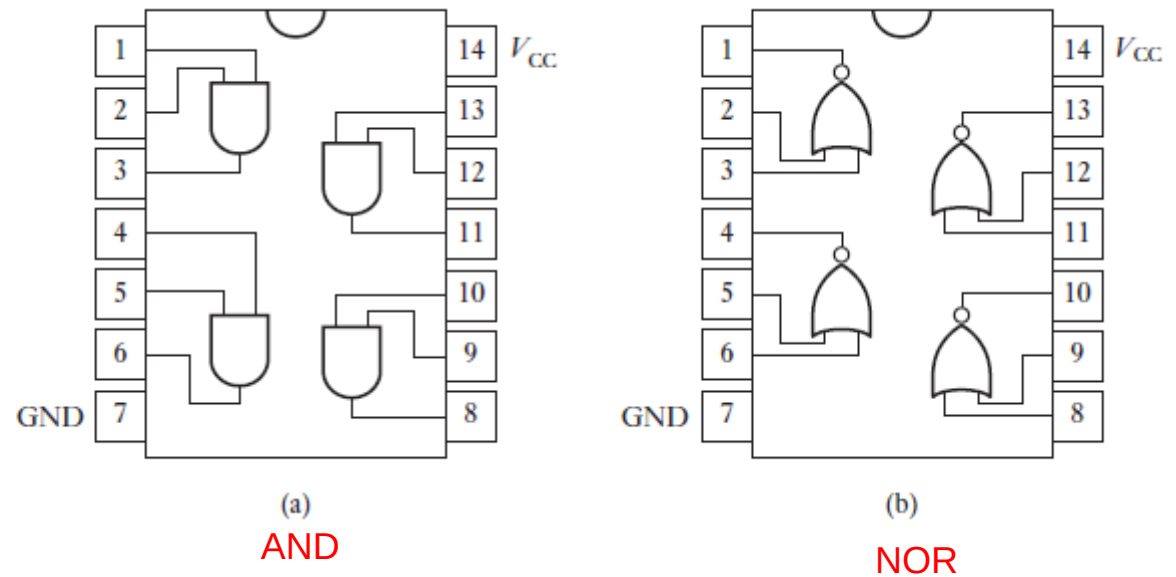
XOR (Exclusive OR)



Inputs		Output <i>Q</i>
<i>A</i>	<i>B</i>	
0	0	0
0	1	1
1	0	1
1	1	0

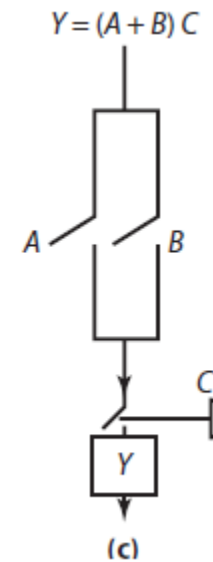
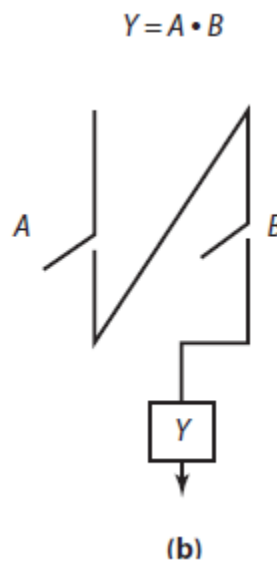
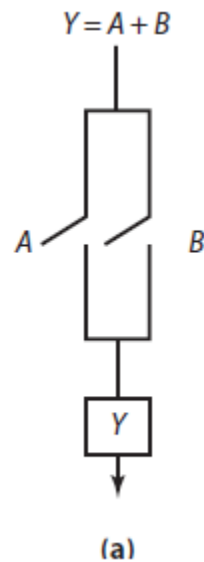
Arrangement

Figure 5.10 Integrated circuits:
(a) 7408, (b) 7402.



Boolean Algebra

- (a) A machine can be operated by either of the two operators, A and B. The power that runs a machine can be connected from either of two locations.
- (b) Due to the safety requirements, the power must be channelled through both stations to operate the machine.
- (c) The final safety regulations allow either station to power the machine only if the operator is out of danger.



Boolean Algebra

- 1 Anything ORed with itself is equal to itself: $A + A = A$.
- 2 Anything ANDed with itself is equal to itself: $A \cdot A = A$.

$$\left. \begin{array}{l} A + A = A \\ A \cdot A = A \end{array} \right\} \begin{array}{l} \xrightarrow{\text{red}} \text{OR} \\ \xrightarrow{\text{blue}} \text{AND} \end{array}$$

$$\left. \begin{array}{l} A + 0 = A \\ A \cdot 1 = A \end{array} \right\}$$

$$\begin{array}{l} A + A' = 1 \\ A \cdot A' = 0 \end{array}$$

$$\left. \begin{array}{l} A \cdot (B + C) = A \cdot B + A \cdot C \\ A + B \cdot C = (A + B) \cdot (A + C) \end{array} \right\} \text{Distributive Law}$$

$$\left. \begin{array}{l} 1 + A = 1 \\ 0 \cdot A = 0 \end{array} \right\}$$

$$\left. \begin{array}{l} A + B = B + A \\ A \cdot B = B \cdot A \end{array} \right\}$$

$$\left. \begin{array}{l} \overline{A \cdot B} = \overline{A} + \overline{B} \\ \overline{A + B} = \overline{A} \cdot \overline{B} \end{array} \right\} \text{De Morgan's theorem}$$

$$\left. \begin{array}{l} A + (B + C) = (A + B) + C \\ A \cdot (B \cdot C) = (A \cdot B) \cdot C \end{array} \right\}$$

De Morgan's Law *

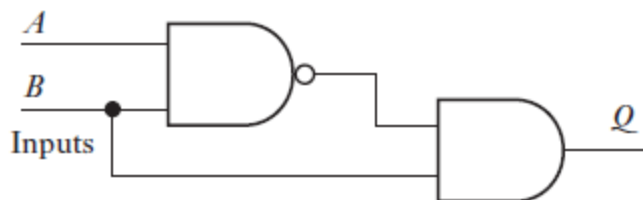
$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

$$\overline{A \cdot B} = \overline{A} + \overline{B}$$

Boolean Algebra to simplify the gates

$$Q = \overline{A \cdot B} \cdot B$$

$$\left. \begin{array}{l} \overline{A \cdot B} = \overline{A} + \overline{B} \\ \overline{A + B} = \overline{A} \cdot \overline{B} \end{array} \right\} \text{De Morgan's theorem}$$

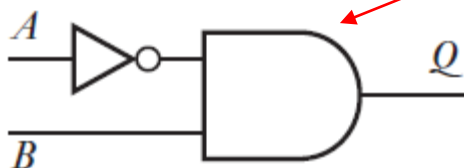


Applying the second law from above gives

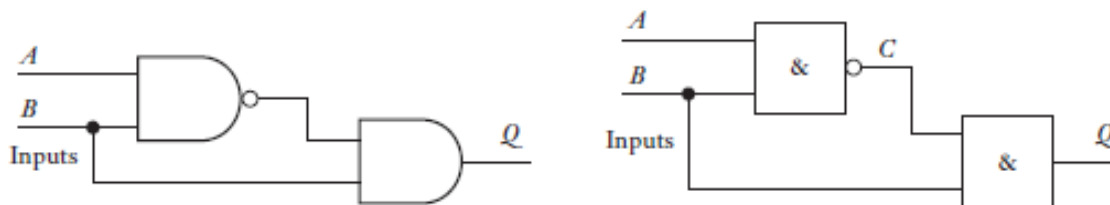
$$Q = (\overline{A} + \overline{B}) \cdot B$$

We can write this as

$$Q = \overline{A} \cdot B + \overline{B} \cdot B = \overline{A} \cdot B + 0 = \overline{A} \cdot B$$



De Morgan's Law *



The Boolean equation for the output in terms of the input is

$$Q = \overline{A \cdot B} \cdot B$$

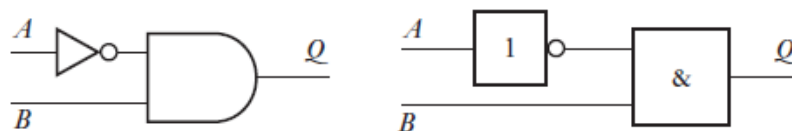
Applying the second law from above gives

$$Q = (\overline{A} + \overline{B}) \cdot B$$

We can write this as

$$Q = \overline{A} \cdot B + \overline{B} \cdot B = \overline{A} \cdot B + 0 = \overline{A} \cdot B$$

Hence the simplified circuit is as shown in Figure C.2.



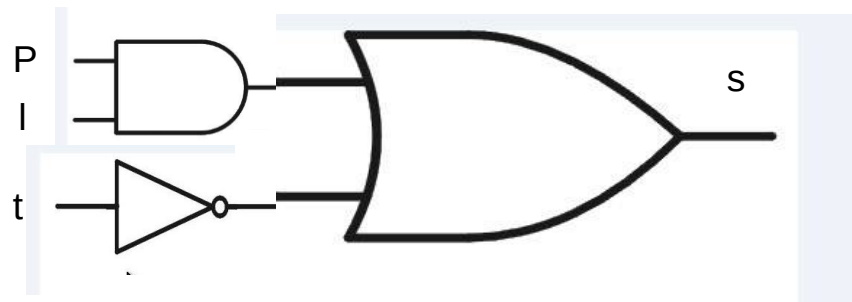
Example

An electronic switch that uses digital logic is to be developed for switching the lights on and off in an art gallery. The switch has to be turned on at 7:00 p.m. and turned off at 6:00 a.m. Also, if there are people in the gallery and the light that enters the gallery from outside is inadequate, the lights have to be turned on regardless of the time of the day.

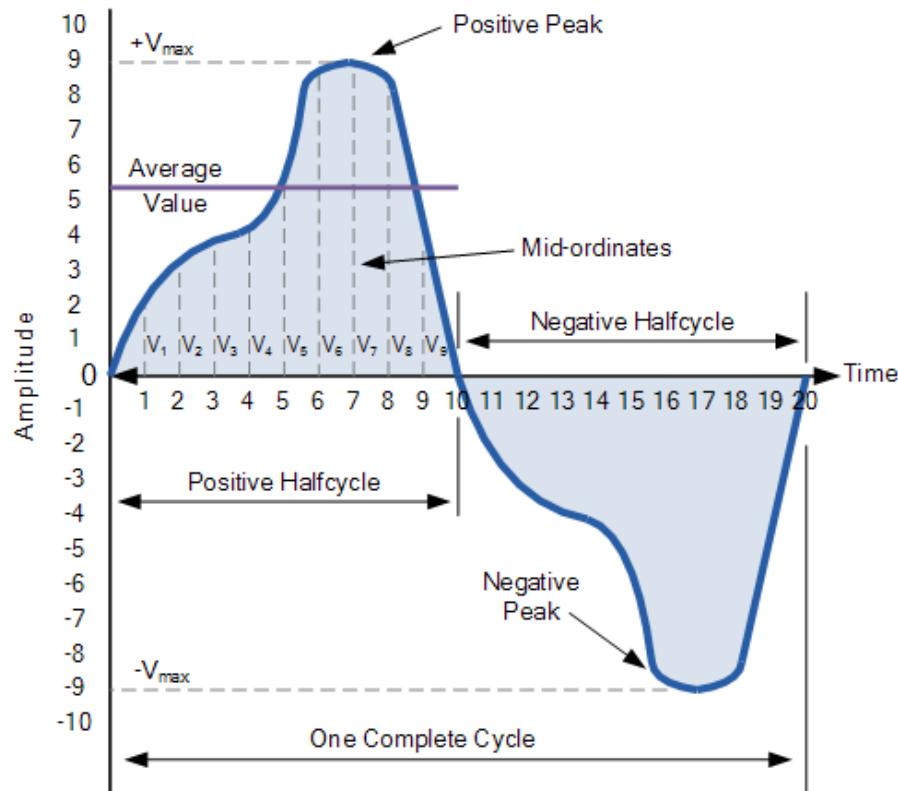
There are 3 sensors (Human, Light, Timer) – Timer goes high if time is between 6.00 AM to 7.00 PM. Human sensor goes high when people are there and light sensor goes high when light is inadequate

The logic state of the people sensor is denoted by p , that of the light sensor is denoted by l , and that of the time sensor

$$s = p \cdot l + \bar{t}$$



Voltage chopping



$$V_{\text{average}} = \frac{V_1 + V_2 + V_3 + V_4 + \dots + V_n}{n}$$

Thyristor (SCRs)

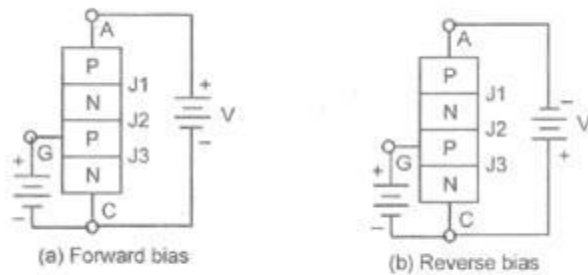
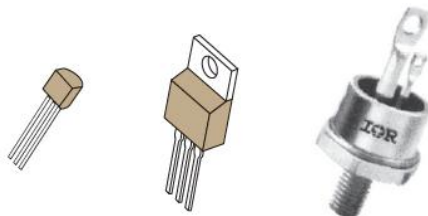
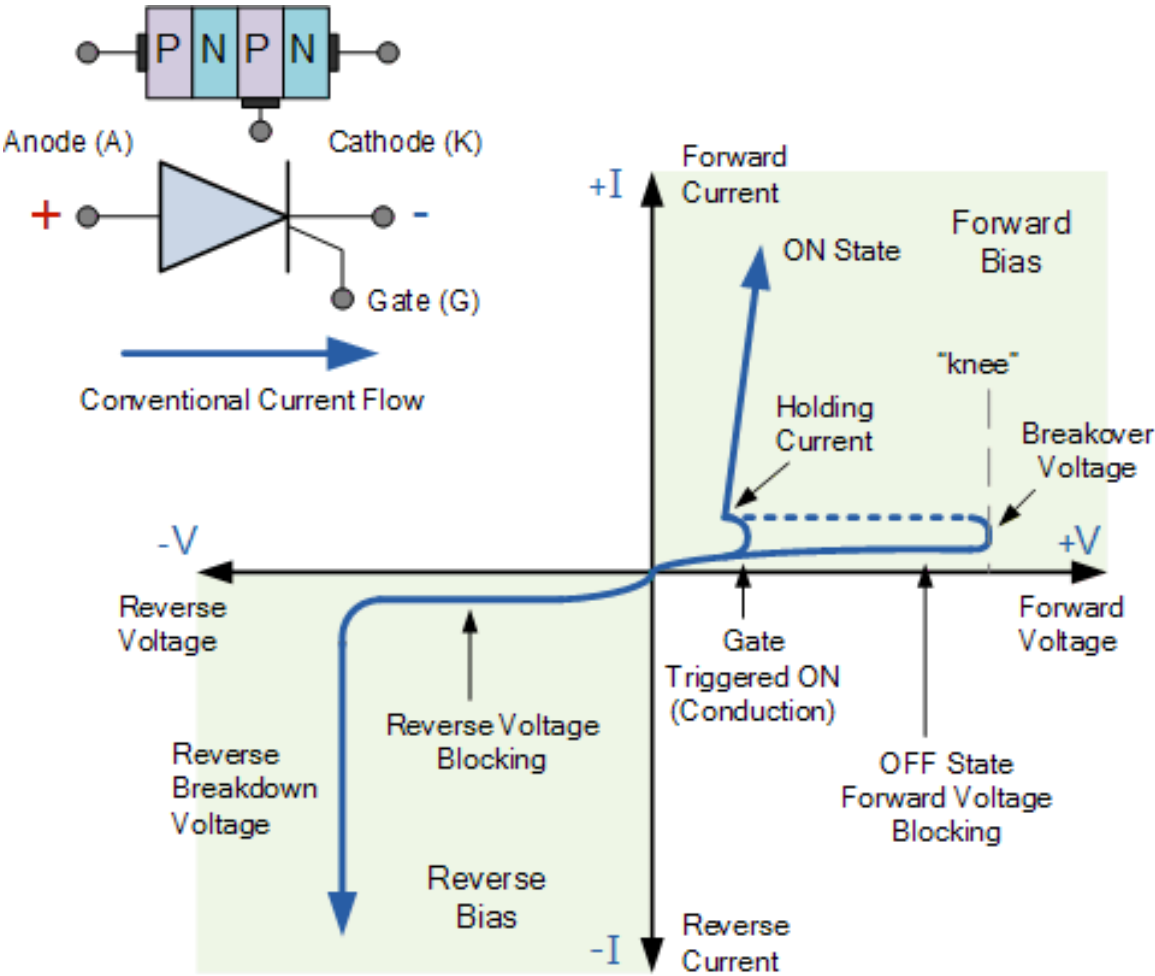
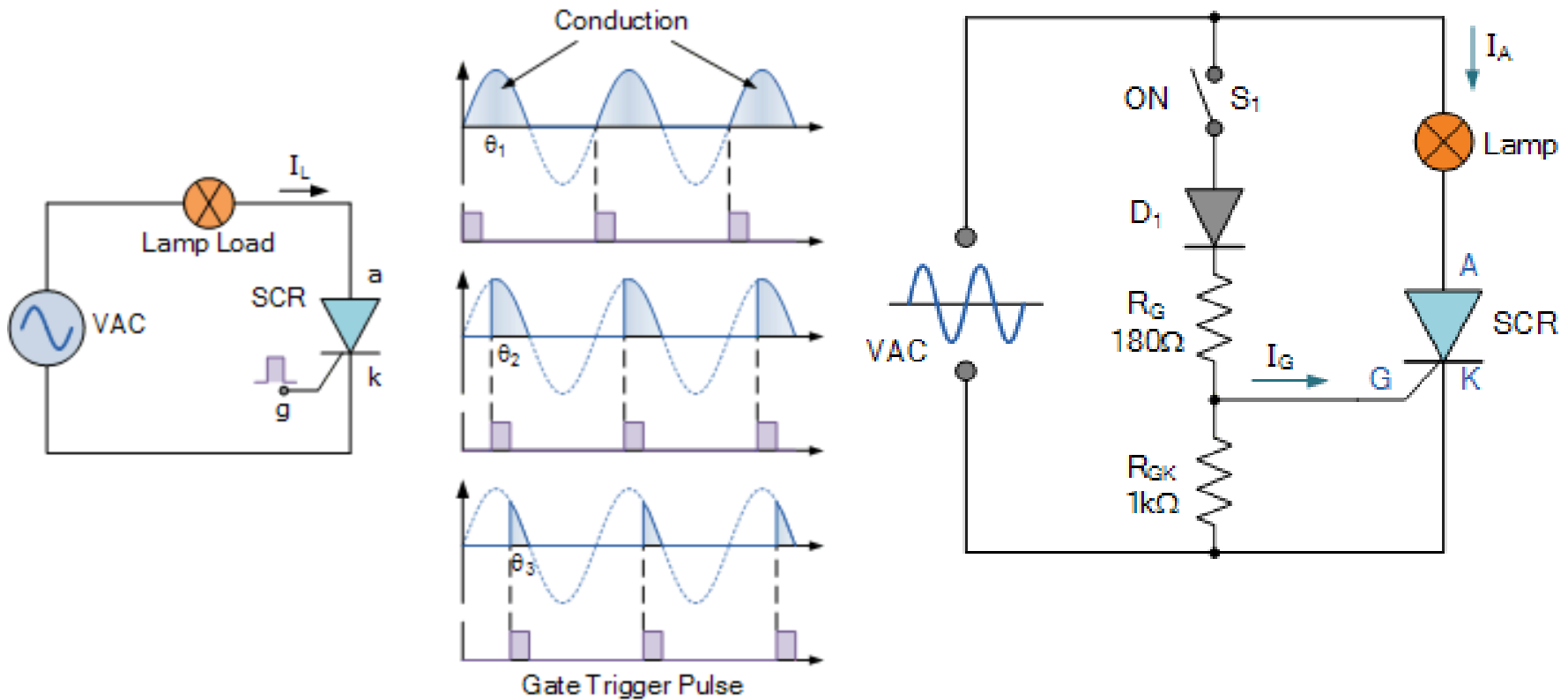


Figure 2 Biasing for SCR

Thyristor (SCRs)



By accurately varying the timing relationship between the Gate pulse and the positive half-cycle, the **Thyristor** could be made to supply any percentage of power desired to the load, between 0% and 50%

Triacs

